**ANNEX 1**

# Horizon Europe (HORIZON)

# Description of the action (DoA)

**Part A**

**Part B**

**DESCRIPTION OF THE ACTION (PART A)**

Associated with document Ref. Ares(2023)8308660 - 05/12/2023

# COVER PAGE

*Part A of the Description of the Action (DoA) must be completed directly on the Portal Grant Preparation screens.*

| **PROJECT**<br>*Grant Preparation (General Information screen) — Enter the info.* | |
|---|---|
| **Project number:** | 101138473 |
| **Project name:** | Sustainable Circular Economy Transition: From Industrial Symbiosis to Hubs for Circularity |
| **Project acronym:** | IS2H4C |
| **Call:** | HORIZON-CL4-2023-TWIN-TRANSITION-01 |
| **Topic:** | HORIZON-CL4-2023-TWIN-TRANSITION-01-37 |
| **Type of action:** | HORIZON-IA |
| **Service:** | HADEA/B/03 |
| **Project starting date:** | fixed date: 1 January 2024 |
| **Project duration:** | 49 months |

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# PROJECT SUMMARY

**Project summary**

*Grant Preparation (General Information screen) — Provide an overall description of your project (including context and overall objectives, planned activities and main achievements, and expected results and impacts (on target groups, change procedures, capacities, innovation etc)). This summary should give readers a clear idea of what your project is about.*

*Use the project summary from your proposal.*

Existing heavily industrialized areas are currently incapable of adopting large-scale industrial symbioses in terms of shared technology/infrastructure use, waste integration, energy and material utilization, as well as expanding through surrounding ecosystems in an inclusive manner for the society despite many of the EU's key strategic priorities in sustainable regional development. Within this context, IS2H4C proposes an ambitious and efficient innovation and action workplan to develop several solutions for the development of Hubs for Circularity (H4C) in diverse industrial areas of process industry surrounded by rural and/or urban settings in the Netherlands, Germany, Spain, and Turkey. The workplan is shaped by development and deployment of most innovative sustainable technologies and infrastructure integration in four demo hubs and is supported by ground-breaking research on societal, governmental, and business innovation for H4C. IS2H4C scales up industrial areas to H4C via implementing systemic change and integrating the surrounding ecosystems to industrial areas. The project implements a digital collaboration platform to manage the resource, infrastructure and information-sharing within H4C via embedding of decision-support modules in the platform. IS2H4C has the ambition of reducing the energy use, waste emissions, and carbon emissions by at least 10%, 20%, and 30% respectively.

IS2H4C will contribute to pave the path towards the development of H4C based on the circularity and resilience requirements of existing and future industrial zones and their surrounding ecosystems by prioritizing resource efficiency, maximizing use of renewable energy, prevention of waste, and promoting industrial/urban/rural symbiosis via reuse and recycling of unavoidable solid, liquid, and gas waste streams. All in all, IS2H4C's biggest ambition is to promote H4C as Europe's future sustainable regional development models.

# LIST OF PARTICIPANTS

**PARTICIPANTS**

*Grant Preparation (Beneficiaries screen) — Enter the info.*

| Number | Role | Short name | Legal name | Country | PIC |
|---|---|---|---|---|---|
| 1 | COO | UT | UNIVERSITEIT TWENTE | NL | 999900833 |
| 2 | BEN | MUL | MONTANUNIVERSITAET LEOBEN | AT | 999898117 |
| 3 | BEN | TUDO | TECHNISCHE UNIVERSITAT DORTMUND | DE | 999848453 |
| 4 | BEN | TEC | FUNDACION TECNALIA RESEARCH & INNOVATION | ES | 999604110 |
| 5 | BEN | INEGI | INEGI - INSTITUTO DE CIENCIA E INOVACAO EM ENGENHARIA MECANICA E ENGENHARIA INDUSTRIAL | PT | 996840968 |
| 6 | BEN | FHG | FRAUNHOFER GESELLSCHAFT ZUR FORDERUNG DER ANGEWANDTEN FORSCHUNG EV | DE | 999984059 |
| 7 | BEN | CIRCE | FUNDACION CIRCE CENTRO DE INVESTIGACION DE RECURSOS Y CONSUMOS ENERGETICOS | ES | 999516907 |
| 8 | BEN | ICCS | EREVNITIKO PANEPISTIMIAKO INSTITOUTO | EL | 999654356 |

**PARTICIPANTS**

*Grant Preparation (Beneficiaries screen) — Enter the info.*

| Number | Role | Short name | Legal name | Country | PIC |
|---|---|---|---|---|---|
| | | | SYSTIMATON EPIKOINONION KAI YPOLOGISTON | | |

| | | | | | |
|---|---|---|---|---|---|
| 9 | BEN | AEE | AEE - INSTITUT FUR NACHHALTIGE TECHNOLOGIEN | AT | 997235370 |
| 10 | BEN | ZLC | FUNDACION ZARAGOZA LOGISTICS CENTER | ES | 999608960 |
| 11 | BEN | EEIP | ENERGY EFFICIENCY IN INDUSTRIAL PROCESSES ASBL | BE | 950972287 |
| 12 | BEN | IMMIB | ISTANBUL MADEN VE METALLER IHRACATCI BIRLIKLERI | TR | 951975558 |
| 13 | BEN | H2HT | COOPERATIE WAARDEMAKERS IN WATERSTOF U.A. | NL | 882117807 |
| 14 | BEN | ACE | CLUSTER DE ENERGIA | ES | 998235925 |
| 15 | BEN | CABB | CONSORCIO DE AGUAS BILBAO BIZKAIA | ES | 925999831 |
| 16 | BEN | ALMELO | COOPERATIE DUURZAAM AADORP U.A. | NL | 882386691 |
| 17 | BEN | KPMG | KPMG & ASSOCIADOS SOCIEDADE DE REVISORES OFICIAS DE CONTAS SA | PT | 895029089 |
| 18 | BEN | TUPRAS | Turkiye Petrol Rafinerileri Anonim Sirketi | TR | 973148233 |
| 19 | BEN | ARCELIK | ARCELIK A.S. | TR | 998821708 |
| 20 | BEN | SKN | SMURFIT KAPPA NERVION SA | ES | 993254005 |
| 21 | BEN | NORTEGAS | NORTEGAS ENERGIA GRUPO SOCIEDAD LIMITADA | ES | 882164949 |
| 22 | BEN | SIDENOR | SIDENOR ACEROS ESPECIALES SL | ES | 956621858 |
| 22.1 | AE | SIDENOR I+D SA | SIDENOR INVESTIGACION Y DESARROLLOSA | ES | 964026353 |
| 23 | BEN | PETRONOR | PETRONOR INNOVACION SOCIEDAD LIMITADA | ES | 904204416 |
| 24 | BEN | CALERA | CALERA DE ALZO SL | ES | 882311807 |
| 24.1 | AE | CALCINOR | CALCINOR SERVICIOS SA | ES | 971282438 |
| 25 | BEN | LTK | LOINTEK INGENIERIA Y TECNICAS DE MONTAJES SL | ES | 938207087 |
| 25.1 | AE | LTK I+D | LOINTEK INVESTIGACION Y DESARROLLO SL | ES | 892189705 |
| 26 | BEN | SBS | SPOUTED BED SOLUTIONS SOCIEDAD LIMITADA | ES | 887121552 |
| 27 | BEN | IH | INFRASERV GMBH & CO. HOCHST KG | DE | 986186100 |
| 28 | BEN | CHX | CROWDHELIX LIMITED | IE | 905205165 |
| 29 | BEN | IRIS | IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA | ES | 998993786 |
| 30 | BEN | WDK | WORKDECK DIGITAL WORKPLACE SL | ES | 888804211 |

Associated with document Ref. Ares(2023)8308660 - 05/12/2023

**PARTICIPANTS**

*Grant Preparation (Beneficiaries screen) — Enter the info.*

| Number | Role | Short name | Legal name | Country | PIC |
|---|---|---|---|---|---|
| 31 | BEN | SOLENCO | SOLAR ENERGY CONVERSION POWER CORPORATION | BE | 917712927 |

# LIST OF WORK PACKAGES

**Work packages**

*Grant Preparation (Work Packages screen) — Enter the info.*

| Work Package No | Work Package name | Lead Beneficiary | Effort (PersonMonths) | Start Month | End Month | Deliverables |
|---|---|---|---|---|---|---|
| WP1 | Project Management | 1 - UT | | 1 | 49 | D1.1 – Management and quality plan report<br>D1.2 – Data Management Plan |
| WP2 | Technology development and technical aspects | 9 - AEE | | 1 | 49 | D2.1 – Technology map with key technologies, technical and economic feasibility, and contribution to synergies, including an assessment of their scalability D2.2 – Design specifications of energy conversion technologies ready for testing D2.3 – Waste & heat recovery technology design specifications usable for testing D2.4 – Design specifications of millichannel reactor that can convert the industrial CO2 into SNG with sufficient quality to be injected into the NG grid D2.5 – Design specifications of e-methanol synthesis unit<br>D2.6 – Design specifications of polymer production process<br>D2.7 – NIPU production process design and specifications<br>D2.8 – Performance analysis results compared to alternative technologies, including recommendations for full-scale implementation |

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|---|---|---|---|---|---|---|
| WP3 | Societal engagement and governance | 3 - TUDO | | 4 | 49 | D3.1 – Map of stakeholders and their interests/needs |

**Work packages**

*Grant Preparation (Work Packages screen) — Enter the info.*

| Work Package No | Work Package name | Lead Beneficiary | Effort (PersonMonths) | Start Month | End Month | Deliverables |
|---|---|---|---|---|---|---|
| | | | | | | D3.2 – Methodology for assessment of nontechnological issues<br>D3.3 – Set of indicators measuring the non-<br>technological impact of hubs D3.4<br>– Management guideline for integrating non-technological aspects |
| WP4 | Modelling and optimisation for H4C development | 1 - UT | | 4 | 49 | D4.1 – Modelling system for the characterisation of resources, energy flows and IS<br>D4.2 – Recommender algorithm for IS collaborations<br>D4.3 – Models for hub's supply chain management<br>D4.4 – Decision support tool for centralized/de-centralized hub<br>D4.5 – Agent-based simulation tool for hub development<br>D4.6 – Impact assessment of H4C from holistic perspective & decarbonisation pathways.<br>D4.7 – Virtual top-down hubs and their use as sustainable regional development models |

| | | | | | | |
|---|---|---|---|---|---|---|
| WP5 | Business and financial models | 17 - KPMG | | 1 | 49 | D5.1 – Guidelines for the True Value Framework<br>D5.2 – IS2H4C Business Model<br>D5.3 – IS2H4C Financial Plan |
| WP6 | Integration and demonstration in hubs | 4 - TEC | | 1 | 49 | D6.1 – Report on hub starting lines D6.2 – Report on hub installation progress and first 100 h of operation |

**Work packages**

*Grant Preparation (Work Packages screen) — Enter the info.*

| Work Package No | Work Package name | Lead Beneficiary | Effort (PersonMonths) | Start Month | End Month | Deliverables |
|---|---|---|---|---|---|---|
| | | | | | | D6.3 – Report on continuous operation & technology demonstration<br>D6.4 – Report of overall hub performance, outcomes, and dissemination |
| WP7 | Digital Collaboration Platform for H4C | 29 - IRIS | | 1 | 49 | D7.1 – Early release (alpha version) with the first layer of functionality<br>D7.2 – Online marketplace platform add-in D7.3 – Extended version of DigitalH4C integrated with the digital tools<br>D7.4 – Report on risk assessment & mitigation – digital platform security D7.5 – Final delivery of DigitalH4C platform |

| | | | | | | |
|---|---|---|---|---|---|---|
| WP8 | Dissemination, communication & exploitation | 11 - EEIP | | 1 | 49 | D8.1 – Dissemination and communication plan<br>D8.2 – Dissemination and communication plan - Update 1<br>D8.3 – Dissemination and communication plan - Update 2<br>D8.4 – Report on dissemination and communication activities<br>D8.5 – Exploitation plan report<br>D8.6 – Exploitation plan report - Update 1<br>D8.7 – Exploitation plan report - Update 2<br>D8.8 – Circular Industry helix “virtual cluster/ community” building |
| WP9 | Ethics requirements | 1 - UT | | 1 | 49 | D9.1 – OEI - Requirement No. 1 |

## Work package WP1 – Project Management

| **Work Package Number** | WP1 | **Lead Beneficiary** | 1. UT |
|---|---|---|---|
| **Work Package Name** | Project Management | | |
| **Start Month** | 1 | **End Month** | 49 |

**Objectives**

The goal of WP1 is to ensure the effective project technical, financial, administrative and legal management, enabling the completion of the objective with the required quality and timely execution.

**Description**

T1.1 Decision-making management (M1–M49) Leader: UT; Participants: All; Deliverable: 1.1
Periodic General Assembly (GA) meetings will be held every six months for progress review, decision making, risks review and conflict resolution. Progress meeting minutes will be produced after each meeting and distributed to GA members. In addition, a summary of the progress meeting will be shared with the PO. For the daily work, the interaction between WPs is fundamental, hence an interaction activity will be included in each WP in addition to the overall management.

T1.2 Administrative and financial management (M1–M49) Leader: UT; Participants: All; Deliverable: 1.1
All administrative tasks will lie in the hands of UT. These include the information/advice to partners on administrative, regulative and financial issues, the organisation of report distribution and archiving, the management of the restricted project area on the internet-based dissemination and communication platform and its implementation in the everyday management workflow. All tasks regarding financial management will lie in the hands of the coordinator, UT. These tasks include the setup and maintenance of financial records, the coordination and control of annual cost claims and certificates on the financial statements (if needed) submitted by all project partners, follow-up of EC payments, distribution of partners' shares and the monitoring and justification of payments. UT will be in charge of reporting and communication with the EC, including the communication of all necessary information, the submission of progress periodic reports, and of financial statements.

T1.3 Contractual management (M1–M49) Leader: UT; Participants: All; Deliverable: 1.1
UT will be responsible for the preparation, collection and maintenance of the EC Grant Agreement and other contractual documents. IS2H4C consortium will conclude a Consortium Agreement (CA), which will include detailed regulations on the consortium's governance structure, responsibilities and duties of the partners, liability towards each other, financial provisions, knowledge management, intellectual property rights, dissemination rules and access rights as well as nondisclosure agreements. Further contracts will be negotiated if necessary.

T1.4 Quality management (M1–M49) Leader: UT; Participants: WDK, All; Deliverable: D1.1
The Project Coordinator (UT) will prepare early in the project a "Management and Quality Plan" (D1.1), supporting the scheduling and monitoring of activities. The plan will include the: (i) implementation of a quality assurance procedure, including deliverables peer review when appropriate; (ii) organisation of an efficient continuous project progress monitoring methodology and supervision of the project tasks, milestones, budget, etc.; (iii) evaluation of the expected impact of issues that might affect the project implementation in terms of delays, quality of work, inferred repercussions on other activities, and other similar factors; and (iv) identification and analysis of risks as well as suggestions for fallback solutions (contingency planning). To digitally facilitate the project execution, monitoring and control, and partner engagement, WDK will host the IS2H4C project on its software in M1, whereby the partners will be able to obtain highlevel visibility of the dynamic evolution of the Gantt, access boards for managing and implementing on their tasks, with embedded risk management tools, and they will also benefit from a secure document repository, communication tools, and on-demand reports.

T1.5 Data management (M1–M49) Leader: UT; Participants: All; Deliverable: D1.2
The data generated will be handled in the best feasible way to maximize impact and inspired by FAIR principles. This will be agreed in the CA. UT will prepare and regularly update the Data Management Plan (DMP, D1.2), which will detail what data will generate, whether and how they will be exploited or made accessible for verification and re-use in the implementation of the other WPs and how the data will be curated and preserved after the completion of the project. UT will also monitor and deal with ethical issues in collaboration with the consortium partners.

## Work package WP2 – Technology development and technical aspects

| Work Package Number | WP2 | Lead Beneficiary | 9. AEE |
|---|---|---|---|
| Work Package Name | Technology development and technical aspects | | |
| Start Month | 1 | End Month | 49 |

**Objectives**

To develop, analyse, and optimise key technologies to maximise industrial synergies and contribute to the circular economy by focusing on energy conversion, waste recovery and utilisation, and CCUS options. Simulation models of technologies will enable the forecasting of system behaviour and optimisation of the circularity pathways on a technical level. A techno-economic analysis of the technologies will be performed closely with the subsequent WPs. The outcomes will provide crucial inputs for the subsequent WPs, ensuring the effective integration of advanced technologies within the project's framework.

**Description**

T2.1 Identification, classification & analysis of key technologies (M1–M6) Leader: MUL; Participants: AEE, INEGI, TEC; Deliverable: D2.1
The aim is to determine the system boundaries of the H4Cs, enabling a technology mapping for each hub as well as the associated industry and infrastructure. Subsequently, key technologies for maximising synergies will be identified, focusing on the technologies further developed within the project and represented by contributing technology providers. The technologies will be described in detail, analysed, and classified in terms of their targeting (e.g., electrification, electrolysis, green gases, CCU/CCS). To evaluate the technologies and their contribution to synergies, KPIs (e.g., specific CO2 emissions, TRLs, useful life, efficiencies, specific CAPEX, and OPEX) will be collected to develop a technology map (D2.1) for the hubs in coordination with WP4 and 6. The technology-related data will be used in WP4 to assess sustainability, efficiency, and circularity gains at the hub level and can be included in the digital collaboration platform in WP7.

T2.2 Development of energy technologies (M6–M30) Leader: UT; Participants: INEGI, SOLENCO, CIRCE; Deliverable: D2.2
The main aim is to develop fully integrated, large-scale, multi-vector energy conversion systems supported by hydrogen that would be deployed in the demo hubs. The aim is to explore how hydrogen can be produced, stored, transported, and used to integrate and optimise the operation of multiple energy vectors, including electricity, heat, and fuels. The techno-economic performance, efficiency, and environmental impact will be investigated and reported (D2.2), with a particular focus on the challenges of hydrogen integration, including grid and infrastructure, storage, transportation, and distribution.

T2.3 Development of waste recovery and utilisation technologies (M6–M30) Leader: AEE; Partners: UT, INEGI, CIRCE, TEC; Deliverable: D2.3
T2.3 will develop two measures for waste recovery and utilisation: 1) A system for the cascading re-use of waste heat for low-temperature heat supply and re-use of wastewater. The potential of such approach will be analysed among the hub participants, developing concepts for the optimised use of industrial waste heat for industrial and commercial use. Solar heat integration and heat pump solutions will contribute to the heat cascade. 2) Replicating an existing Direct Contact Membrane Distillation (DCMD) pilot container system. The technology introduces a critical link to the circularity of nitrogen extracted from wastewater. A pilot container will be adapted to the conditions of the potential operation site. Design specifications will be reported (D2.3). 3) Aiming at getting a higher C-content solid fuel, microwave-assisted HTC will be the first stage of the process to produce a suitable hydrochar that will be further fed into the gasification process to produce the syngas. CIRCE will conduct experimental lab-scale trails of MW-assisted HTC of sewage sludge and gasification of hydrochar, to validate this valorisation route. Experimental conditions of MW-HTC will be optimised to maximise the yield towards hydrochar and the solid's C-content. Different gasification tests will be implemented varying the operating conditions to promote the hydrogen production in a fluidized bed reactor (2 kg/h capacity). Due to the hydrochar production limitations of the MW-HTC reactor (1 L capacity), commercial hydrochar with similar physical-chemical characteristics will be used as gasification feed.

T2.4 Development of carbon capture & methanation technologies (M6–M30) Leader: LKT; Participants: TEC, CALCINOR, CIRCE; Deliverable: D2.4
CO2 capture through innovative membranes and PSAs (Pressure Swing Adsorption) will be developed with the support

of TEC and deployed in Calcinor with a novel millichannel technology to convert the CO2 into synthetic methane (with enough quality to directly inject into the gas grid). The CO2 generated at Calcinor will be captured and stored. For the captured CO2, it will be investigated which technologies based on polymeric membranes and PSA (pressure swing adsorption) using specific zeolites to retain this CO2 should be developed (TRL5). The technologies developed in IS2H4C will treat 5Nm3/h of combustion gases. The CO2 captured (1Nm3/h) will be converted into synthetic natural gas (SNG) using a novel reactor (millichannel reactor TRL5) and will be brought to TRL7 to produce 1Nm3/h of SNG from 1 Nm3/h of captured CO2. The technologies will be designed and constructed (LKT) to upgrade the system to TRL7. The CO2 capture equipment and a container will be installed and tested at Calcinor in WP6 based on the reported design specifications (D2.4).

T2.5 Producing e-methanol from captured carbon (M6-M42), Leader: FHG; Participants: TUP, IH, CIRCE; Deliverable: D2.5
The objective of this task is the development of a Methanol synthesis based on CO2/H2 or CO2-rich synthesis gases. Laboratory and pilot units for the synthesis of Methanol as well as the necessary downstream operations to separate the Water/Methanol mixtures and analyse products and side products, are used. The synthesis gas will be varied in its composition to be able to use as many industrial residual gas flows as possible. With the technical equipment, Methanol samples up to 50 kg/week can be provided based on project-specific synthesis gases. A system analysis and optimisation for cross-industrial approaches and techno-economic analysis will be provided (D2.5). For system models, process models and process simulation from the overall methanol production pathway and its unit operations up to detailed reactor models for dynamic methanol synthesis based on fluctuating gas streams that vary in composition and volume flow will be used.

T2.6 Synthesis of dicarbamates from CO2, methanol and diamines (M6-M42) Leader: FHG; Deliverable: D2.6
The synthesis of the dicarbamates is developed in high-pressure reactors. The optimal reaction conditions in terms of catalysts used and experimental parameters are carried out on a laboratory scale with a volume of 63 mL. Based on the laboratory-scale experiments, the synthesis of dicarbamates is carried out on a kg scale as a building block to produce polyurethanes (design specifications in D2.6).

T2.7 Designing and developing NIPU production from captured carbon (M6–M42). Leader: FHG; Participants: TUP; Deliverable: 2.7
CO2-based carbamates will be used to synthesise non-isocyanate polyurethanes (NIPUs). Material properties are to be defined by industrial partners to suit the application of NIPU materials in the demonstrators. The NIPU recipes will be developed on a lab scale (50 g) to envisage the desired material properties. Thermal and mechanical properties of the synthesised NIPUs will be characterised. Successful recipes will be scaled up to the technical centre scale (5 kg) to yield three different types of NIPU material (design specifications in D2.7).

T2.8 Performance analysis & optimisation via simulations (M25-M49) Leader: UT; Participants: AEE; Deliverable: D2.8 The main objective of this task is to analyse and optimise the performance of the developed technologies and processes via simulations. This task aims to identify potential optimisation opportunities for the developed technologies and processes. The milestones for this task are developing simulation modelsfor the technologies and processes, and identifying potential optimisation opportunities. The outputs of this task include (i) the simulation results ; (ii) test data and analysis, optimisation plans and recommendations, and a report detailing the findings and recommendations for optimisation; and (iii) design modifications to improve performance and recommendations for full-scale implementation (D2.8).

## Work package WP3 – Societal engagement and governance

| **Work Package Number** | WP3 | **Lead Beneficiary** | 3. TUDO |
|---|---|---|---|
| **Work Package Name** | Societal engagement and governance | | |
| **Start Month** | 4 | **End Month** | 49 |
| **Objectives** | | | |

| |
|---|
| Identify relevant stakeholders and their connection to H4Cs. These include industrial companies within the hub, citizens living in the surrounding of the hub, policy makers & the general public. WP3 considers these stakeholders and identifies non-technical (regulatory, ethical, societal) challenges and barriers that arise in connection with the establishment of hubs. It will implement a series of tasks through social living labs and involve them from the beginning of the project |
| at local and EU level. It will allow the definition of needs, exchanging knowledge, definition of best strategies based on the stakeholders' needs. |

| **Description** |
|---|

T3.1 Addressing stakeholders' needs and interests(M4–13) Leader: TUDO; Partner: UT, Hub representatives (TUDO, ACE, SIDENOR, SIDENOR I+D, ARCELIK, UT), IRIS; Deliverable: D3.1

Subtask 3.1.1 Define relevant stakeholder groups and needs (M4-M13) (Leader: TUDO; Participants: UT).
TUDO will develop a theoretical framework which includes the relevant stakeholder groups and non-technological topics (regulatory, political, ethical and societal) to be addressed. By means of a stakeholder analysis, relevant stakeholders that are (directly/indirectly) affected by innovation are mapped. It will analyse the industrial, rural, and urban settings and actors within and in the surrounding ecosystems of the hubs. The regional actors (identified as key stakeholders) will be assigned to the theoretical framework. Based on desk research and interviews (at least one per key stakeholder group of each hub), the general needs, interests, interactions and potential challenges, barriers and conflicting interests of diverse stakeholders will be identified. In a second step, more specific interests and needs of stakeholders will be identified, e.g., the main barriers faced by industrial agents for the deployment of the different decarbonisation technologies analysed in WP2 and the enablers that would allow to overcome them. It also ensures the engagement and addresses the concerns of "Not in my back yard" (NIMBY) citizens as well as enable the societal engagement of "prosumers" (link to WP8). UT will use tools suggested at Constructive Technology Assessment (CTA) toolbox to broaden technology development and implementation by including various aspects and actors. Part of this task is also a benefit analysis of the hub concept for households by providing cheaper (district) heating for small communities (e.g., implementing positive energy districts). This provides links to WP 2&5 (T2.1 and T5.5).

Subtask 3.1.2 Increasing trust and acceptance (M4-M13) (Leader: UT; Participants: TUDO, hub representatives.
By developing a standardisation framework on the quality of waste streams, it shows how to facilitate the use of waste and increase trust among companies on the use of waste in production activities. Facilitation of trust among stakeholders is needed to increase the acceptance of circular products derived from circular streams. Further, in this task, it deploys long-term policymaking (state-of-the-art of circular economy policymaking aligned with the 5-stage policy cycle based ) to impede greenwashing and to spread the share of truthful information about circular resource streams and circular products.

Subtask 3.1.3 System thinking: Socio-technical configuration, multi-level perspective and system dynamics (M4-M13) (Leader: UT; Participants: TUDO, Hub representatives).
UT will map the socio-technical system/configuration (e.g., framework of regulations and policies; distribution of skills of workers; conformity with standards; environmental considerations and specific requirements – link to WP4) as part of the multi-level perspective. Three levels are covered: Regime (standard/established system), landscape (characteristics of the system that cannot be changed easily like climate or demographic change, dependence on finite resources) including broader societal, economic and environmental developments, and niche (protected space within socio-technical system from with new innovations and models emerge (link to WP5). It addresses how innovation impact is diffused among the society. This will be done, for instance, by means of a social network analysis.

T3.2 Establishing the working structure for integrating topics (M8–M42) Leader: TUDO; Partners: ICCS, KPMG, TUDO, TUPRAS, TEC, ALMELO; Deliverable: 3.2
Expert discussion panels will be established to raise and discuss the general non-technological issues of IURS ensured by living labs that will be taken up and specified by the regional hubs.

Subtask 3.2.1 Establishing expert discussion panels (M8-M42) (Leader: KPMG; Participants: TUDO, ICCS, hub representatives).
Based on web and desk research, general non-technological issues of IURS will be raised. Current challenges will be identified when it comes to the implementation of the IS2H4C solutions. Expert discussion panels will be established to enable them to discuss transversal topics, namely green industry; regulatory and political; ethical; and societal topics. Each discussion panel will be chaired by a project partner with an expertise for the mentioned topics (e.g., Green Industry – KPMG, Societal engagement – TUDO), taking place in different locations (Netherlands, Spain, Germany, and Turkey). KPMG will lead this sub-task, building upon the outcomes of the Intelligent Cities Challenge (ICC) project and leveraging from the know-how within the "Green Economy and Local Green Deals" thematic theme. KPMG will collaborate with local project partners for each expert discussion panel implementation, and support in the engagement of international and regional stakeholders. Results will feed T3.3 where insights will be applied to the demo hubs.

Subtask 3.2.2 Applying insights on non-technological topics to the hubs (M4-M13) (Leader: TUDO; Participants: ICCS) Deliverable D 3.2

In the hubs as regional pilot cases, living labs composed of regional stakeholders will be established and work on the regulatory, political, ethical and societal topics. The living labs will be fed by results of T3.2.1. These will be compared with the specific interests and needs of the regional pilot cases (fed by results from T3.1). Potential solutions for

identified challenges will be developed (D3.2). Technical developers will present proposed solutions, mock-ups and prototypes in the living lab, so ideas for technological and non-technological solutions can be integrated.

T3.3 Development of socio-economic indicators to measure regional impact of hubs (M6–M49) Leader: UT; Participants: TUDO, TUPRAS, ACE; Deliverable: D3.3
Indicators will be developed to measure the process of stakeholder engagement and non-technological impact (D3.3). Stakeholder engagement will be measured by the Societal Readiness Level (SoReL). To determine the SoReL, it develops indicators to measure the impact of the solutions in terms of value for stakeholders to determine the regional (direct, indirect, induced) impact of hubs ensuring the resilience and the sustainability of the region considering the principles of circularity and priorities of circular economy pyramid and SDGs. Additionally, UT develops (multi-)regional inputoutput (MRIO) models to measure the regional impact of hubs. For that, it uses regional input-output tables to monitor the initial state in each hub region and to set-up a regional impact framework for each hub region. Also, it formulates potential trade tables between a hub region and surrounding regions in order to set-up a multi-regional input-output analysis framework. Then it finds environmental interventions of hub regions based on different governance scenarios and apply enterprise input-output model to the demo H4Cs and regional and multi-regional input-output analysis to the regions where the demo hubs are located. Thanks to the input-output models (achieved employment, re-distribution of income, attracted investment in region, income shift from unsustainable to sustainable sectors, etc.) socioeconomic and sustainability indicators can be computed. Based on these indicators, progress of non-technological issues will be monitored covering the extent to which stakeholders are involved, the achieved value for stakeholders and how these indicators contribute to an improved SoReL.

T3.4 Developing a management guideline for integrating non-technological aspects into the hubs (M37–M49) Leader: TUDO; Participants: ICCS, ACE, TUPRAS, UT; Deliverable: D3.4
Based on methodology, best practices, lessons learned and other findings from the hubs, a management guideline will be developed that integrates non-technological aspects into the (technological) project solutions for the follower hubs (D3.4). It will show how requirements of different stakeholders will be raised, how they will be fulfilled/addressed and how relevant stakeholders will be involved in developing solutions. The guideline will advise how the different aspects can be integrated into a coherent methodology to ensure viable solutions for H4C taking up results from T3.2 on how to embed technological solutions into a regional environment.

## Work package WP4 – Modelling and optimisation for H4C development

| **Work Package Number** | WP4 | **Lead Beneficiary** | 1. UT |
|---|---|---|---|
| **Work Package Name** | Modelling and optimisation for H4C development | | |
| **Start Month** | 4 | **End Month** | 49 |

**Objectives**

Develop methodology to identify, coordinate, and evaluate IS collaborations. Actors and companies are characterized based on their resource and geolocation properties, and an intelligent matchmaking system is developed to recommend IS opportunities. H4Cs are optimized by ensuring that material resources flow efficiently via logistics and supply chain analytics, and that energy resources are utilized & exchanged in the hub and with the energy networks efficiently. It also analyses how H4Cs can successfully evolve from IS networks to full circular hubs and studies their impact on decarbonisation. Methodology will include state-of-the-art techniques from AI, operations research, and agent-based simulation.

**Description**

T4.1 Modelling and geolocation of resources, energy, and industrial symbiosis (M4–M28). Leader: TEC; Participants: AEE, MUL, ACE, SIDENOR I+D; Deliverable: D4.1
The modelling structure that identifies the main criteria and factors for the maximisation of IS synergies will be developed (D4.1). The characterisation of industries and elements (technologies, local resources, etc., from T2.1) in terms of energy and materials production and demand, and synergetic opportunities, will be carried out with geo-referenced information. TEC will integrate data about the energy and products/subproducts (e.g., H2, CO2, O2, CH4, feedstock such as lignin,

slags, wastewater, urban waste), transport and production infrastructure options (e.g., natural gas/bioCH4 pipelines, Hydroduct, in-situ production). AEE will model the integration of heat production and demand potential in the system and the surroundings to maximise urban-industrial symbiosis (especially industrial waste heat integration to district heating). By leveraging technology-specific analyses from T2.1, MUL will investigate how flexible energy flows of industrial energy systems can maximise different levels of symbiosis (industrial-industrial, urban-industrial). Companies and resources are classified into a structured digital database for easy mapping and analysis.

T4.2 Matchmaking for industrial symbiosis (M6–M30). Leader: UT; Participants: ICCS, IMMIB, KPMG, ACE; Deliverable: D4.2
T4.1 provides input to this task where potential IS synergies are evaluated based on partners compatibility, resource locations and availability, transportation distances, infrastructure requirements, economic viability, and potential environmental impact reduction. To exploit these synergies and improve the efficiency of IS identification, UT will develop a matchmaking framework for IS based on recommender algorithms to match potential partners and recommend alternatives of symbiotic exchanges. It facilitates the exploration of available options while filtering a series of items to reduce each actor's information burden at a managerial level. ICCS, IMMIB and KPMG will support this task by providing their expertise on matchmaking and identification of synergies as input to the recommender algorithms (e.g., involving entities from the innovation and industrial ecosystem in Widening countries). The resulting recommender system (D4.2) will be embedded in the digital information platform developed in WP7 by taking inputs from other WPs.

T4.3 Hub logistics and supply chain management (M6–M37). Leader: ZLC; Participants: UT, ICCS, ACE; Deliverable: D4.3
Using the inputs from T4.1 and T4.2, ZLC will develop supply chain management models (D4.3) to optimize the logistics operations of the hubs' participants. These models will embed advanced quantitative analytics to improve capacity planning (production and processing), demand forecasting and shaping, inventory management, network design (production facilities, distributors/retailers, transport infrastructure), sustainable supply chain management strategies (carbon footprint accounting and decarbonisation), supply chain contracts (pricing decisions, incentive mechanisms), sales and operations. UT will support this task with ample experience in developing AI and optimisation methods for logistics processes. Moreover, a tracing conceptual framework to support data definition and information management across the chains will be developed by ICCS, including operations, materials, products, services and their attributes (state, origins, constitution, quality, footprint, availability etc.). Units of raw materials and products (assembly of raw material) will be defined, and methodologies for their digital representation will be introduced to contribute to the digital modelling of T4.1. A conceptual Digital Product Passport will be developed based on existing schemes and regulations (e.g., REACH for chemicals, BOMs, LCA). It will enable the comparison of flows' properties and patterns with regulation, environmental and quality criteria, their characterisation and standardisation, supporting the establishment of performance, circularity, and sustainability metrics.

T4.4 Energy system operation strategies and optimisation (M6–M49). Leader: UT; Participants: MUL, TEC; Deliverable:
D4.4
To get a holistic perspective on how a hub and its participants should operate, UT will investigate centralised and decentralised energy management perspectives. The centralised perspective represents the hub as a coordinating actor that optimises the distribution of energy flows across its components and develops unified bidding strategies towards the national energy markets to maximise its overall efficiency. The decentralised perspective optimises instead the operational decisions and bidding strategies of individual assets within the hub (e.g., renewable energy producers, hydrogen producers, energy consumers) to maximise their own performance when interacting with the hub and the national grids. These complementary perspectives establish two different energy system optimisation models for H4Cs. MUL will focus on the design and operation optimisation for implementing energy key technologies (WP2) in industrial energy systems to forecast industrial energy flows (e.g., electricity for hydrogen production, waste heat for district heating) and flexibility options of industrial processes (e.g., peak shaving) used as input to the energy flow optimisation. The developed methodologies will build on state-of-the-art mathematical optimisation and AI techniques such as stochastic programming, reinforcement learning, network flow algorithms, neural networks, and heuristic optimisation. The resulting algorithms (D4.4) will be included in the digital collaboration platform in WP7.

T4.5 Agent-based analysis of H4C (M13–M49). Leader: UT; Participants: MUL, AEE, ACE; Deliverable D4.5
As IS networks evolve towards H4Cs on a larger scale, the complexity increases not only in the dimension of physical exchanges but also regarding stakeholder roles, coordination patterns, and evaluation methods. This task identifies, categorises, and summarises the existing evolving mechanisms of such an expansion based on the inputs from WP 2, 3, 4 and 5. For this, UT will develop an agent-based simulation tool (D4.5) that enables the analysis of future scenarios of hub evolution. Since a circularity hub is an inherently complex adaptive system, agent-based simulation can capture the

behavioural patterns of stakeholders as agents (companies, households, financial institutions, governmental bodies, etc.) and foresee how H4Cs systemically evolve. This enables to understand how agent interactions at the bottom level will

lead to the emergence and expansion of H4Cs at the top level. This will help to derive theories for hubs' evolution and shed light on the future expansion strategies of hubs. MUL will support by defining agents for industrial systems and by integrating the relevant technologies identified in WP2. As a simulation scenario, AEE will support the impact analysis on heating demand related to surrounding ecosystems. It will demonstrate and evaluate the systematic evolvement of hubs triggered by individual decisions of adopting green energy technology and integrating energy infrastructure.

T4.6 Impact assessment & decarbonisation pathways for H4C (M25–M49) Leader: TEC; Participants: UT, IRIS; Deliverable: D4.6
This task develops the methodology for impact assessment of the energy and GHG emissions reduction opportunities and the economic effects of H4Cs. Outputs from T4.1-T4.4 will be used to evaluate the impact of hubs on longterm decarbonisation pathways, considering prospective energy evolution and prices. Long-term scenarios will allow evaluating hubs' effect on the entire regional energy system and make projections for carbon emission reductions. The evaluation results will be also taken as iterative feedback in T4.2-T4.4 to improve the recommender's accuracy and enhance the decision support on operational supply chain analytics and energy management, obtaining a bi-directional assessment approach. Once the calculation and modelling approach is closed, the impact assessment of the specific demo H4C of the projects will be carried out (D4.6).

Task 4.7 Top-down design of virtual hubs and their integration on expansion pathways for long-term impact (M25–M49). Leader: UT; Participants: TEC, Hub representatives, IRIS, ACE, CIRCE; Deliverable: D4.7
UT and TEC will analyse an expansion scenario beyond each hub both at regional and interregional levels using techniques such as location analysis while considering the strategic importance of existing and to-be-implemented infrastructure. These infrastructures include logistical corridors available in river/road transportation (e.g., connecting the Basque hub to Zaragoza hub –REDOL project being run by CIRCE–), existing natural gas pipelines (e.g., connecting the Dutch hub to the Groningen hydrogen valley –Heavenn project–), and to-be-developed H2 transportation infrastructure (e.g., planned H2 infrastructure from Portugal/Spain to the Netherlands/Germany, –H2MED project–). In the future, H4Cs in the proximity of existing natural gas infrastructure and the H2 backbone of Europe will get interconnected. Using existing industrial symbioses databases (such as MAESTRI) and the outcomes from bottom-up hub development from the demonstration hubs, virtual H4Cs will be developed and potential strategies for opening these hubs in the proximity of above-mentioned infrastructure will be assessed. This task strategically positions H4Cs as future sustainable and resilient regional development models all over Europe and delivers a strategy report on sustainable regional development considering future prospective scenarios (D4.7).

## Work package WP5 – Business and financial models

| **Work Package Number** | WP5 | **Lead Beneficiary** | 17. KPMG |
|---|---|---|---|
| **Work Package Name** | Business and financial models | | |
| **Start Month** | 1 | **End Month** | 49 |
| **Objectives** | | | |
| WP5 aims at designing and developing an impact valuation model, using the “True Value Methodology”, to monetise and assess the potential of impact valuation as a project screening tool in IS2H4C investment vs. other existing screening methodologies. Economic, environmental, and social impact indicators (WP3 and WP4) will be applied to the pilots being demonstrated. Valuation factors will be applied from KPMG’s existing database and external sources including input/output data, and the regional input-output models considered in WP3. | | | |
| **Description** | | | |

Task 5.1 Objective setting, scoping and baselining (M1-M6) Leader: KPMG; Participants: Hub Leaders, UT, IRIS; Deliverable: D5.1
The KPMG True Value methodology will help estimate the Region's Hub "true" earnings, understand future earnings at risk, and create societal value by assessing the H4C externalities to the Region. KPMG will work with hub leaders and participants to develop the True Value Framework's objectives and economic, environmental, and social metrics. The future thorough analysis must analyse all aspects of the present business models, including their products or services, client groups, pricing plans, distribution routes, market strategies, and revenue models, using data from the project and WPs. This analysis compares and proposes different business models using coopetition concepts (e.g., 1-1 business and M-M business networks) to reduce risk for businesses and their financial evaluation by minimising costs and increasing

credit capabilities. Organisational/product focus, value-chain boundary, value viewpoint, impacts and/or dependencies, baseline and time ranges, and topic selection are also important scoping aspects. A maximum of four additional indicators per trial (focused on H2, carbon capture, wastewater treatment) will be selected by the hub leaders and KPMG to create a "Theory of Change" for effect monitoring. Based on predicted impact, data availability, and value maturity, these indicators will be chosen. True Value analysis results are only meaningful when contextualized. The baseline, which usually represents the industry average or archetype, compares data and helps users see where societal value is made or destroyed. The most important baselines for each pilot type and hub will be formulated together (D5.1). This will provide balanced contextualisation of results.

Task 5.2 Measuring and valuation (M6–M49) Leader: KPMG; Participants: UT, Hub Leaders; Deliverable: D5.2, D5.3
The valuation methodology is developed based on the selected indicators from T5.1 and applied to the four Hubs. Decisions will be made upon the measurement and valuation approach based on KPMG and the partners' experience, True Value database, scientific publications, available data from the hubs and other relevant sources.

Subtask 5.2.1 Designing, Deploying and Tunning Valuation Models (M6-M49) (Leader: KPMG; Participants: TUDO, UT, Hub Leaders).
The valuation framework will be developed case by case, in which the most material economic, environmental, and societal costs and benefits are considered. The indicators can include, among others, (i) Economic - Job creation/salaries paid; supplier spend applied to the regional input-output models to be developed in WP3; taxes paid; (ii) Environmental Resource use; energy use, GHG emissions; waste, water use, land use and biodiversity impact; and (iii) Social - Societal contribution relevant to the project; community enhancement; health and safety incidents. For the selected indicators, the valuation approach, required input information and other relevant information will be detailed in the True Value measurement framework. Through the pilots, the selected indicators will be monitored and evaluated, to validate the methodology and provide examples of how the True Value approach can benefit the proposition (and improvement strategy). The valuation framework, based on the four hub outcomes, can be scaled, and grown further to other valuation cases. This step will result in the quantification of the indicators in scope and a complementary scalable calculation model.

Subtask 5.2.2 Design, Testing and Deploying AI Models (M6-M49) (Leader: UT; Participants: KPMG, Hub Leaders) It identifies variables that can be considered in Artificial Intelligence (AI) frameworks that aim to assess financial investment, credit risk evaluation, and valuation of sustainable and circular business opportunities. These include not only individual enterprises but also process/energy technologies and infrastructure being deployed in H4C. Then, it uses fair and accurate AI frameworks capable of assessing the risk level of retail and commercial customers in H4C settings and accurately predict the valuation of H4C's assets. This task provides businesses, society, and the government with transparent tools such as explainable machine learning algorithms capable of analysing financial investments. It also suggests a policy (to banks and financial institutions) to be considered for H4C companies' credit risk assessment and investment options, considering factors that financial analysts and traditional models typically disregard. Moreover, information related to (co)financing (cash-flow simulations with all necessary parameters) for infrastructure, products, and services (per region) will be gathered and provided as input for the digital collaboration platform (WP7).

Task 5.3 Designing IS impact measurement approach (M13-M49) Leader: KPMG; Participants: TUDO, Hub Leaders; Deliverable: D5.2, D5.3
It will focus on developing and designing the most adequate IS impact measurement approach through the True Value methodology. Thus, the following approach will be implemented by: (1) Defining the hypothesis for the relationships between IS and economic, environmental, and social impact, using the impact chain approach; (2) Selecting the 2 or 3 strongest causal relationships based on expert judgement; (3) Strengthening the evidence base for these relationships; (4) Deciding on the "best approach for action (A4A)": based on the strength of the scientific evidence, assess the feasibility of quantitative inclusion of IS impacts in the True Value Framework; (5) Quality controlling the proposed approach by an expert panel, leveraging from the engagement in WP3 actions; It will result in a validated impact measurement procedure for IS, capable of being scaled to future H4C across EU.

Task 5.4 Develop a comprehensive financial plan for the H4C business models (M37–M49) Leader: KPMG; Participants: UT, Hub Leaders, IRIS; Deliverable: D5.3
The main goal of this task is to develop a comprehensive financial plan (D5.3) (i.e., financial canvas) for the implementation of the new business models. The financial plan includes projections of revenue, expenses (OPEX and CAPEX), cash flow, risk and investment indicators, and profitability for multiple years. This plan will help determine the financial viability of H4C's proposed business models and ensure they have the necessary resources to be implemented successfully. In this plan, outputs from the AI models (T5.2.2) will be considered along with, a sensitivity analysis will be conducted throughout simulations of different scenarios. To support the development of the financial

plan, a tailored market research will be conducted, where the target markets will be defined; competitors will be selected and analysed;

and demand will be calculated, which will result in the development of a market strategy. The financial plan will be updated as the business model evolves and added information becomes available and will also account for inputs from the consortium partners, advisors, & investors. The goal is to increase the likelihood of market success to implement the models envisioned in this proposal. For this purpose, a tailored diagnostic of potential investment sources, including funding opportunities.

Task 5.5 Business & financial models for society (M6-M49) Leader: UT; Participants: KPMG, Hub Leaders, SIDENOR I+D, TUDO; Deliverable: D5.2, D5.3
This task involves creating, simulating, and deploying financial and business models (D5.2) for IS and the CE. These models account for both financial and non-financial costs of materials, goods, and services. The financial costs and benefits include capital and operational expenditures, revenue generation, and potential cost savings from reduced waste and improved efficiency. However, the non-financial costs and benefits would be equally important to consider because they would have a direct effect on society. These could include environmental impacts like reduced carbon emissions, improved air and water quality, and decreased resource depletion, as well as social impacts like increased job creation, improved health outcomes, and increased community engagement. Given the cost-benefit analysis of each hub and its technologies, this task aims to identify business opportunities that could be used to develop a financial and business model that aligns with CE and IS principles and maximises societal benefits. After completing this task, businesses and end users of materials (e.g., energy, cement, paper) or services (e.g., logistic costs) will be able to simulate scenarios in which they buy products (or services) from H4C that offer financial incentives for sustainable choices. These models would also identify implementation hurdles and countermeasures, providing end-users and enterprises with a financial analysis canvas tool for investment sharing and simulation of economic benefits for financing and deploying hubs-related new infrastructures, products, and services.

## Work package WP6 – Integration and demonstration in hubs

| **Work Package Number** | WP6 | **Lead Beneficiary** | 4. TEC |
|---|---|---|---|
| **Work Package Name** | Integration and demonstration in hubs | | |
| **Start Month** | 1 | **End Month** | 49 |
| **Objectives** | | | |
| Field work takes place in WP6 to deploy the a-priori decided technologies in the hubs. WP6 aims at advancing the industrial zones into H4C via applying the technologies in the operational environment under the principals of circularity and IS and integrating the existing and new infrastructure. Project's real impact occurs in the WP6. | | | |
| **Description** | | | |

T6.1 Definition of the current state-of-the-art and validation of success (M1-18) Leader: KPMG. Participants: Hub representatives (TUDO, ACE, SIDENOR, SIDENOR I+D, ARCELIK, UT), tech-providers for each hub; Deliverable: D6.1
The four hubs have vastly different starting points from where the respective transformation into a H4C takes place. Defining those starting points from a techno-economic point of view will be the line of comparison for all executed efforts to achieve the targets. This includes three parts: 1) Executive summary of the hub starting points in terms of technology, CO2 emissions and energy consumption are valuable process reports from which to compare and position upcoming tasks. 2) A Definition of requirements of the industrial plants for the implementation of the technologies is the follow-up estimation based on the first report. 3) An outlook at the success of each undertaken measure and the impact on the overall system will work as a dissemination report filled with the defined success criteria over all tasks in this WP (D6.1).

T6.2 Commissioning, set up and initial testing of technologies and infrastructure (M13-M49). Leader: TEC; Participants: Hub representatives, tech-providers for each hub; Deliverable: 6.2
Basque Hub: (1) Carbon capture, purification, and CO2 conversion to Synthetic Natural Gas (SNG) in lime industry. PSA, polymeric membranes, and a multichannel reactor based on process intensification technologies will be used (TRL from 5 to 7). Once the plant is installed, several tests will be carried out for more than 500 hours using real gas generated at Calcinor. The behaviour and performance of the process will be evaluated by carrying out tests under different conditions. Modifications may be made to the plant to improve its operation. The activity and deactivation of the zeolites and the catalyst will also be evaluated. (2) Substitution of natural gas air combustion by hydrogen oxy-fuel combustion in the application of ladle preheating in Sidenor (TRL increases from 6 to 8). The thermal profile of the ladle should be similar as the obtained by natural gas combustion. (4) Carbonation of steel slags from Sidenor with Calcinors industry's CO2 by SBS. The CO2 from Calcinor will be compressed and transported to SBSs' facilities to carry out the carbonation of Sidenor slags and produce construction products for the cement industry. These products will be analysed and their main characteristics and properties will be provided.
Dutch Hub: (1) Building H2 pipelines for injecting H2 produced by H2Hub (from solar energy –and from to-beimplemented wind energy- via an already implemented 70 kW electrolyser) to (2) feed 5 houses located in nearby Aadorp village supporting their heating demand by H2. (3) Stabilisation of the green hydrogen generation by 70kW of wind turbines and energy storage. (4) O2, by-product of electrolysis, used for purification of waste water in a circular process, (5) (treated water) to be used back in the electrolysis process (4-5 requires two pipelines construction). (6) Use H2 trucks to supply H2 to a crematory replacing natural gas. Synergy 1: to be done by Cogas/Coteq (local grid operator); Synergy 2: to be supported by Solenco, Almelo Energie; Synergy 3: to be done by H2Hub; Synergy 4-5: to be done by wastewater treatment plant and H2Hub.
Turkish Hub: (1) Use CC technology to capture the carbon emission of Tu TUPRAS pras using Tup TUPRAS ras' MOF4AIR CO2 capture unit installed in previous H2020 Project. (2) Put the captured carbon in reaction with green H2 produced by electrolysis fed by renewable energy (already being implemented onsite or power purchase agreement (PPA) contract) to produce e-methanol as an intermediate product. (3) Produce non isocyanate polyurethane (NIPU) from captured carbon and e-methanol (technology developed by Fraunhofer). (4) Use the NIPU in the production of refrigerators by Arçelik.
German Hub; (1) Initiate CC for the chemical industry and erect carbon infrastructure on site; Infraserv Höchst, CC technology provider, (2) Initiate Methanol Synthesis after CC on site; Infraserv Höchst, e-methanol technology provider. This task will report on the installation and first 100h of operation (D6.2) based on the initial tests from T6.2.

T6.3 Monitoring of operational performance and evaluation (M19-M49). Leader: ZLC; Participants: Hub representatives, tech-providers for each hub, TEC, UT, ICCS, IRIS; Deliverables D6.3, D6.4
To monitor and validate the operations in the hubs to visualize the resource and energy reductions in close cooperation with WP4 (T4.6) and WP7 (T7.5). The performance of each hub use case will be evaluated, particularly from the objectives and KPI perspective to evaluate performance and illustrate how the project outcomes can be leveraged for each use case. It will deliver such methodology and procedures that will be implemented in the hubs. Sensing technologies will be considered and tested for data gathering to the hubs to support monitoring of the streams and the tracing module to be developed in T7.5. It will enable implementation results to be analysed, based on more extended data collection and on feedback from the users. Final KPIs measurements/analysis and validation will be performed. Based on the lessons learned from the evaluation phase, a decision is made regarding the continuation of the pilot or the initiation of a new updated iteration to explore a new idea that stemmed out from the Community of Practice. The results will be reported in different reports: based on the initial test from T6.2, operation and demonstration (D6.3) and overall hub performance and outcomes (D6.4).

## Work package WP7 – Digital Collaboration Platform for H4C

| **Work Package Number** | WP7 | **Lead Beneficiary** | 29. IRIS |
|---|---|---|---|
| **Work Package Name** | Digital Collaboration Platform for H4C | | |
| **Start Month** | 1 | **End Month** | 49 |
| **Objectives** | | | |
| To develop a user-friendly, scalable, and adaptable (to the needs of different industries and hubs) digital collaboration platform for H4C (DigitalH4C). The platform consists of a modular suite of tools, features, and data analytics to facilitate IS identification, coordination, and evaluation. Beyond a physical platform, it aims to derive a theory of IS-oriented digital platform development from the implementation evidence collected across the hubs. This WP (1) guides the design of digital H4C in a stepwise manner, and (2) reveals the underlying mechanism of how a digital platform orchestrates diverse analytical components and contributes to the emergence of H4C. Therefore, connecting to WP6, the knowledge and experiences acquired from this work package can be generalised for future cases in different contexts. Overall, DigitalH4C creates a digital collaborative space for H4C leading to a digitalised and resilient Circular Economy. | | | |
| **Description** | | | |

T7.1 DigitalH4C platform architecture design (M1-M13) Leader: IRIS, Participants: ICCS, UT Deliverable: D7.1
A secure digital collaboration platform (web and mobile applications) will be built based on a microservices architecture that will ensure scalability and modularity. First, a design blueprint of the platform will be provided based on a formal reference architecture. It demonstrates a coherent and comprehensive system architecture by (1) suggesting the optimal setup of technological integration, (2) articulating the interdependencies of identified decision support components, and (3) providing a knowledge basis for platform development, implementation, and evolvement. Proceeding in parallel with other WPs, this task builds a communication foundation where the platform design is visible, transparent, and reachable.

T7.2 Online marketplace (M13-M30). Leader: UT, Participants: IRIS, ICCS; Deliverable: D7.2
To further build on the opportunities of IS, an online marketplace add-in will be integrated to the platform (D7.2). The marketplace module will be adapted from the existing Houseful & SHAREBOX (know-how) platform. Digital contracts could automatically be generated and ensure efficient IS once sharing conditions, such that the quality standards and transportation logistics, are met. The digital contracts will also facilitate the tracking and verification of resource flows, enabling greater transparency and accountability in the exchange process. Hence, it creates a collaborative, efficient, and trustful marketplace for hub actors. The financial analytical tools by WP5 will be added into the marketplace component ensuring robust business transactions.

T7.3 Virtual collaboration tools (M13-M30) Leader: WDK, Participants: IRIS, UT, ICCS; Deliverable: 7.3
WDK will adapt its current collaboration software with workflows designed for the management and implementation of IS projects, which will be integrated with the DigitalH4C in T7.1. Its key function is to simplify and accelerate the collaboration procedures among diverse stakeholders. It is a project workflow management tool that streamlines complex information exchanges and improves the efficiency of multi-actor communications. In addition to flexible and configurable workflows, it will also use secure communication tools (messaging, chat rooms, and forums) to enable communication with each other.

T7.4 Integration of intelligent matchmaking and decision support components (M25-M42) Leader: ICCS, Participants: IRIS, UT, INEGI; Deliverable: D7.3
The matchmaking and recommender algorithms for IS and decision support tools of WP4 will be integrated and deployed in DigitalH4C (D7.3). The integration helps to identify potential IS matches based on their resource needs and capacities (matching supply and demand using the platform database), and assess them considering factors as availability, logistics, and compatibility of resources. Integrated with other functional components, it performs the overarching functionality of the platform accelerating IS implementation in demo hubs. The real-time data associated with IS exchanges from industrial partners will be used to test and improve the delivered algorithms through customised user interfaces. Overall, the integration of these components establishes a decision-making basis for digitalised IS. Inputs from WP3 (stakeholders list) and WP5 (business models and financial plans) will also be evaluated and integrated in the platform to dynamise and enrich it.

T7.5 Integration of monitoring and evaluation components (M25-M42) Leader: ICCS, Participants: UT, IRIS; Deliverable: D7.3
Analytics tools for measuring the environmental and economic impacts and tracking the symbiotic resource flows of IS collaborations will also be integrated in the platform. First, the methodology of consequential LCA will be tailored for demo hubs evaluating the environmental consequences of adopting different business strategies. The analytical suite of LCA combines the outputs of functional components from other WPs to prepare a computational basis for LCA in the form of structured input-output datasets. Built upon the LCA component, this task will enable the real-time visualisation and tracking of the IS synergies. Combined with LCA, the monitoring tool helps participants to interpret the results and deliver corresponding strategies regarding coordination improvement. Based on T4.3, a traceability module will be developed and tested in specific flows of hubs (T6.3) potentially enhancing synergies tracking on LCA component and T7.4 tools.

T7.6 Cyber security, data protection and compliance (M28-M49) Leader: UT, Participants: IRIS, WDK, KPMG; Deliverable: D7.4
This task will ensure a secure-by-design approach to cybersecurity for information sharing, and there will be early input into T7.1 for ensuring the architecture of the platform has robust security features, such as: a secure data transfer protocol (HTTPS or SSL/TLS) for data encryption during transit; robust access control system (two-factor authentication, password policies, and role-based access control); data encryption (of data at rest) using robust encryption algorithms such as AES; Data Backup and Recovery; monitoring and alerting system that continuously monitors the system for any suspicious activities or potential security breaches; Incident Response Plan outlining the steps to be taken to mitigate the damage and restore normal operations in the event of a security breach. This task will oversee the implementation of these cybersecurity measures on the platform and will also perform a series of security audits should be conducted

to identify and mitigate potential vulnerabilities in the system (D7.4). In addition to cybersecurity, this task will also ensure compliance with GDPR.

T7.7. Evaluation & version release deployment of Digital H4C (M37-M49) Leader: IRIS, Participants: UT, WDW; Deliverable: D7.1, D7.5
The User Interface and User Experience (UI/UX) will be defined following a process of concept, low-fidelity wireframes, high-fidelity wireframes, and final designs. Once final UI/UX designs are agreed upon (in a co-creation process with end-users), the back-end and front-end development will commence. Echoing with the reference architecture, the entire work package will be managed in iterative Sprints (according to the Scrum methodology) each culminating in functionality releases (D7.1 and D7.5). It will follow internal Quality Assurance and debugging to the end-users for the demonstration in WP6. The evaluation feedback will be analysed by the development team iteratively so that the platform can be optimised and refined. Practically, an Application Programming Interface (API) will be built and documented allowing connectivity with third-party data sources, applications, and services in an open digital ecosystem. Modules and algorithms from existing platforms developed by IRIS (Houseful; SHAREBOX) will be incorporated into the platform. Possible standard cloud platform modules (from IRIS' moap platform builder) will be selected, deployed, and adapted to ensure rigorous version release. Also, the platform will provide training and support in the form of case studies and best practices to help users get familiar with it and with IS concepts.

## Work package WP8 – Dissemination, communication & exploitation

| **Work Package Number** | WP8 | **Lead Beneficiary** | 11. EEIP |
|---|---|---|---|
| **Work Package Name** | Dissemination, communication & exploitation | | |
| **Start Month** | 1 | **End Month** | 49 |

**Objectives**

WP8's activities are centred on effectively communicating project findings, results, and outcomes to targeted audiences and promoting the project to a broader audience. The objective of this WP is to design and implement strategies to engage stakeholders and pursue collaborations with other pertinent initiatives. In addition, it entails supporting the implementation of key exploitable results (KERs) and ensuring their protection through adequate intellectual property rights (IPR) measures. Thus, this WP seeks to maximize the IS2H4C's impact and create opportunities for further exploitation and dissemination.

**Description**

T8.1 Dissemination & communication plan & visual identity (M1-M49) Leader: EEIP; Participants: All; Deliverable: D8.1, D8.2, D8.3
A detailed Dissemination and Communication Plan (D&C) will be produced by EEIP at the beginning of the project (D8.1), based on the preliminary indications given in Section 2.2 and in collaboration with all the consortium partners. It will present the visual identity (logo, colours and fonts chosen to identify the project) and outline the project's audiences, key messages, and communication channels for the dissemination, including roles and responsibilities. The different updates of the plan (D8.2 and D8.3) will offer the monitoring of the different dissemination and communications activities carried out by all partners, evaluated against its KPIs. A project's website will be created in order to disseminate the knowledge generated by the project to the scientific community and interested stakeholders to improve future studies in the field and allow further exploitation outcomes. It will also allow the public to consult non-confidential information about the project (description, benefits, cost effectiveness, etc.). Partners will publish a description of their role, the project description, and a link to their respective company or other websites. The project website will serve as a reference point, maintaining links to other related projects and events worldwide. This task will also be focused on the preparation of appropriate materials (factsheets, leaflets, roll-ups) and the dissemination of the project´s results to external target groups. This includes communication through social media platforms (LinkedIn, Twitter), where dedicated accounts will be set up for the project. A monthly social media plan will allow for the regular posting of relevant contents, considering project activities as well as European and international initiatives. To boost the dissemination efforts, it will rely on various multipliers such as: partners websites and social media channels. Additionally, a broad range of external network websiteswill be used to disseminate and take advantage of multiplication effect. A report on dissemination and communication activities will be provided by EEIP (D8.4)

T8.2 Collaborative ecosystem: building the circular industry helix (M1-M49) Leader: CHX; Participants: All; Deliverable: D8.8

CHX will build the already existing virtual cluster/community, called the Circular Industry Helix, on the CHX platform that will act as the backbone of the commercial dissemination and exploitation strategies developed (D8.8). The following activities will take place: (a) By M6, update of the forum on the CHX Platform: include a description about the IS2H4C and communicate the helix news to the membership base. (b) Profile >150 new organisations (key stakeholders and end-users at researcher and organisational level) in the CHX platform (M37): IS2H4C partners, members from relevant existing helixes, and stakeholders defined, participants of other relevant projects, and relevant actors across EU/International (Section 2.2). (c) Deliver impact in line with the Helix strategy: including supporting dissemination events and building collaboration through the platform. (d) Organise 1 Helix event (M49). The Helix will be managed by a Helix Manager, information about the project and the members, and will have a resources tool with links to the project website & social media (Facebook, Twitter and LinkedIn), and any dissemination and communication material (presentations, scientific publications, etc.) produced during the project. To ensure sustainability, CHX will commit to maintaining the helix for at least 2 years after the project ends (more details in Section 2.2). In addition to CHX task, strategic stakeholder engagement actions lead by other IS2H4C partners include:

(i) Policymakers and standardisation organisations: This particular group will be approached by attending local and international events (e.g., ACR+ project's events, CCRI forums and hybrid events). The representatives of each H4C will be in charge of this engagement, each attending at least 2 events. (ii) Research and academia: In order to have a better understanding of IS structure and logistics, UT has developed a Sustainable and Resilient Circular Economy Laboratory (SRC-Lab) which contains two serious online games. The first game simulates an IS network environment where students and researchers can control, monitor and observe the formation and evolution of an IS network via trying to overcome the challenge of coopetition in an experiential learning setting. The second game is the Circular Economy Transition game which has a story embedded in a virtual circular hub composed of companies, houses, and a municipality. The game setting allows the players to experience the challenge of surviving with the limited resources provided to the city not sufficient to run the entire game. Thus, the players need to implement circularity solutions not to let the hub bankrupt via experiencing the role of households, company managers, and policymakers. Four demo hubs of IS2H4C will be embedded in this game setup in order to produce a digital twin of these cities and will be provided to students/company managers. Complementary to the game, in-site visits will be arranged for the students to learn about real-life operations in the H4Cs. (iii) Private/Public companies: This group will be approached through networking events (two webinars) and in-house presentations, but also through collaborations with EU projects (see T8.3 below). (iv) General public: IS2H4C aims to develop outreach events for the surrounding communities to learn about the benefits that IS initiatives could have in their lives (liked to WP3 activities). Such events include: (i) site visits to nearby H4C; (ii) developing "Hub Days" which will be held once a year and will include relevant information; and finally, (v) a Hackathon/workshop(s) about H4C, CC technologies andH2 production, among other topics.

T8.3 Clustering with other EU projects and initiatives (M1-M49) Leader: CHX; Participants: All; Deliverable: D8.4 A clustering plan will be included in the DEC plan (D8.1, details in Section 2.2). The following activities will be developed in order to maximise cross-project communication, to exchange knowledge and experiences and to facilitate synergies and strategic partnering with other European organisations: (a) Identification of the projects linked and relevant to IS2H4C including sister projects and other EC-chaired or funded initiatives in the relevant domains and knowledge networks set up by EC services (details in clustering plan, Section 2.2). (b) First contact with the project directors. Coordinators will be invited to the IS2H4C Kick-off meeting. (c) Development of a Joint Clustering Board and Plan (with the sister projects), including the following activities: Creation of a common identity (logo), mailing list, promotion of other projects in IS2H4C website, yearly meetings, IS2H4C events. A virtual clustering event will be organised (M13) to encourage partners to create consortia and develop new ideas for future funding opportunities. Project partners will also explore possible on-going clustering assistance offered by the EU, such as the Horizon Results Booster. During the project, EEIP will continue to screen the most interesting initiatives to establish direct communication channels with them, so joint technical, dissemination and communication activities can be agreed. Then, the different activities will be executed during the project and reported under D&C activity report (D8.4).

T8.4 Exploitation plan and IPR management (M1-M49) Leader: UT; Participants: All; Deliverable: D8.5, D8.6, D8.7 T8.4 will 1) identify intangible and tangible project KERs and prepare exploitation plans for them; 2) outline strategies for exploitation of project results in close collaboration with project partners after the project period. The predefined KERs will be evaluated against project outcomes. An initial exploitation plan (D8.5 updated in D8.6 and D8.7) will be developed according to the steps described in Section 2.2 (Exploitation plan). A process of IPR Management will be established so that valuable knowledge generated is protected and the adequate dissemination and exploitation measures can be put in place. The IPR strategy will be coordinated by UT as Project Coordinator. The specific tasks include: i) IPR Background and FTO analysis, ii) IPR Joint Strategy, iii) IPR Monitoring. IPR Management will be included as

part of the exploitation plan and the mandatory final Result Ownership List (ROL) included in the final project reporting period.

| |
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| T8.5 Impact analysis (M1-M49) Leader: CHX; Participants: All Deliverable: 8.4<br>CHX, in collaboration with all partners, will analyse the IS2H4C dissemination impact and outputs by using data science. Dissemination impact analysis: to study classic measures of reach, such as numbers of article citations, by using repositories of academic research papers. It will use “topic analysis” to understand degrees of interdisciplinary dissemination. It will also analyse reporting in news articles, blog posts and social media. Measuring IS2H4C outputs: It will use techniques such as: a) aggregating additional and external data sources to inform on broader project, b) modelling to determine causality of outputs, and c) statistical analysis to conclude on the robustness of outputs measurements. |

## Work package WP9 – Ethics requirements

| **Work Package Number** | WP9 | **Lead Beneficiary** | 1. UT |
|---|---|---|---|
| **Work Package Name** | Ethics requirements | | |
| **Start Month** | 1 | **End Month** | 49 |
| **Objectives** | | | |
| The objective is to ensure compliance with the 'ethics requirements' set out in this work package. | | | |
| **Description** | | | |
| This work package sets out the 'ethics requirements' that the project must comply with. | | | |

# STAFF EFFORT

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Participant** | | | | | | | | | | |
| 1 - UT | | | | | | | | | | |
| 2 - MUL | | | | | | | | | | |
| 3 - TUDO | | | | | | | | | | |
| 4 - TEC | | | | | | | | | | |
| 5 - INEGI | | | | | | | | | | |
| 6 - FHG | | | | | | | | | | |
| 7 - CIRCE | | | | | | | | | | |
| 8 - ICCS | | | | | | | | | | |
| 9 - AEE | | | | | | | | | | |
| 10 - ZLC | | | | | | | | | | |
| 11 - EEIP | | | | | | | | | | |
| 12 - IMMIB | | | | | | | | | | |
| 13 - H2HT | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|
| 14 - ACE | | | | | | | | | | |
| 15 - CABB | | | | | | | | | | |
| 16 - ALMELO | | | | | | | | | | |
| 17 - KPMG | | | | | | | | | | |
| 18 - TUPRAS | | | | | | | | | | |
| 19 - ARCELIK | | | | | | | | | | |

**Staff effort per participant**

*Grant Preparation (Work packages - Effort screen) — Enter the info.*

| Participant | WP1 | WP2 | WP3 | WP4 | WP5 | WP6 | WP7 | WP8 | WP9 | Total Person-Months |
|---|---|---|---|---|---|---|---|---|---|---|
| 20 - SKN | | | | | | | | | | |
| 21 - NORTEGAS | | | | | | | | | | |
| 22 - SIDENOR | | | | | | | | | | |
| 22.1 - SIDENOR I+D SA | | | | | | | | | | |
| 23 - PETRONOR | | | | | | | | | | |
| 24 - CALERA | | | | | | | | | | |
| 24.1 - CALCINOR | | | | | | | | | | |

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|---|---|---|---|---|---|---|---|---|---|---|
| 25 - LTK | | | | | | | | | | |
| 25.1 - LTK I+D | | | | | | | | | | |
| 26 - SBS | | | | | | | | | | |
| 27 - IH | | | | | | | | | | |
| 28 - CHX | | | | | | | | | | |
| 29 - IRIS | | | | | | | | | | |
| 30 - WDK | | | | | | | | | | |
| 31 - SOLENCO | | | | | | | | | | |
| **Total Person-Months** | | | | | | | | | | |

# LIST OF DELIVERABLES

**Deliverables**

*Grant Preparation (Deliverables screen) — Enter the info.*

*The labels used mean:*

*Public — fully open (⚠ automatically posted online)*
*Sensitive — limited under the conditions of the Grant Agreement*
*EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444*

| Deliverable No | Deliverable Name | Work Package No | Lead Beneficiary | Type | Dissemination Level | Due Date (month) |
|---|---|---|---|---|---|---|
| D1.1 | Management and quality plan report | WP1 | 1 - UT | R — Document, report | SEN - Sensitive | 3 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D1.2 | Data Management Plan | WP1 | 1 - UT | DMP — Data Management Plan | PU - Public | 6 |
| D2.1 | Technology map with key technologies, technical and economic feasibility, and contribution to synergies, including an assessment of their scalability | WP2 | 2 - MUL | R — Document, report | PU - Public | 6 |
| D2.2 | Design specifications of energy conversion technologies ready for testing | WP2 | 1 - UT | R — Document, report | PU - Public | 30 |
| D2.3 | Waste & heat recovery technology design specifications usable for testing | WP2 | 9 - AEE | R — Document, report | PU - Public | 30 |
| D2.4 | Design specifications of millichannel reactor that can convert the industrial CO2 into SNG with sufficient quality to be injected into the NG grid | WP2 | 4 - TEC | R — Document, report | SEN - Sensitive | 30 |
| D2.5 | Design specifications of e-methanol synthesis unit | WP2 | 6 - FHG | R — Document, report | SEN - Sensitive | 42 |
| D2.6 | Design specifications of polymer production process | WP2 | 6 - FHG | R — Document, report | SEN - Sensitive | 42 |

**Deliverables**

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*EU classified —RESTREINT-UE/EU-RESTRICTED, CONFIDENTIEL-UE/EU-CONFIDENTIAL, SECRET-UE/EU-SECRET under Decision 2015/444*

| Deliverable No | Deliverable Name | Work Package No | Lead Beneficiary | Type | Dissemination Level | Due Date (month) |
|---|---|---|---|---|---|---|
| D2.7 | NIPU production process design and specifications | WP2 | 6 - FHG | R — Document, report | SEN - Sensitive | 42 |
| D2.8 | Performance analysis results compared to alternative technologies, including recommendations for full-scale implementation | WP2 | 1 - UT | R — Document, report | PU - Public | 49 |
| D3.1 | Map of stakeholders and their interests/needs | WP3 | 3 - TUDO | R — Document, report | SEN - Sensitive | 13 |
| D3.2 | Methodology for assessment of non-technological issues | WP3 | 3 - TUDO | R — Document, report | PU - Public | 15 |
| D3.3 | Set of indicators measuring the nontechnological impact of hubs | WP3 | 1 - UT | R — Document, report | PU - Public | 18 |
| D3.4 | Management guideline for integrating nontechnological aspects | WP3 | 3 - TUDO | R — Document, report | PU - Public | 49 |
| D4.1 | Modelling system for the characterisation of resources, energy flows and IS | WP4 | 4 - TEC | R — Document, report | PU - Public | 25 |
| D4.2 | Recommender algorithm for IS collaborations | WP4 | 1 - UT | OTHER | SEN - Sensitive | 30 |
| D4.3 | Models for hub's supply chain management | WP4 | 10 - ZLC | OTHER | SEN - Sensitive | 37 |
| D4.4 | Decision support tool for centralized/decentralized hub | WP4 | 1 - UT | OTHER | SEN - Sensitive | 42 |

**Deliverables**

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| Deliverable No | Deliverable Name | Work Package No | Lead Beneficiary | Type | Dissemination Level | Due Date (month) |
|---|---|---|---|---|---|---|
| D4.5 | Agent-based simulation tool for hub development | WP4 | 1 - UT | OTHER | SEN - Sensitive | 49 |
| D4.6 | Impact assessment of H4C from holistic perspective & decarbonisation pathways. | WP4 | 4 - TEC | R — Document, report | PU - Public | 40 |
| D4.7 | Virtual top-down hubs and their use as sustainable regional development models | WP4 | 1 - UT | R — Document, report | PU - Public | 49 |
| D5.1 | Guidelines for the True Value Framework | WP5 | 17 - KPMG | R — Document, report | SEN - Sensitive | 6 |
| D5.2 | IS2H4C Business Model | WP5 | 17 - KPMG | OTHER | SEN - Sensitive | 42 |
| D5.3 | IS2H4C Financial Plan | WP5 | 17 - KPMG | R — Document, report | SEN - Sensitive | 49 |
| D6.1 | Report on hub starting lines | WP6 | 17 - KPMG | R — Document, report | PU - Public | 18 |
| D6.2 | Report on hub installation progress and first 100 h of operation | WP6 | 4 - TEC | R — Document, report | PU - Public | 37 |
| D6.3 | Report on continuous operation & technology demonstration | WP6 | 10 - ZLC | R — Document, report | PU - Public | 49 |

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|---|---|---|---|---|---|---|
| D6.4 | Report of overall hub performance, outcomes, and dissemination | WP6 | 10 - ZLC | R — Document, report | PU - Public | 49 |
| D7.1 | Early release (alpha version) with the first layer of functionality | WP7 | 29 - IRIS | OTHER | SEN - Sensitive | 13 |
| D7.2 | Online marketplace platform add-in | WP7 | 1 - UT | OTHER | SEN - Sensitive | 30 |

**Deliverables**

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| Deliverable No | Deliverable Name | Work Package No | Lead Beneficiary | Type | Dissemination Level | Due Date (month) |
|---|---|---|---|---|---|---|
| D7.3 | Extended version of DigitalH4C integrated with the digital tools | WP7 | 8 - ICCS | OTHER | SEN - Sensitive | 42 |
| D7.4 | Report on risk assessment & mitigation – digital platform security | WP7 | 1 - UT | R — Document, report | SEN - Sensitive | 49 |
| D7.5 | Final delivery of DigitalH4C platform | WP7 | 29 - IRIS | R — Document, report | PU - Public | 49 |
| D8.1 | Dissemination and communication plan | WP8 | 11 - EEIP | R — Document, report | PU - Public | 6 |
| D8.2 | Dissemination and communication plan - Update 1 | WP8 | 11 - EEIP | R — Document, report | PU - Public | 25 |
| D8.3 | Dissemination and communication plan - Update 2 | WP8 | 11 - EEIP | R — Document, report | PU - Public | 49 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D8.4 | Report on dissemination and communication activities | WP8 | 11 - EEIP | R — Document, report | PU - Public | 49 |
| D8.5 | Exploitation plan report | WP8 | 1 - UT | R — Document, report | SEN - Sensitive | 6 |
| D8.6 | Exploitation plan report - Update 1 | WP8 | 1 - UT | R — Document, report | SEN - Sensitive | 25 |
| D8.7 | Exploitation plan report - Update 2 | WP8 | 1 - UT | R — Document, report | SEN - Sensitive | 49 |
| D8.8 | Circular Industry helix “virtual cluster/ community” building | WP8 | 28 - CHX | R — Document, report | PU - Public | 49 |
| D9.1 | OEI - Requirement No. 1 | WP9 | 1 - UT | ETHICS | SEN - Sensitive | 2 |

## Deliverable D1.1 – Management and quality plan report

| | | | |
|---|---|---|---|
| **Deliverable Number** | D1.1 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Management and quality plan report | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 3 | **Work Package No** | WP1 |
| **Description** | | | |
| Management and quality plan report | | | |

| | | | |
|---|---|---|---|
| **Deliverable Number** | D2.2 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Design specifications of energy conversion technologies ready for testing | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 30 | **Work Package No** | WP2 |
| **Description** | | | |

| |
|---|
| Design specifications of energy conversion technologies ready for testing |

## Deliverable D1.2 – Data Management Plan

| | | | |
|---|---|---|---|
| **Deliverable Number** | D1.2 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Data Management Plan | | |
| **Type** | DMP — Data Management Plan | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 6 | **Work Package No** | WP1 |
| **Description** | | | |
| Data Management Plan | | | |

## Deliverable D2.1 – Technology map with key technologies, technical and economic feasibility, and contribution to synergies, including an assessment of their scalability

| | | | |
|---|---|---|---|
| **Deliverable Number** | D2.1 | **Lead Beneficiary** | 2. MUL |
| **Deliverable Name** | Technology map with key technologies, technical and economic feasibility, and contribution to synergies, including an assessment of their scalability | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 6 | **Work Package No** | WP2 |
| **Description** | | | |
| Technology map with key technologies, technical and economic feasibility, and contribution to synergies, including an assessment of their scalability | | | |

## Deliverable D2.2 – Design specifications of energy conversion technologies ready for testing

## Deliverable D2.3 – Waste & heat recovery technology design specifications usable for testing

| | | | |
|---|---|---|---|
| **Deliverable Number** | D2.3 | **Lead Beneficiary** | 9. AEE |
| **Deliverable Name** | Waste & heat recovery technology design specifications usable for testing | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 30 | **Work Package No** | WP2 |
| **Description** | | | |
| Waste & heat recovery technology design specifications usable for testing | | | |

| | | | |
|---|---|---|---|
| **Deliverable Number** | D2.6 | **Lead Beneficiary** | 6. FHG |
| **Deliverable Name** | Design specifications of polymer production process | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 42 | **Work Package No** | WP2 |

## Deliverable D2.4 – Design specifications of millichannel reactor that can convert the industrial

## CO2 into SNG with sufficient quality to be injected into the NG grid

| | | | |
|---|---|---|---|
| **Deliverable Number** | D2.4 | **Lead Beneficiary** | 4. TEC |
| **Deliverable Name** | Design specifications of millichannel reactor that can convert the industrial CO2 into SNG with sufficient quality to be injected into the NG grid | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 30 | **Work Package No** | WP2 |
| **Description** | | | |
| Design specifications of millichannel reactor that can convert the industrial CO2 into SNG with sufficient quality to be injected into the NG grid | | | |

## Deliverable D2.5 – Design specifications of e-methanol synthesis unit

| | | | |
|---|---|---|---|
| **Deliverable Number** | D2.5 | **Lead Beneficiary** | 6. FHG |
| **Deliverable Name** | Design specifications of e-methanol synthesis unit | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 42 | **Work Package No** | WP2 |
| **Description** | | | |
| Design specifications of e-methanol synthesis unit | | | |

## Deliverable D2.6 – Design specifications of polymer production process
## Deliverable D2.7 – NIPU production process design and specifications

| | | | |
|---|---|---|---|
| **Deliverable Number** | D2.7 | **Lead Beneficiary** | 6. FHG |
| **Deliverable Name** | NIPU production process design and specifications | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 42 | **Work Package No** | WP2 |
| **Description** | | | |
| NIPU production process design and specifications | | | |

| | | | |
|---|---|---|---|
| **Deliverable Number** | D3.2 | **Lead Beneficiary** | 3. TUDO |
| **Deliverable Name** | Methodology for assessment of non-technological issues | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 15 | **Work Package No** | WP3 |
| **Description** | | | |
| Design specifications of polymer production process | | | |

## Deliverable D2.8 – Performance analysis results compared to alternative technologies, including recommendations for full-scale implementation

| | | | |
|---|---|---|---|
| **Deliverable Number** | D2.8 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Performance analysis results compared to alternative technologies, including recommendations for full-scale implementation | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP2 |
| **Description** | | | |
| Performance analysis results compared to alternative technologies, including recommendations for full-scale implementation | | | |

## Deliverable D3.1 – Map of stakeholders and their interests/needs

| | | | |
|---|---|---|---|
| **Deliverable Number** | D3.1 | **Lead Beneficiary** | 3. TUDO |
| **Deliverable Name** | Map of stakeholders and their interests/needs | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 13 | **Work Package No** | WP3 |
| **Description** | | | |
| Map of stakeholders and their interests/needs | | | |

## Deliverable D3.2 – Methodology for assessment of non-technological issues Deliverable D3.3 – Set of indicators measuring the non-technological impact of hubs

| | | | |
|---|---|---|---|
| **Deliverable Number** | D3.3 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Set of indicators measuring the non-technological impact of hubs | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 18 | **Work Package No** | WP3 |
| **Description** | | | |
| Set of indicators measuring the non-technological impact of hubs | | | |

| |
|---|
| **Description** |
| Methodology for assessment of non-technological issues |

## Deliverable D3.4 – Management guideline for integrating non-technological aspects

| | | | |
|---|---|---|---|
| **Deliverable Number** | D3.4 | **Lead Beneficiary** | 3. TUDO |
| **Deliverable Name** | Management guideline for integrating non-technological aspects | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP3 |
| **Description** | | | |
| Management guideline for integrating non-technological aspects | | | |

## Deliverable D4.1 – Modelling system for the characterisation of resources, energy flows and IS

| | | | |
|---|---|---|---|
| **Deliverable Number** | D4.1 | **Lead Beneficiary** | 4. TEC |
| **Deliverable Name** | Modelling system for the characterisation of resources, energy flows and IS | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 25 | **Work Package No** | WP4 |
| **Description** | | | |
| Modelling system for the characterisation of resources, energy flows and IS | | | |

## Deliverable D4.2 – Recommender algorithm for IS collaborations

| | | | |
|---|---|---|---|
| **Deliverable Number** | D4.2 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Recommender algorithm for IS collaborations | | |
| **Type** | OTHER | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 30 | **Work Package No** | WP4 |

## Deliverable D4.3 – Models for hub's supply chain management

| | | | |
|---|---|---|---|
| **Deliverable Number** | D4.3 | **Lead Beneficiary** | 10. ZLC |
| **Deliverable Name** | Models for hub's supply chain management | | |
| **Type** | OTHER | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 37 | **Work Package No** | WP4 |
| **Description** | | | |
| Models for hub's supply chain management | | | |

| |
|---|
| **Description** |
| Recommender algorithm for IS collaborations |
| **Description** |

| |
|---|
| Impact assessment of H4C from holistic perspective & decarbonisation pathways. |

## Deliverable D4.4 – Decision support tool for centralized/de-centralized hub

| | | | |
|---|---|---|---|
| **Deliverable Number** | D4.4 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Decision support tool for centralized/de-centralized hub | | |
| **Type** | OTHER | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 42 | **Work Package No** | WP4 |
| **Description** | | | |
| Decision support tool for centralized/de-centralized hub | | | |

## Deliverable D4.5 – Agent-based simulation tool for hub development

| | | | |
|---|---|---|---|
| **Deliverable Number** | D4.5 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Agent-based simulation tool for hub development | | |
| **Type** | OTHER | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 49 | **Work Package No** | WP4 |
| **Description** | | | |
| Agent-based simulation tool for hub development | | | |

## Deliverable D4.6 – Impact assessment of H4C from holistic perspective & decarbonisation pathways.

| | | | |
|---|---|---|---|
| **Deliverable Number** | D4.6 | **Lead Beneficiary** | 4. TEC |
| **Deliverable Name** | Impact assessment of H4C from holistic perspective & decarbonisation pathways. | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 40 | **Work Package No** | WP4 |

## Deliverable D4.7 – Virtual top-down hubs and their use as sustainable regional development models

| | | | |
|---|---|---|---|
| **Deliverable Number** | D4.7 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Virtual top-down hubs and their use as sustainable regional development models | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP4 |
| **Description** | | | |
| Virtual top-down hubs and their use as sustainable regional development models | | | |

## Deliverable D5.1 – Guidelines for the True Value Framework

| | | | |
|---|---|---|---|
| **Deliverable Number** | D5.1 | **Lead Beneficiary** | 17. KPMG |
| **Deliverable Name** | Guidelines for the True Value Framework | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 6 | **Work Package No** | WP5 |
| **Description** | | | |
| Guidelines for the True Value Framework | | | |

## Deliverable D5.2 – IS2H4C Business Model

| | | | |
|---|---|---|---|
| **Deliverable Number** | D5.2 | **Lead Beneficiary** | 17. KPMG |
| **Deliverable Name** | IS2H4C Business Model | | |
| **Type** | OTHER | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 42 | **Work Package No** | WP5 |
| **Description** | | | |
| IS2H4C Business Model | | | |

## Deliverable D5.3 – IS2H4C Financial Plan

| | | | |
|---|---|---|---|
| **Deliverable Number** | D5.3 | **Lead Beneficiary** | 17. KPMG |
| **Deliverable Name** | IS2H4C Financial Plan | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 49 | **Work Package No** | WP5 |

## Deliverable D6.1 – Report on hub starting lines

| | | | |
|---|---|---|---|
| **Deliverable Number** | D6.1 | **Lead Beneficiary** | 17. KPMG |
| **Deliverable Name** | Report on hub starting lines | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 18 | **Work Package No** | WP6 |
| **Description** | | | |
| Report on hub starting lines | | | |

| |
|---|
| **Description** |
| IS2H4C Financial Plan |

## Deliverable D6.2 – Report on hub installation progress and first 100 h of operation

| | | | |
|---|---|---|---|
| **Deliverable Number** | D6.2 | **Lead Beneficiary** | 4. TEC |
| **Deliverable Name** | Report on hub installation progress and first 100 h of operation | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 37 | **Work Package No** | WP6 |
| **Description** | | | |
| Report on hub installation progress and first 100 h of operation | | | |

## Deliverable D6.3 – Report on continuous operation & technology demonstration

| | | | |
|---|---|---|---|
| **Deliverable Number** | D6.3 | **Lead Beneficiary** | 10. ZLC |
| **Deliverable Name** | Report on continuous operation & technology demonstration | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP6 |
| **Description** | | | |
| Report on continuous operation & technology demonstration | | | |

## Deliverable D6.4 – Report of overall hub performance, outcomes, and dissemination

| | | | |
|---|---|---|---|
| **Deliverable Number** | D6.4 | **Lead Beneficiary** | 10. ZLC |
| **Deliverable Name** | Report of overall hub performance, outcomes, and dissemination | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP6 |

## Deliverable D7.1 – Early release (alpha version) with the first layer of functionality

| | | | |
|---|---|---|---|
| **Deliverable Number** | D7.1 | **Lead Beneficiary** | 29. IRIS |
| **Deliverable Name** | Early release (alpha version) with the first layer of functionality | | |
| **Type** | OTHER | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 13 | **Work Package No** | WP7 |
| **Description** | | | |
| Early release (alpha version) with the first layer of functionality | | | |

| |
|---|
| **Description** |
| Report of overall hub performance, outcomes, and dissemination |
| **Description** |

| |
|---|
| Report on risk assessment & mitigation – digital platform security |

## Deliverable D7.2 – Online marketplace platform add-in

| | | | |
|---|---|---|---|
| **Deliverable Number** | D7.2 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Online marketplace platform add-in | | |
| **Type** | OTHER | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 30 | **Work Package No** | WP7 |
| **Description** | | | |
| Online marketplace platform add-in | | | |

## Deliverable D7.3 – Extended version of DigitalH4C integrated with the digital tools

| | | | |
|---|---|---|---|
| **Deliverable Number** | D7.3 | **Lead Beneficiary** | 8. ICCS |
| **Deliverable Name** | Extended version of DigitalH4C integrated with the digital tools | | |
| **Type** | OTHER | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 42 | **Work Package No** | WP7 |
| **Description** | | | |
| Extended version of DigitalH4C integrated with the digital tools | | | |

## Deliverable D7.4 – Report on risk assessment & mitigation – digital platform security

| | | | |
|---|---|---|---|
| **Deliverable Number** | D7.4 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Report on risk assessment & mitigation – digital platform security | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 49 | **Work Package No** | WP7 |

## Deliverable D7.5 – Final delivery of DigitalH4C platform

| | | | |
|---|---|---|---|
| **Deliverable Number** | D7.5 | **Lead Beneficiary** | 29. IRIS |
| **Deliverable Name** | Final delivery of DigitalH4C platform | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP7 |
| **Description** | | | |
| Final delivery of DigitalH4C platform | | | |

| |
|---|
| **Description** |
| Dissemination and communication plan - Update 2 |

## Deliverable D8.1 – Dissemination and communication plan

| | | | |
|---|---|---|---|
| **Deliverable Number** | D8.1 | **Lead Beneficiary** | 11. EEIP |
| **Deliverable Name** | Dissemination and communication plan | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 6 | **Work Package No** | WP8 |
| **Description** | | | |
| Dissemination and communication plan | | | |

## Deliverable D8.2 – Dissemination and communication plan - Update 1

| | | | |
|---|---|---|---|
| **Deliverable Number** | D8.2 | **Lead Beneficiary** | 11. EEIP |
| **Deliverable Name** | Dissemination and communication plan - Update 1 | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 25 | **Work Package No** | WP8 |
| **Description** | | | |
| Dissemination and communication plan - Update 1 | | | |

## Deliverable D8.3 – Dissemination and communication plan - Update 2

| | | | |
|---|---|---|---|
| **Deliverable Number** | D8.3 | **Lead Beneficiary** | 11. EEIP |
| **Deliverable Name** | Dissemination and communication plan - Update 2 | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP8 |

## Deliverable D8.4 – Report on dissemination and communication activities

| | | | |
|---|---|---|---|
| **Deliverable Number** | D8.4 | **Lead Beneficiary** | 11. EEIP |
| **Deliverable Name** | Report on dissemination and communication activities | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP8 |
| **Description** | | | |
| Report on dissemination and communication activities | | | |

## Deliverable D8.5 – Exploitation plan report

| | | | |
|---|---|---|---|
| **Description** | | | |
| Exploitation plan report - Update 2 | | | |
| **Deliverable Number** | D8.5 | **Lead Beneficiary** | 1. UT |

| | | | |
|---|---|---|---|
| **Deliverable Name** | Exploitation plan report | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 6 | **Work Package No** | WP8 |
| **Description** | | | |
| Exploitation plan report | | | |

## Deliverable D8.6 – Exploitation plan report - Update 1

| | | | |
|---|---|---|---|
| **Deliverable Number** | D8.6 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Exploitation plan report - Update 1 | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 25 | **Work Package No** | WP8 |
| **Description** | | | |
| Exploitation plan report - Update 1 | | | |

## Deliverable D8.7 – Exploitation plan report - Update 2

| | | | |
|---|---|---|---|
| **Deliverable Number** | D8.7 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | Exploitation plan report - Update 2 | | |
| **Type** | R — Document, report | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 49 | **Work Package No** | WP8 |

## Deliverable D8.8 – Circular Industry helix “virtual cluster/ community” building

| | | | |
|---|---|---|---|
| **Deliverable Number** | D8.8 | **Lead Beneficiary** | 28. CHX |
| **Deliverable Name** | Circular Industry helix “virtual cluster/ community” building | | |
| **Type** | R — Document, report | **Dissemination Level** | PU - Public |
| **Due Date (month)** | 49 | **Work Package No** | WP8 |
| **Description** | | | |
| Circular Industry helix “virtual cluster/ community” building | | | |

## Deliverable D9.1 – OEI - Requirement No. 1

| | | | |
|---|---|---|---|
| **Deliverable Number** | D9.1 | **Lead Beneficiary** | 1. UT |
| **Deliverable Name** | OEI - Requirement No. 1 | | |
| **Type** | ETHICS | **Dissemination Level** | SEN - Sensitive |
| **Due Date (month)** | 2 | **Work Package No** | WP9 |

| **Description** |
| --- |
| Ethics Advisor_<br>The proposal raises several and intersectorial ethics issues. Although they do not represent complex and serious ethics issues, it is recommended that the Consortium is supported by an external independent ethics advisor, since the applicants do not show sufficient awareness on the ethics issues raised by the project: for instance, the ethics issues table is empty and the ethics self-assessment doesn't provide sufficient information. The Ethics Advisor should have expertise at least on involvement of humans in the research, on personal data, on environmental ethics and on AI ethics and should be consulted on such issues. The external independent Ethics Advisor must be appointed at Month 2 and provide a report at the end of each reporting period. |

# LIST OF MILESTONES

**Milestones**

*Grant Preparation (Milestones screen) — Enter the info.*

| Milestone No | Milestone Name | Work Package No | Lead Beneficiary | Means of Verification | Due Date (month) |
|---|---|---|---|---|---|
| 1 | Technology Map | WP2 | 2-MUL | A technology map with key technologies and their contributions to industrial synergies is available. | 6 |
| 2 | Energy Conversion Technologies | WP2 | 1-UT | Detailed design specifications, technical drawings, and performance data for the developed energy conversion technologies | 30 |
| 3 | Waste recovery and utilisation technologies | WP2 | 9-AEE | M2.3 Waste recovery and utilisation technologies 2 30 Detailed design specifications, technical drawings, and performance data for the developed waste recovery technologies and a pilot container at TRL7 demonstrating DCMD technology | 30 |
| 4 | Carbon capture & methanation technologies | WP2 | 4-TEC | A millireactor that can convert the industrial CO2, captured from Calcinor, into SNG with sufficient quality to be injected into the NG grid | 30 |
| 5 | Polymer production from captured carbon | WP2 | 6-FHG | Specifications of NIPU synthesized from captured carbon are available. | 30 |
| 6 | Expert panels and living labs | WP3 | 3-TUDO | A working structure for non-technological issues is established. | 25 |
| 7 | Digital tools prototype | WP4 | 1-UT | Prototype of decision support tools for matchmaking and hub management, for integration into DigitalH4C | 30 |

| | | | | | |
|---|---|---|---|---|---|
| 8 | Final version of digital tools | WP4 | 1-UT | Final version of decision support tools for matchmaking and hub management, for integration into DigitalH4C | 49 |

**Milestones**

*Grant Preparation (Milestones screen) — Enter the info.*

| Milestone No | Milestone Name | Work Package No | Lead Beneficiary | Means of Verification | Due Date (month) |
|---|---|---|---|---|---|
| 9 | Monitoring of indicators | WP5 | 17-KPMG | Regular and close analysis of the data being collected from the defined indicators, ensuring the information collected does not suffer major deviations. General overview in two stages (after 18 and 36 months of data monitoring) | 37 |
| 10 | Validated developed business model | WP5 | 17-KPMG | The developed business model will be tested on a regular basis before being finalised. 6 months prior to its conclusion, the model will undergo its main stress test, going through all 4 Hubs. | 37 |
| 11 | Status quo levelled | WP6 | 4-TEC | Hubs - standardized and comprehensive overview on available technologies, energy flows and CO2 emissions, relevant for this project, is generated. | 13 |
| 12 | Requirements for innovation defined | WP6 | 4-TEC | Requirements for all innovation aspects and planned installations are brought together in | 18 |
| 13 | All hubs achieve the first 100 h of operation | WP6 | 4-TEC | In all hubs, the first newly installed plants run for the first 100 h. | 37 |
| 14 | All innovation activities in operation | WP6 | 10-ZLC | In all hubs, all planned installation and operation activities have been kicked off | 40 |

| | | | | | |
|---|---|---|---|---|---|
| 15 | Process evaluation is completed | WP6 | 10-ZLC | Data gathering as per task T6.3 enables the use of the newly installed evaluation and analysis tool. | 25 |
| 16 | Preliminary release of DigitalH4C | WP7 | 29-IRIS | Submission of Deliverable D7.1 | 13 |
| 17 | Final demonstration of DigitalH4C | | 29-IRIS | Functional DigitalH4C and submission of Deliverable D7.5 | 49 |
| 18 | Website and social media up and running | WP8 | 11-EEIP | Website is operational with basic information about the project available by M3, to be enriched as the project progresses; social media accounts created (LinkedIn and Twitter) | 3 |

**Milestones**

*Grant Preparation (Milestones screen) — Enter the info.*

| Milestone No | Milestone Name | Work Package No | Lead Beneficiary | Means of Verification | Due Date (month) |
|---|---|---|---|---|---|
| 19 | 150 members Circular helix | WP8 | 28-CHX | To check number of helix members in CHX Platform | 37 |
| 20 | Stakeholder Map | WP3 | 3-TUDO | A stakeholder map with their interests, needs, target conflicts is available | 13 |
| 21 | Management guideline | WP3 | 3-TUDO | Drafted management guideline for integrating non-technological issues is available. | 42 |
| 22 | Defined hub models and input requirements | WP4 | 1-UT | The models for hub logistics planning and centralized/decentralized energy management are defined, with their requirements, constraints, and input data | 18 |

# LIST OF CRITICAL RISKS

| **Critical risks & risk management strategy**<br>*Grant Preparation (Critical Risks screen) — Enter the info.* | | | |
|---|---|---|---|
| **Risk number** | **Description** | **Work Package No(s)** | **Proposed Mitigation Measures** |
| 1 | Technical difficulties or delays during the development phase | WP1 | Regular progress monitoring and availability of backup solutions or alternative technologies in case of unexpected issues. |
| 2 | Inaccurate identification of key technologies, resulting in suboptimal outcomes | WP2 | Regular consultation with stakeholders and experts to ensure accurate identification and prioritisation of technologies. |
| 3 | Regular consultation with stakeholders and experts | WP2 | Close collaboration with industrial partners to ensure effective technology adaptation and customisation. |

| **Critical risks & risk management strategy**<br>*Grant Preparation (Critical Risks screen) — Enter the info.* | | | |
|---|---|---|---|
| **Risk number** | **Description** | **Work Package No(s)** | **Proposed Mitigation Measures** |
| | to ensure accurate identification and prioritisation of technologies. | | |
| 4 | Unavailability of accurate data for the simulations. | WP2 | Collaboration with experts in the field to ensure the accuracy of the simulation data and training project participants on simulation software. |
| 5 | Technical difficulties and the high cost of the polymer production process | WP2 | Collaboration with experts in the field to ensure the selection of the most cost-effective polymer and process |
| 6 | Poor stakeholder engagement in the hubs | WP3 | Hub representatives could enable identification and engagement of stakeholders (based on existing collaborations in the regions) |

| | | | |
|---|---|---|---|
| 7 | Digital database not ready to be used as input for matchmaking algorithm | WP4 | The matchmaking algorithm and decision support tools will first be tested using publicly available and/or artificial data. |
| 8 | Decision support tools delayed for the integration into DigitalH4C | WP4 | Since the development of the digital tools is conducted in an agile and iteratively manner, with versions that are extended step-by-step, the testing can be started with a prototype with limited functionality |
| 9 | The complexity of hub modelling is so high that the decision support tools cannot capture all aspects in one model | WP4 | Split the planning problem into smaller sub problems that can be optimised in parallel and combined again to an overall solution. |
| 10 | Market volatility and geopolitical uncertainty | WP5 | Economic and political ecosystem is in constant shift. In order to anticipate unexpected changes, contingency plans will be set, together with the coordinator and Hub leaders, and guidelines will be defined, with specific procedure to tackle/mitigate identified risks. |
| 11 | Legal and regulatory constraints | WP5 | Global presence of partners as KPMG enables the consortium to be informed and aware of legal and regulatory barriers in the countries of the different Hubs, and also allows an immediate tackle to any changes which may occur. |
| 12 | Confidentiality issues | WP6 | While all participating parties have the best intentions in mind for developing and nurturing the H4C approach, strict confidentiality rules might increase difficulty while trying to share operational data which is unavoidable for the success. |

**Critical risks & risk management strategy**

*Grant Preparation (Critical Risks screen) — Enter the info.*

| Risk number | Description | Work Package No(s) | Proposed Mitigation Measures |
|---|---|---|---|
| 13 | Adaptation modules in DigitalH4C platform due to source-coding differences | WP7 | DigitalH4C will embed different modules through APIs for separate modules hence nullifying the risk |
| 14 | Lack of sufficient stakeholders' traffic in the platform decreasing the platforms usability | WP7, WP8 | Involvement of different hubs within the consortia and specific tasks to encourage and massively involve stakeholders and end users using digital platforms and digital games reduces the accomplishment of this risk |

# PROJECT REVIEWS

**Project Reviews**

*Grant Preparation (Reviews screen) — Enter the info.*

| **Review No** | **Timing** (month) | **Location** | **Comments** |
|---|---|---|---|
| RV1 | 18 | On spot or online | |
| RV2 | 36 | On spot or online | |
| RV3 | 48 | On spot or online | |

**HISTORY OF CHANGES**

| Version | Date | Changes |
|---|---|---|
| 1.0 | 25/08/2023 | • Footer replaced with the following information 'Project: 101138473— IS2H4C – Part B – Page number (starting at 1 for Part B)'.<br>• History of changes added in first page of Annex 1 Part B.<br>• Table of contents added.<br>• Across the document, short names of each partner now match their short name in their registration in PDM.<br>• Section 1. Elimination of participant list and International Advisory Board (page 1).<br>• Section 1. Figure 1 – modification – to add water sector in Basque Hub (page 4).<br>• Section 1.1 Objectives and Ambition – Objective 1- Details regarding the state of the art of IS2H4C's technologies and Hub concepts (answering GAP Comment #Excellence) (page 6-9).<br>• Section 1.1. Objective 2: inclusion of an external Github link to visualize scale up impacts of the synergies in the hubs (page 11)<br>• Section 1.1 Objectives and Ambition – Objective 2 – inclusion of a new subobjective (page 10).<br>• Section 1.2. Methodology – Phase 2- Explanation the usage of AI in IS2H4C (answering GAP Comment #Excellence) (page 16-17).<br>• Section 1.2. Methodology – Description of the IS2H4C demonstration hubs – updated figure 5 (page 19)<br>• Section 2.1. Modification and explanation of details regarding the % of emissions reduced (answering GAP Comment #2.1) (page 25).<br>• Section 2.1. Details added for the external barriers and requirements per KPI -. (answering GAP Comment #2.1) (page 26, 27, 28).<br>• Section 2.2 Description of the business cases for each IS2H4C's Hubs (answering GAP Comment #2.2) (page 32-33).<br>• Section 2.2. Intellectual property management – Change of entity in charge UT as main manager (page 34).<br>• Section 3. Tables 3.1.a, 3.1.b, 3.1.c, 3.1.d., 3.1.e, 3.1.f. moved to Part A.<br>• Section 3. In tables 3.1.g and h, number of the beneficiaries are now included along their name (page 38-40).<br>• Section 3. Addition of new text above table 3.1.h (Purchase costs) and 3.1.g (subcontracting costs) (answering GAP Comments #1.4 and #4.4) (page 38).<br>• Part A Milestones: ○ added 2 new milestones (MS20 (M12) & MS21 (M42)) in WP3 ○ and a new milestone MS22 (M18) in WP4– Information attached in the portal.<br>• Part A Critical Risks:<br>  ○ Risk "Difficulty in adapting technologies to specific requirements" deleted<br>  ○ Risk "Regular consultation with stakeholders and experts" added<br>• Section 4: Addition of new Section 4 – Ethics self-assessment (answering GAP Comment #3<br>• Part A Work Packages: ○ WP 2 description: Added TEC to T2.1 and 2.3 as contributing partner. Corrected T2.4 1.1 NM3/h to 1 Nm3/h. ○ WP 3 description: Added Hub representatives to contributing partners for T3.2:, TUPRAS, TEC, ALMELO. |
| 2.0 | 17/10/2023 | • Part A, Work Package descriptions ○ Mentioned Deliverables in titles of Task descriptions ○ Replaced abbreviation SRL with SoRel in T3.3 description • Part B changes |

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|---|---|---|
| | | • Section 1.2. Methodology - Figure 4 update with removal of PROV partner (page 18)<br>• Section 2.1.2, Key Impact Pathway 1: changed two sentences according comments PO on first version (page 25)<br>• Section 2.2. Exploitation plan: Wording corrected as requested by the PO (Page 31)<br>• Section 2.2. Exploitation plan: deletion of partial text according to PO's comments (Page 32)<br>• Section 2.2. Exploitation plan: addition of text for strategic plan for all hubs based on PO's comments (Page 32)<br>• Section 2.3. Summary – correction of table size and inclusion of Dutch and Turkish hub KPIs (page 35).<br>• Section 3.1.1. Figure 11 inclusion of WP9-Ethics (page 36).<br>• PM efforts table updated after redistributing PROV tasks to other beneficiaries (Page 36)<br>• Deletion of CFS cost from INEGI (page 38)<br>• Deletion of CFS cost from IRIS (page 39)<br>• Section 4: Addition of a text regarding the external ethics advisor (page 44)<br>Removing PROV from project:<br>• Part A Work Packages: ○ WP3 description: T3.1 Hub representatives TUDO, ACE, ARCELIK and UT replace PROV; T3.2 TUDO replaces PROV; T3.3 TUDO, UT, TUPRAS, ACE replace role of PROV; T3.4 ACE, TUPRAS, UT, TUDO replace role of PROV<br>○ WP6: TEC replace PROV as lead. T6.1: KPMG replaces PROV; T6.2: TEC replaces PROV; T6.3: ZLC replaces PROV<br>• Part A Work Packages Effort:<br>• WP1: PROV -2; total Effort now 92 (was 94)<br>• WP2: PROV -3; total Effort now 312 (was 315)<br>• WP3: PROV -3, UT +6, TUDO +6, ACE +3, , ARCELIK +8; total Effort now 170,5 (was 150,5)<br>• WP4: PROV -4, ZLC +6; total Effort now 329 (was 327)<br>• WP5: PROV -5, TEC +5,5, KPMG +2; total Effort now 153 (was 150,5)<br>• WP6: PROV -48, UT +5, TEC +20, ZLC +2, KPMG +10; total Effort now 564,5 (was 575,5)<br>• WP7: PROV -4; total Effort now 198 (was 202)<br>• WP8: PROV -2,5, TEC +1; total Effort now 137 (was 138,5)<br>• In total 10 PMs have been added compared to the proposal.<br>• Part A Deliverables: ○ D6.1 lead: KPMG replaces PROV ○ D6.4 lead: ZLC replaces PROV<br>• Part A Milestones: ○ MS11 lead: TEC replaces PROV ○ MS12 lead: TEC replaces PROV |

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| 3.0 | 23/10/2023 | Adaption of project duration from 48 to 49 months<br>• Part A Work packages<br>• Part A Milestones<br>• Part B Section 3.1.1 table 8 "List of work packages" (page 37)<br>• Part B Section 3.1.1 "gantt chart" (page 38)<br>Adaption of deliverable deadlines and milestones in line with the extension of project duration from 48 to 49 months<br>• Part A Work packages<br>• Part A Milestones<br>• Part B Section 3.1.1 "gantt chart" (page 38) |
| | | Addition of corresponding deliverables to the task descriptionsAssociated with document Ref. Ares(2023)8308660 - 05/12/2023<br>• Part A Work packages<br>• Part B Section 3.1.1 "gantt chart" (page 38)<br>Adaption of TRL overview: "IS2H4C's technology details regarding the current level of TRL and achievement of TRL milestones during the project", to indicate the current maturity and the targeted maturity and TRL levels of the individual technologies.<br>• Part B Section 1.1 Objectives and Ambition (page 7) Revision of the Ethics Self-Assessment<br>• Part B Section 4 The Ethics Self-Assessment has been revised according to the provided structure (page 45) |
| 4.0 | 06/11/2023 | Revision of Hub Representatives<br>• Part B Section 1.2 Methodology & Overall Concept the mentioning of PROVADIS Hochschule as partner of the German Hub has been deleted (page 19) and SIDENOR I+D SA has been added to the list of hub representatives for the Basque Hub (page 20)<br>• Part A, Work Package descriptions:<br>• WP3 T3.1: added SIDENOR and SIDENOR I+D to list of Hub representatives<br>• WP4 T4.1: added SIDENOR I+D<br>• WP5 T5.5: added SIDENOR I+D<br>• WP6 T6.1: added names of Hub representatives between brackets as elucidation. |

# Sustainable Circular Economy Transition: From Industrial Symbiosis to Hubs for Circularity CLUSTER 4 — CLIMATE NEUTRAL, CIRCULAR AND DIGITISED PRODUCTION

**HORIZON-CL4-2023-TWIN-TRANSITION-01-37:** Hubs for circularity for near zero emissions regions applying industrial symbiosis and cooperative approach to heavy industrialized clusters and surrounding ecosystems (Processes4Planet partnership) (IA)

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Heavily industrialized clusters are important sources of jobs and wealth in the EU. However, despite the ongoing efforts of EU policies, industries are lagging in the implementation of Industrial Symbiosis (IS) to transit to more resource- and energy-efficient setups. Furthermore, citizens living in the clusters' proximity suffer from a polluted environment. Despite the EU's strategic priorities in sustainable development, human-centred IS awaits to be designed, implemented, and expanded. Currently, the stateof-the-art of IS falls short of such a demand. Initiatives like SPIRE (now Processes for Planet: P4Planet) have researched and demonstrated pilot concepts of IS and even Industrial-Urban-Rural Symbiosis (IURS), achievable via closing energy, resource, and data loops through infrastructure sharing. In this regard, P4Planet has designed the concept Hubs for Circularity (**H4C**)[1]. **H4C** provide spaces where diverse actors, such as businesses, governments, researchers, and civil society, collaborate to accelerate the Circular Economy (CE) transition. They are nexuses for knowledge and resource sharing, cross-industrial collaboration, and sustainable technology innovation. **H4C** can take various forms, from physical spaces, such as IS Networks (ISNs) and co-working clusters, to virtual platforms such as online communities and digital marketplaces. They can also serve as a resource for governmental bodies, providing evidence-based research and recommendations for CE policymaking. Overall, **H4C** pave the way for IS leading to a net-zero CE. In light of advanced IS implementation, the "From **I**ndustrial **S**ymbiosis to **H**ubs for **C**ircularity: **IS2H4C"** project will deliver a systemic approach of IS to Hubs for Circularity, helping P4Planet materialising its goals.

Next to conventional IS achieved through the exchange of materials, water, and energy, **IS2H4C** promotes infrastructure-, service, and carbon-based IS to transform existing industrial areas into ISNs. These ISNs are integrated with surrounding rural and/or urban ecosystems to form physical **H4C**. The development of physical **H4C** is facilitated by a digital collaboration platform (Digital**H4C**). Digital**H4C** fosters the bottom-up emergence of ISNs by real-time resource monitoring and tracking, dynamic supplydemand matchmaking, efficient and secure information-sharing, and transparent marketplace trading. It also performs effective evaluation of top-down coordination strategies based on a multi-dimensional approach combining attributional and consequential Life Cycle Assessment (LCA). **IS2H4C** demonstrates four first-of-a-kind hubs with business cases and exploitation strategies in the Netherlands, Germany, Spain, and Turkey, populated by various industrial sectors (Fig. 1). Cutting-edge green technologies are deployed and demonstrated to realise IS in demo hubs, including wind and solar power generation, electrolysis for hydrogen, carbon capture and utilisation, waste heat capture and distribution, heat pumps, hydrogen use in built environment, and oxycombustion. The technology innovation and integration elevate the sustainable industrial performance and boost the socioeconomic wellbeing in the surrounding ecosystems. In a nutshell, **IS2H4C** is a real-life demonstrator of **H4C** in scale. It establishes best IS practices towards a climate neutral, circular, and digitised future.

[1] Hubs4Circularity (**H4C**s) | A.SPIRE. (n.d.). https://www.aspire2050.eu/p4planet/hubs4circularity

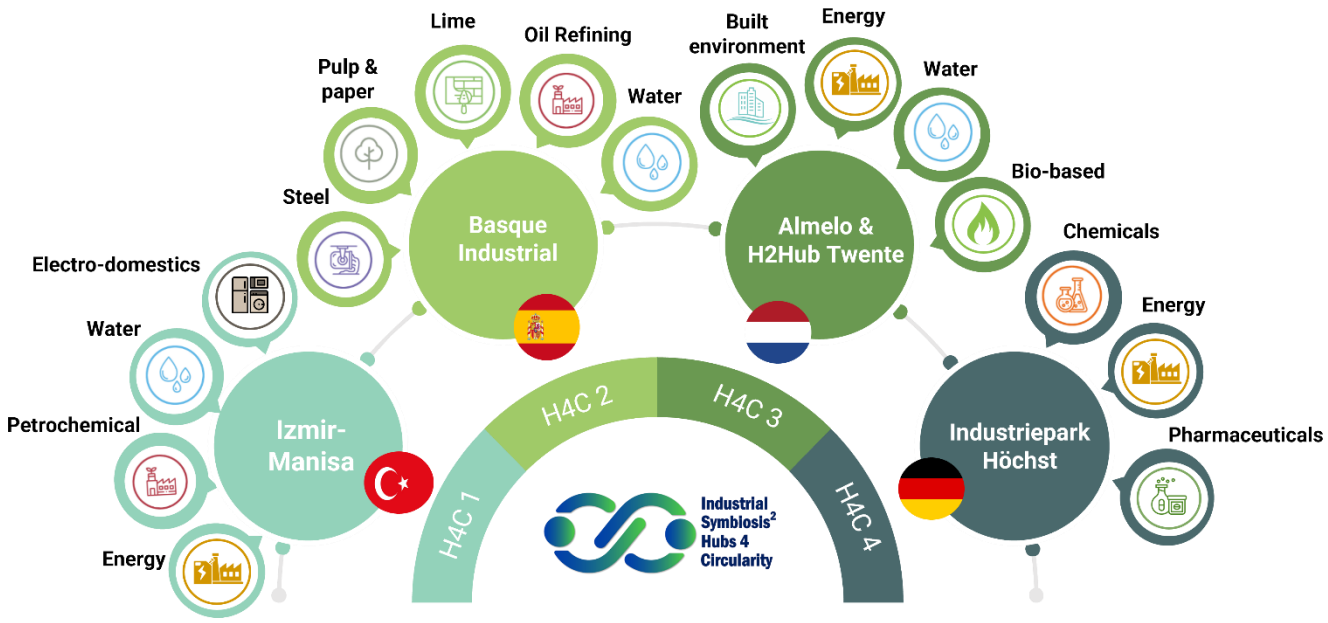


*Figure 1. **IS2H4C**'s overall concept*

## 1.1. OBJECTIVES AND AMBITION

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**IS2H4C**'s mission is to pave the way towards the development of **H4C** based on the circularity and resilience requirements of existing and future industrial zones and their surrounding ecosystems. To achieve this mission, **IS2H4C** prioritizes resource efficiency, use of renewable energy, and waste prevention, and promotes IURS via reuse and recycling of unavoidable solid, liquid, and gas waste streams via deployment of innovative sustainable technologies and via systemic integration of existing and new infrastructure. Importantly, **IS2H4C** ensures societal and governmental engagement to **H4C** via securing acceptance and trust among citizens based on clear communication, transparency, and inclusion, by proposing sustainable business and financial models, as well as respecting the planetary health boundaries. **IS2H4C**'s biggest ambition is to become a reference for promoting **H4C** as Europe's future sustainable regional development model.

The consortium formulated six operational objectives with specific Key Performance Indicators (KPIs) to precisely monitor the technological readiness evolution, the project's advancement beyond the State of the Art (SoA), and to operationally implement **IS2H4C**'s mission.

### Objective 1. Deploy and demonstrate near commercial scale industrial symbiosis through innovative technologies

Each hub has strong individualized and already-implemented technologies that aim to comply with the EU Green Deal guidelines[2]. **IS2H4C** scales up and integrates recent sustainable technologies (at different TRLs) to the involved industries and infrastructure to exploit the hubs' vast room for resource and energy-use optimisation and IURS implementation. **IS2H4C** demonstrates how **H4C** emerge from bottom-up[3], operate, and evolve in four locations (populated with different process industries and rural/urban

[2] EC (2023). The EU Green Deal. Re: https://eur-lex.europa.eu/legal-content/EN/TXT/?qid=1576150542719&uri=COM%3A2019%3A640%3AFIN [3] Blomsma, F., Brennan, G., (2017). The Emergence of Circular Economy. *J. Ind. Ecol.* 21, 603-614. https:// doi.org/10.1111/jiec.12603

combinations) and brings these **H4C** to a systemic TRL7/TRL8. **IS2H4C** proposes two types of synergies: (1) **17 synergies technologically demonstrated in the project** and (2) **24 synergies to be studied for hub expansion in the project**. Furthermore, scale-up and further expansion will be supported by on-site matchmaking workshops and IS recommenders embedded in the digital **H4C** via deploying **Information Technologies**. For the first category, **IS2H4C** proposes the following technology and infrastructure deployment and implementations in the involved **H4C**:

1. **Basque Hub**: contains oil refining, steel, pulp and paper and lime industries, and the public wastewater treatment plant located in the highly industrialized area of the Basque Country. Planned synergies: (i) **oxycombustion** with $O_2$ and **$H_2$** in the steel industry **(**potentially from the forthcoming electrolysis of the oil refining sector). (ii) **$CO_2$** via carbon capture (**CC**) by the lime industry and used **(CCU)** to produce **$CH_4$** (and potentially for the forthcoming production of e-fuels in the oil sector). (iii) **Carbonation** of steel slags with the $CO_2$ obtained from the lime industry (**CCU)** to produce construction products for the cement industry.
2. **Dutch Hub**: Twente area contains a small-scale hydrogen hub formed by approximately 20 process industry companies connected to a nearby village, Aadorp, which is planned to evolve into a positive energy district via becoming independent from natural gas and fossil-based electricity. Planned synergies: (i-ii-iii) **Green $H_2$** obtained from **electrolysis** based on **solar** energy and (to-be-implemented) **wind** energy is used in **five houses** in built environment in Aadorp via (to-be-implemented) **$H_2$** pipelines. (iv) **Energy storage** is implemented to stabilize the feed of the green energy generation mentioned in synergies i-ii to tackle supply uncertainty. (v) **$H_2$** is transported to **three crematoria** in Twente by $H_2$ tube trailers & substitutes **natural gas**. (vi-vii) **$O_2$** from electrolysis will be sent to the **wastewater treatment unit** located 100 meters away from the electrolyser for **water purification** & **treated water** will be sent to electrolyser for $H_2$ production via (to-be-implemented) **two pipelines**.
3. **Turkish Hub**: Izmir-Manisa area contains petrochemical and household appliances companies, located nearby an industrialized port area on the Aegean coast. Planned synergies: (i-ii) Green **$H_2$** obtained from **electrolysis** based on the use of **renewable energy** and captured **$CO_2$** from oil refining activities via **CC adsorption** technology are used for **e-methanol** production. (iii-iv) Captured **$CO_2$** is also used in the production of **non-isocyanate polyurethane (NIPU)** which substitutes traditional **polyurethane** in the production of **refrigerators**.
4. **German Hub**: Industriepark Höchst contains chemicals and pharmaceuticals industry, located in Frankfurt. Planned synergies: (i-ii) **$CO_2$** obtained from (to-be-implemented) **CC** technology implemented in a power generation unit and **$H_2$** obtained from existing electrolysis (facilitating industrial waste heat use) are both used in **e-methanol** production replacing traditional methanol in the chemical and pharmaceutical industry.

**All hubs:** Next to hub-specific solutions, **sensing technology** will be used to monitor the content and flows of one sample resource in each hub. The retrieved data will be embedded in digital product pass to achieve resource traceability and data interoperability to improve the performance of IS recommendations in the digital collaboration platform (Digital**H4C**).

| Specific activities, KPIs and validation means |
|---|
| - Identify (energy) infrastructure ready to be used and facilitate its use for multiple unpractised users<br>- Analyse infrastructure integration scenarios and simulate new synergies<br>- Upgrade existing processes and co-design new processes to facilitate coupling and/or versatile/shared use of resources by multiple users from diverse sectors |

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|---|
| - Develop technologies from TRL3-5 to at least TRL7 and deploy them in demo hubs or directly apply technologies with at least TRL7 in the hubs<br>- Maximize renewable energy use, utilize green hydrogen as energy carrier or alternative material, and implement material/water-/energy-/carbon-/service-based IS models |

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- Implement an agent-based modelling and simulation framework[3] and simulate bottom-up emergence and evolution scenarios considering that **H4C** are complex adaptive systems
- Analyse the scenarios in the four locations during and beyond project period to propose systemic scale-up and expansion strategies for these hubs aligned with the operational objective#8 of Processes4Planet partnership

**KPIs:** 2 synergies in the German hub, 4 synergies in the Turkish hub, 4 synergies in the Basque hub, 7 synergies in the Dutch hub will be implemented thanks to the development and deployment of sustainable technologies and integration of existing and new infrastructure. These total of **17 synergies** are direct technology deployment and demonstrations of the **IS2H4C**. Furthermore, there are **24 hub expansion synergies** explained (for sake of brevity) in the description of the **IS2H4C** hubs in the methodology section. Next to these, there will be new synergies thanks to the engagement of other actors of the hubs to planned symbioses (who are not direct partners of **IS2H4C** but located in hubs). **IS2H4C** will reach out to a total of at least **150 companies** in all hubs via ad-hoc matchmaking **workshops** and the **marketplace** embedded in the digital **H4C**. Furthermore, new synergies will be triggered thanks to the adoption of (already developed and to-be-further-improved) **two IS recommenders**[4,5]. Considering that one IS recommender performs at 33% success rate[6] and one synergy involves at least two companies (potential of 75 synergies involving 150 companies), it will trigger at least **25 extra synergies**. Assuming to increase **recommender performance to at least 50%**, it can achieve **40 extra synergies**. **IS2H4C** will produce simulation scenarios for hub expansion and scale-up, via developing **one agent-based simulation tool.** Depending on parameters and number of influential factors[6], it can produce thousands of different simulation results. **4 expansion and scale-up strategy reports** will be produced, one for each hub, customised under different development scenarios based on simulation results.

**Validation**: 11 technologies to be deployed: water electrolysis (green $H_2$, TRL9), CC membrane technology (TRL7 to TRL9), CC adsorption technology (TRL 6 and higher), NIPU from $CO_2$ (from TRL3 to TRL7), e-methanol from $CO_2$ (TRL8), synthetic natural gas from captured $CO_2$ (from TRL5 to TRL7), oxycombustion (from TRL6 to TRL8), steel slag carbonisation (from TRL5 to TRL7), waste heat recovery (from TRL5 to TRL7), oxygen use in wastewater treatment (from TRL8 to TRL9), use of water from wastewater treatment in electrolysis (from TRL6 to TRL8), sensing technology (from TRL7 to TRL8), and IS recommenders (from TRL5 to TRL7). Activities below TRL7 (such as lab experiments and desk research, e.g., as technology modelling) are attributed to WP2, and activities for field work for technology deployment (TRL7 & higher) are attributed to WP6.

**5 houses** test **$H_2$** in built environment using **newly developed $H_2$ pipelines** (WP6).

| *IS2H4C's technology details regarding the current level of TRL and achievement of TRL milestones during the project* | | |
|---|---|---|
| **Technology** | **Current TRL** | **Targeted TRL** |
| Water electrolysis (green H2, TRL9) – (TUPRAS) | Currently, there is no water water electrolyzer in Tupras. | Tupras will buy commercial water electrolyzer(TRL8+). Tupras will invest more than the project requires. |
| CC adsorption technology (TRL 6 and higher) (TUPRAS, TEC & LKT) | Tupras has CO2 adsorption unit TRL6 through EU funded MOF4AIR project. | Since e-methanol production requires high purity (99+%) CO2. Tupras will consider either investment of new unit (Amine based CO2 capture, TRL9) or additional purification unit for MOF4AIR. |
| NIPU from CO2 (from TRL3 to TRL7) (FHG) | NIPU polycondensation is performed on lab scale up to 1kg per batch. CO2-based monomers for NiPU are synthesized in high pressure reactors on lab scale up to 0.5 kg per batch. | Within the project, the scale-up of monomer synthesis and NiPU polycondensation to 10 kg per batch in stirred tank reactor and verify reproducibility for industrial applicability shall be realized. |

[3] EC Processes4Planet. (2023): https://ec.europa.eu/research-and-innovation/sites/default/files/bmr-2022/ec_rtd_bmr-2022-processes4planet-fiche.pdf

[4] van Capelleveen, G., et al., 2021. Exploring recommendations for circular supply chain management through interactive visualisation. *Decis. Support Syst.*, *140*, 113431.

[5] van Capelleveen, G., et al., 2018. The influence of knowledge in the design of a recommender system to facilitate industrial symbiosis markets. *Environ. Model. Softw.*, *110*, pp.139-152.

[6] Castiglione, C., et al., 2021. A holistic technological eco-innovation methodology for industrial symbiosis development. *Sustain. Prod. Consum.*, *28*, pp.1538-1551.

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| e-methanol from CO2 (TRL8) (FHG & CIRCE) | Fraunhofer UMSICHT owns several laboratory and pilot units for the synthesis of Methanol as well as the necessary downstream operations to separate the Water/Methanol mixtures. The capacity covers up to 2 kg methanol per hour. | Within the project, the plants will be used to produce methanol based on the gas compositions from the real industrial processes provided by the industrial partners, e.g. Tupas. |
| Synthetic natural gas from captured CO2 (from TRL5 to TRL7) (TEC) | The milli-channel reactor for the conversion of 0,5 Nm3/h of CO2 to synthetic natural gas, has been already tested in a relevant environment (with Calcinor's combustion gas) and has demonstrated its viability (TRL5). | The main goal is to develop and demonstrate the viability of 1Nm3/h pilot plant using real and continuous combustion gas coming from the industrial site (Calcinor) and thus in an industrial operational environment (TRL7). This objective will be achieved by carrying out a capture and cleaning system for the industrial gas generated at Calcinor (designed by Lointek) and by designing the reaction system capable of treating 1 Nm3/h of CO2 an 4Nm3/h of H2 continuously (Tecnalia). These two systems will be integrated and will work continuously. |
| oxycombustion (from TRL6 to TRL8) (TEC) | Sidenor has already tested the effects of small quantity of H2 in the steel ladle preheating and thus demonstrated its applicability (replacement of NG use) in a relevant scale (TRL6). | The substitution of natural gas air combustion by hydrogen and including oxy-fuel combustion in the application of ladle preheating will allow to demonstrate the applicability of H2 and O2 in the replacement of fossil fuel in the steel sector (TRL8). H2 and O2 trials monitoring and tests will be performed for this purpose |
| steel slag carbonisation (from TRL5 to TRL7) (TEC) | The patented High Efficiency Contact HECO® technology is being adapted for mineralization of diverse alkaline waste streams; among others, for stainless steels. The device is a fluidised bed reactor type, which consists in a 360mm Ø conical/cylindrical metallic body, fed with material through its upper mouth, with a CO2 gas flow inlet through the lower part, and a sampler in its lower part. An internal draft tube, placed in the gas inlet mouth, favours the formation of the distinctive characteristic movement (spouting), which resembles the movement of water from a fountain. This type of movement greatly maximizes the gas/solid contact rate during the reaction, increasing the speed of the matter and energy transfer. An additional internal device (confiner) generates a recirculation of gas and solid internal to the reactor, thereby increasing the contact per unit of time, and then the CO2 uptake.<br><br>Currently, SBS provides a pilot plant reactor validated in relevant environments (TRL5) but improvements in efficiency must be performed.<br><br>It has not been used yet for carbonatation of stainless steel slags. | When dealing with direct gas/solid reaction-based technologies for alkaline waste mineralization, there are still unresolved optimization challenges: i) the maximization of the gas/solid contact for a maximum CO2 uptake; ii) the optimal humidity distribution through solids and gas under industrial flue-gas conditions with variable conditions (flow rate, temperature, pressure, humidity, chemistry, waste nature), iii) the continuous monitoring of the relevant mineralogical transformation; iv) minimal process complexity and v) minimization in energy use. The goal will be to validate and demonstrate the performance of the carbonated stainless steel slags for construction applications, and thus its potential use in an industrial sector environment (TRL7). |

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| Waste heat recovery (from TRL5 to TRL7) (H2HUB, SOLENCO, ALMELO) | Technology is validated in Scandinavia at TRL 5. | We aim to demonstrate this technology using the geothermal heat storage on site. This geothermal source will be used with water-water heat pumps to recover the waste heat from the H2-producten. The two-usage scenario's will be to heat the office and factory areas as well as to keep the stack's inflow at the correct temperature for optimal operation.(TRL 7) |
| Use of **oxygen** in wastewater treatment (from TRL6 to TRL8) (H2HUB, SOLENCO, ALMELO) | TRL 6 is curre ntly proven in another environment (Waterschap Drents Delta). Overijsselse Drents Waerschap Delta Overijsselse a Vechtstromen nd information abo interchange for future implen | In close cooperation with Waterschap Vechtstromen we will implement a realworld application of the proven concept, including the two pipelines for oxygen and wastewater transport. |
| Sensing technology (from TRL7 to TRL8) | Sensing technologies (prototypes or products) demonstrated in an operational environment will be used along for real time data exchange in a secure way with the IS2H4C digital collaboration platform. | The sensing technologies will be completed and qualified next to hub-specific solutions, used to monitor the content and flows of sample resources in the largescale real environment of hubs. The retrieved data will be embedded to achieve resource traceability and data interoperability to improve the performance of IS recommendations in the digital collaboration platform. |
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**Information technologies influencing synergy implementation and hub scale-up/expansion:** recommenders, marketplace, and conceptual digital product passports will be developed in the form decision-support tools (WP4) which are going to be embedded in the digital **H4C** (WP7).

## State of the art - Recommender Systems Technology for Industrial Symbiosis Implementation

**Introduction:**

Recommender systems have emerged as valuable tools in the field of industrial symbiosis, aiding industries in identifying potential resource-sharing opportunities and fostering collaboration.

Industrial symbiosis is the practice of collaborative resource sharing among industries to optimize resource utilization and reduce waste. Recommender systems in this context act as decision support tools, suggesting potential symbiotic partners, resources, or processes to enhance sustainability and circularity.

**Current State of the Art:**

- **TRL Assessment (Current TRL: 4-5):**

Recommender systems for industrial symbiosis are currently at TRL 4-5. Research and development have yielded promising results in controlled environments and pilot projects, but widespread implementation and scaling are limited.

- **Key Features and Capabilities:**
- **Data Integration:** Recommender systems integrate various data sources, including resource inventories, production processes, and environmental impact data, to identify symbiotic opportunities.
- **Machine Learning Algorithms:** Advanced machine learning algorithms, including collaborative filtering, content-based filtering, and hybrid approaches, are employed to analyze and predict resource-sharing possibilities.
- **User Interaction:** Many systems offer user-friendly interfaces, allowing industries to interact with the recommendations, provide feedback, and customize their preferences.
- **Performance Metrics:** Evaluation metrics, such as precision, recall, and F1-score, are used to assess recommendation quality.
- **Case Studies:** Several successful case studies showcase the potential of recommender systems in reducing waste, optimizing resource use, and improving the sustainability of industrial processes.

**Steps to Advance to TRL 7:**

- **TRL 6 - Prototype Testing:**

- **Prototype Deployment:** Conduct field tests and pilot implementations in industrial settings to validate system performance, user acceptance, and scalability.
- **Integration with Existing Systems:** Ensure compatibility and integration with existing industrial data systems and enterprise resource planning (ERP) platforms.
- **Performance Optimization:** Continuously refine machine learning algorithms to improve recommendation accuracy and response time.
- **TRL 7 - Full-Scale Deployment:**
- **Scalability:** Develop the architecture and infrastructure necessary for large-scale deployment, accommodating diverse industries and resource types.
- **Interoperability:** Establish standardized data formats and interfaces to facilitate cross-industry collaboration and data sharing.
- **Regulatory Compliance:** Ensure compliance with data privacy regulations and industry-specific standards.
- **User Training and Support:** Provide comprehensive training and support resources to help industries maximize the system's benefits.
- **Performance Monitoring:** Implement robust monitoring and reporting tools to track the environmental and economic impact of symbiotic collaborations.

**Conclusion:**

Recommender systems for industrial symbiosis hold great promise in advancing sustainability and circular economy goals. While they are currently at TRL 4-5, a structured approach involving prototype testing (TRL 6) and full-scale deployment (TRL 7) can enable the technology to be reliably deployed and scaled in real-world industrial contexts. Advancing the TRL of these systems is contingent on addressing scalability, interoperability, regulatory compliance, user support, and performance monitoring challenges, ultimately driving the transition towards more sustainable and circular industrial practices.

## State of the Art in Hubs for Circularity:

Industrial symbiosis, particularly in the form of Hubs for Circularity, has made significant progress in advancing sustainability and circular economy goals.

Industrial symbiosis has evolved from isolated initiatives to well-established collaborative networks. Hubs for Circularity bring Associated with document Ref. Ares(2023)8307660 - 05/12/2023

together diverse stakeholders, including industries, governments, research institutions, and NGOs, fostering stronger cooperation and resource-sharing. Advanced digital technologies and data analytics have been integrated into circular hubs. This enables real-time resource tracking, demand-supply matching, and predictive maintenance, optimizing resource flows and minimizing waste.

Companies within circular hubs have embraced circular design principles, leading to innovative product designs, materials, and production processes. This has resulted in products that are more easily recycled, remanufactured, or refurbished. Governments have recognized the importance of circular hubs and have introduced supportive policies and incentives. These include tax incentives for circular practices, extended producer responsibility (EPR) regulations, and waste reduction targets.

Circular hubs have advanced in extracting value from waste streams. Technologies for recycling, upcycling, and recovering resources from waste have become more sophisticated, reducing environmental impact and creating economic opportunities. Circular hubs have extended their influence along supply chains, encouraging suppliers to adopt sustainable practices and customers to engage in responsible consumption. This holistic approach promotes sustainability throughout the product lifecycle.

**Remaining Challenges and Steps Towards Full-Scale Circular Hubs:**

1. **Scaling Up:** While circular hubs have demonstrated success in specific regions, scaling up this model remains a challenge. Achieving widespread adoption requires overcoming logistical, regulatory, and financial barriers.

2. **Standardization and Interoperability:** Establishing common standards for resource exchange platforms and data sharing is crucial for seamless collaboration. Ensuring interoperability among digital systems is essential to connect various stakeholders effectively.

3. **Behavioral Change:** Shifting mindsets and behaviors at all levels of the supply chain and society is a complex task. Education, awareness campaigns, and incentives are needed to promote circular thinking and practices.

4. **Investment and Funding:** Securing long-term investment and funding for circular hubs is essential. Public-private partnerships and innovative financing models are necessary to support the transition to circularity.

5. **Regulatory Alignment:** Ensuring that regulations support circular economy principles and remove obstacles to circular practices is crucial. Regulatory frameworks need to evolve in sync with the circular economy's goals.

6. **Measurement and Reporting:** Developing standardized metrics and reporting mechanisms for circularity is vital to assess the impact of circular hubs accurately. This will facilitate transparency and accountability.

7. **Cross-Sector Collaboration:** Encouraging collaboration among industries from different sectors, not just within specific hubs, is essential for a holistic circular economy transition.

In summary, industrial symbiosis and Hubs for Circularity have made significant strides in promoting sustainability and circularity. However, achieving full-scale circular hubs requires overcoming challenges related to scaling up, standardization, behavioral change, investment, regulatory alignment, measurement, and cross-sector collaboration. Continued innovation, cooperation, and strategic planning are essential to realizing the full potential of circular hubs in creating a sustainable and circular future.

## Objective 2: Ensure resilience and sustainability of H4Cs considering planet, people, and profit

**IS2H4C** will adopt circularity principles, safeguarding environmental constraints, social needs, and business requirements. **IS2H4C** integrates bottom-up feedback with top-down coordination mechanisms

[7] Bocken, N. M. P., et al., 2016, Product design and business model strategies for a circular economy. *J. Ind. Prod. Eng.*, 33(5), 308–320.

that connect human needs, inter-organisational collaboration, and (inter-)regional governance. Circularity principles will align with the concept of Reduce-Reuse-Recycle in the form of narrowing resource loops and closing resource loops[8]. This results in a reduction of resource (material/water/energy) consumption, emissions of waste material/wastewater/waste energy, and GHGs ($CO_2$, CO, $CH_4$, NOx, etc.). Also, planetary and societal health in **H4C** is improved via deploying the most innovative resource-efficient technologies and enhancing systemic change. **IS2H4C** assesses the social, economic, and environmental impact of introducing innovations through measure metrics and consequential LCA[8]. Consequential LCA is ideally suited for **IS2H4C** because it allows studying system expansion and carries out more comprehensive results compared to attributional LCA[9].

**Specific activities, KPIs and validation means**

- Prepare a resilience plan for **H4C** supported by circularity strategies to enable hub actors to tackle short-term shocks such as material and/or energy shortages
- Design optimized resource flows in **H4C** considering the Earth's resource capacity & limitations imposed by regulations
- Implementing the objective functions for common good of hubs such as maximum renewable energy use, fossil fuel savings, captured carbon, profit, etc., or minimum resource consumption, waste discharge, carbon emissions, health impacts, infrastructure cost, etc.
- Achieve at least **30% carbon, 10% energy, and 20% overall resource (material/water/energy) reduction** in four demo **H4C** locations, aligned with the expectations of the **Fit-for-55 plan of the EC**[10]
- The analysis of the biogenic CO2 from the wastewater treatment plant to produce bio-fuels.

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[8] Consequential-LCA (2015). The ISO 14040 standards for consequential LCA. Retrieved April 2023 from www.consequential-lca.org

[9] Weidema B P (2014). Has ISO 14040/44 failed its role as a standard for LCA? *J. Ind. Ecol.* 18(3):324-326

[10] EC (2022). Fit for 55 Package. (2023): https://www.europarl.europa.eu/RegData/etudes/BRIE/2022/733513/EPRS_BRI(2022)733513_EN.pdf

**KPIs: 1 integrated circular hub management plan** based on resource/energy optimisation scenarios including a resilience plan based on future projections of supply and demand shocks for all hubs. The environmental impacts of the future strategies are described in a four-year period as follows:

**1. Basque Hub:** The percentages are calculated in respect to either the given or estimated total $CO_2$ production of each industry in the Basque Hub. (1) Petronor's main contribution to the hub is the potential supply of $H_2$ and $O_2$ to the rest of the industrial processes. The potential annual carbon reduction achieved by circular strategies will be around 5,506 tons (based on 2.9 million $m^3$ of natural gas reduction). (2) Smurfit Kappa: potentially using the paper sludge to produce biomethane could result in 37KT of $CO_2$ reduction (8%). Also, by using the paper waste for bio-ethanol production with a potential of generating 206l of bioethonal per 1000kg of paper waste and lignin for biofuels production, could result in a new and very interesting business for these future bio-refineries. They could supply biofuels to other industries and endorse circularity and decorbanisation.(3) Calcinor: Methanation results in a $CO_2$ reduction of 9.3 tons of $CO_2$ (1.40%). (4) Sidenor: Using $O_2$ and $H_2$ for reheating the furnaces results in 127,567 tons reduction of $CO_2$ (14.79%).

**2. Dutch Hub:** (1) Wastewater treatment plant: The expected amount of total avoided $CO_2$ emission is 306 tons. It will save energy to treat around 400 million litres of wastewater. Approximately 8,000 tons of $O_2$by-product can be reused thanks to treated water send to the electrolysis reducing energy consumption by 18%. Meanwhile, the reuse of waste heat in the offices will avoid 34% of energy waste. (2) Solar & wind energy: It will avoid 1,740 tons of $CO_2$. This benefits the surrounding residential areas to reduce their carbon emissions by 65%. (3) Hydrogen fuelling stations: The stations with a capacity of 200-1,000 kg/day leading to a total carbon reduction of around 19,274 tons (assuming 70kW electrolyser). The full-scale implementation of 5MW electrolyser will generate 2 tons of hydrogen/day avoiding around 20,000 tons of $CO_2$/year based on five stations.

**3. German Hub:** The implementation plan will contribute to material reduction of 52,500 tons of methanol/year and $CO_2$ reduction of 157,500 tons/year on average, when fully scaled up. Over the course of the project, it will lead to an overall hub performance of 210,000 tons of methanol reduction and 630,000 tons of $CO_2$ reduction. Further, 557.8 GWh/year energy input reduction is achieved. For long-term implementation, the production of synthetic ethylene, acetic acid and methane can contribute further to achieve the resource and emission reduction as follows: (1) Methane: 190,000 tons/year in total material inputs and 570,000 tons/year in $CO_2$. (2) Ethylene: 460,000 tons/year in total material inputs and 1,380,000 tons/year in $CO_2$. (3) Acetic acid: 80,000 tons/year in total material inputs and 240,000 tons/year in $CO_2$. Additionally, the future expansion of heat pump systems and ethylene production will result in annual $CO_2$ reduction of 570,000 tons under the scenario with the maximum capacity. This corresponds to 21% resource reduction, 10% energy reduction and 45.25% $CO_2$ reduction.

**4. Turkish Hub:** $H_2$ manufacturing unit will generate 750 tons $H_2$/year with a minimal capacity of 5MW. When scaling up to the full scale of 20 MW in the future implementation plan, the hydrogen input will be 3,000 tons $H_2$/year. 1,000 tons of captured $CO_2$/year and 3,000 tons $H_2$/year produce 1,000 tons of e-methanol/year replacing traditional methanol. Traditional methanol production causes 3 kg $CO_2$/kg methanol. Therefore, total savings thanks to e-methanol production are 4,000 tons/year (1,000 tons $CO_2$ reduction thanks to carbon capture and 3,000 tons $CO_2$ reduction thanks to replacement of traditional methanol).

Around 0.2 kg captured $CO_2$ is used in production of 1 kg of NIPU and 10 kg NIPU is required for production of one fridge, leading to 2 kg $CO_2$ captured by one fridge. 1,000 tons of captured $CO_2$/year allows production of 5,000 tons of NIPU resulting in production of 500,000 fridges. 1kg of NIPU replaces 1kg of traditional polyurethane. 1 kg polyurethane production causes 6.3 kg $CO_2$ emissions. Accordingly, the total savings thanks to NIPU production are: 32,500 tons/year (1,000 tons $CO_2$ reduction thanks to carbon capture and 5,000*6.3=31,500 tons thanks to polyurethane replacement). Summing up first and second scenarios: 35,500 tons of $CO_2$ reduction, 1,000 tons of methanol reduction, and 5,000 tons of polyurethane reduction are achieved. Manisa factory of ARCELIK produces 500,000 fridges/year consuming 5,000 ton polyurethane which means that 100% resource saving is achieved. Production of 1kg traditional methanol and 1 kg traditional polyurethane consume an average of 3 kg and 8 kg natural gas, respectively. Accordingly, 9,000 tons natural gas from methanol and 40,000 tons of natural gas from

polyurethane production are saved, with a total of 49,000 tons of natural gas savings. To accurately estimate the potential environmental impacts of **H4C**, UT team undertook a preliminary evaluation analysis. The methodologies of Greenhouse Gases Equivalencies Calculator were used and relevant configuration data was retrieved from industrial partners.

For the sake of simplicity, only the approximate end results are mentioned above. The detailed calculations are placed in

**GitHub repository**[12] **as reference materials.**

The scale up computations are provided in the following link: https://github.com/Yifei-Yu-UTwente/IS2H4C

**Validation:** Deliverables and tasks related to hub management plan (WP4), deployment of technologies in demo hubs (WP6), and measurement of impacts via the integrated LCA module (WP7).

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**Objective 3: Facilitate the market penetration of H4C through novel financial schemes and social innovation approaches unlocking public and private investment**

**IS2H4C** includes the two key enablers for the success of the EU Green Deal to unlock non-technical aspects of **H4C**: **social innovation (objective 3a)** and **financial innovation (objective 3b)**. **IS2H4C** will implement a social innovation action, giving the attention to the local community actors and involving them from the beginning at local and EU level. This will safeguard the definition of stakeholder needs, knowledge exchange, and definition of strategies to enhance the integrity of **H4C**. **IS2H4C** will provide each hub with a customized business case comprised of innovative business and financial models for sustainability and circularity via distributing investment costs and risks among the involved actors and will ensure the replication of these models in diverse geographical contexts. **H4C**'s implementation and scalability across EU are differentiated from existing business models and capitalise on emerging trends and technologies in the circularity and ISN settings. This analysis will help to make wellinformed decisions about business models, increasing chances of success on a digital collaboration platform.

**Specific activities, KPIs and validation means**

**Specific activities for social innovation (3a)**

- Define **a living lab** of key regional stakeholders at each hub to analyse the local setting and regional actors through the implementation of at least 30 interviews (at least one per key stakeholder group per hub) to inclusively address the regulatory, political, ethical, and societal topics
- Establish **expert discussion panels:** 4 thematic panels - one per each hub involved and topic: green industry, regulatory and political, ethical, and societal
- Achieve societal engagement and education via considering market needs for businesses, increasing **user/consumer acceptance** of circular products/services, providing skills training for workers, and educating all relevant stakeholders in innovative educational/experiential environments
- Propose a **business case** that addresses the user/customer needs, explores new market opportunities, increases competitive market advantage of involved parties in a cost-effective manner, and ensures consumer acceptance of products/technologies developed within the project
- Offer skills training for workers in participating companies to accelerate the uptake of the products/services/technologies embedded, so that an upscaling can be achieved via ***EU Social Fund plus (ESF+)*** and a shift in skills development for sustainability/circularity can be achieved by ***EU labour force***

**Specific activities for financial innovation (3b)**

- Establish innovative circular business models in which environmental and social externalities are accounted
- Challenge the ***"coopetition"*** nature of the IS business models[13] where companies need **coope**ration to achieve cost savings (e.g., reduce waste discharge cost) but also have compe**tition** to pay the scant part of additional costs of running IS (so called 3T costs - waste **t**reatment, waste **t**ransportation, and **t**ransaction costs)
- Propose **de-risking strategies** for required financial investments including **risk-sharing** and **incentivisation** scenarios to reduce financial barriers for IS cases involving establishment and integration of infrastructure
- Support investment plan unlocking the financial resources aligned with ***Processes4Planet (objective#9)***
- Ensure the **scalability, commercialisation, and deployment** plans to be supported by funding instruments such as ***Invest EU, Life,*** etc., via simulating different cash-flow scenarios

**KPIs:** 30 stakeholder interviews, 4 thematic panels, 4 business models through True Value Methodology[14], 4 financial plans
(one per business model) which will include projections of revenue, expenses (OpEx and CapEx), cash flow, and profitability, 4 investment scenario (one per hub) aligned with 4 expansion and scale-up strategy reports and 4 societal acceptance events to foster society-government-industry collaboration

**Validation:** Investment & financial risk management (WP5) and regulatory, and ethical management and societal engagement in surrounding ecosystems (WP3). Both WPs are in tune and solid considering the leadership of KPMG as financial expert in the financial sustainability projects as one of the main

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| consultancies at global level, support of TUDO with their expertise in social innovation and regulatory frameworks & UT's vast expertise in circular business models and institutional/human learning. |

[12] https://github.com/Yifei-Yu-UTwente/IS2H4C.git

[13] **Yazan, D**. M., et al., 2020. Learning strategic cooperative behavior in industrial symbiosis: A game-theoretic approach integrated. *Bus. Strat. Env.*, 29(5), 2078–2091. [14] KPMG. (n.d). Valuing your impacts on society. https://assets-kpmg-com/content/dam/kpmg/ke/pdf/sustainability/KPMG-True-Value-brochure-electronicversion-pdf

## Objective 4: Develop a digital collaboration platform for information-sharing and smart decision-support

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The project will implement a user-friendly, scalable, and adaptable (to the needs of different industries and regions) digital collaboration platform for **H4C** (Digital**H4C**) consisting of a modular suite of tools, features, and data analytics to facilitate the efficient and effective sharing of resources among participating organisations. Digital**H4C** will allow the participating companies to digitally manage shared infrastructure, exchange information with each other regarding resource/waste/energy flows, observe real-time operations of energy management within the hubs, matchmake for symbiotic opportunities and engage with local communities in surrounding areas thanks to the developed smart decision-support tools to be operated within the platform in a cyber-secure environment. **IS2H4C** will secure the basis for integrated management and preservation of confidentiality of sensitive data through a cybersecure-by-design approach[15]. The platform will also ensure the use of digital modelling tools and sensing systems as a basis for dynamic resource management across the implementation of an innovate methodology toolbox to evaluate the sustainability impact achieved by the **H4C** interventions through consequential LCA.

| Specific activities, KPIs and validation means |
|---|
| - Develop a digital collaboration platform where hub actors can exchange information, monitor real-time business cases, and explore IS opportunities efficiently<br>- Establish smart decision-support tools[16] aligned with user needs to be integrated to the digital collaboration platform, namely recommenders for symbiosis matchmaking[17], algorithms for supply chain management, and algorithms for centralized and decentralized energy management[18] within the hubs<br>- Implement consequential LCA in the platform to quantify environmental, economic, and social impact of the project actions<br>- Ensure that the implementation uses a common semantic for data based on international standard models so that interoperability and confidentiality of information exchanges are ensured |
| **KPIs: (a) 8 smart decision-support modules** (2 IS recommenders, 1 agent-based simulation tool, 1 consequential LCA module, 1 Digital Product Passport framework fed by data retrieved via sensing technologies, 1 supply chain management tool 2 energy market management tools) **(b) One modular, scalable, secure micro-services-based web application** that can be interfaced via desktops, tablets and smartphone devices, and which is open (Application Programming Interface) for integration with third party services **(c) System Organisations Control (SOC 2) compliance:** fulfilment of criteria for managing customer data based on **five "trust service principles":** security, availability, processing integrity, confidentiality and privacy.<br>**(d)** At least **150 users** accessing the platform at M48<br>**Validation**. All the decision-support tools are developed in WP4. Activities related to the development of Digital**H4C** are performed in WP7, D7.1, 7.2 ,7.3, 7.4 & 7.5. Several partners (IRIS, WORKDECK, UT, ICCS) are already experts on designing, building, deploying, and maintaining digital platforms and |

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| have the expertise to deliver interoperable solutions according to specific criteria and needs and supporting the most efficient decision-making needs such as the SHAREBOX platform[19]. |

## Objective 5: Achieve a standardized top-down H4C design methodology, applicable as a reference regional development model

**IS2H4C** will investigate how **H4C** can be top-down designed in the extended form of eco-industrial parks using co-location strategies in a virtual setting and utilize such **H4C** as **interoperable, expandable, and interconnected regional development models** in accordance with Sustainable Development Goals (SDGs).

| Specific activities, KPIs and validation means |
|---|
| - Use existing symbiosis experiences data from all over the world (including public symbioses databases such as MAESTRI[20]) to design **virtual H4Cs** under the concept of "**perfect symbiosis**"[21] where no traditional inputs and wastes leak out of the **H4C**s as a theoretical optimum state<br>- Approximate the four demo **H4C** locations to this theoretical optimum via using location optimisation techniques[22], social network analysis, nature-inspired solutions such as food web analysis.[23]<br>- Report the analysis on potential use of existing and to-be-implemented infrastructure (such as transportation pathways, natural gas pipelines, or $H_2$ pipelines) as expansion corridors for **H4C** and utilisation of **H4C** as an **interconnected and interregional functional unit for European sustainable development** |
| **KPIs: (a) 4 virtual H4C prototypes** will be produced with a mix of hypothetical components such as process industry companies, service companies, energy/water infrastructure, houses, and living space in urban/rural areas. **(b) 4 regional development scenarios** will be reported building upon virtual **H4C** prototypes combined with real regional settings of the demo hubs. **(c) 1 EU level H4C expansion plan**, prepared considering the surrounding ecosystems of the existing natural gas pipelines, new European $H_2$ backbone, and transportation infrastructure.<br>**Validation:** Tasks planned for virtual **H4C** development in WP4 are executed by UT, expert in the field of design of ISNs, transformation of industrial zones into eco-industrial parks, and use of ISNs as regional development models both in urban and rural settings. Each hub representative (TEC, ALMELO, PROV, and TUPRAS) will support the deliverables of regional development reports. |

[15] Johnsson, D. B., Deogun, D., & Sawano, D., 2019. Secure By Design. Manning Publications. ISBN 9781617294358

[16] **Yazan, D**. M., et al., 2022. Decision-Support Tools for Smart Transition to Circular Economy. *Smart Ind. Better Manag.* (Vol. 28), pp. 151-169.

[17] van Capelleveen, G., et al., 2021. Exploring recommendations for circular supply chain management through interactive visualisation. *Decis. Support Syst.,* 140, 113431.

[18] **Trivella, A.** & Mohseni-Taheri, 2022. Meeting Corporate Renewable Power Targets. *Manag. Sci. 69, pp. 491-512*

[19] A.Spire aisbl. (2018, December 6). SHAREBOX: An Online Platform for Enabling the Next Generation of Industrial Symbiosis [Video]. YouTube.

[20] MAESTRI. (2020). Symbiosis Space: Library of Case Studies & Exchanges Database. Retrieved April 2023

[21] **Yazan, D**. M., et al., 2017. The design of industrial symbiosis: an input–output approach. *J. Clean. Prod.*, 129, 537–547.

[22] Genc, O., et al., 2020. Circular eco-industrial park design inspired by nature: An integrated non-linear optimization, location, and food web analysis. *J. Environ. Manage.*, *270*, p.110866.

[23] Genc, O., et al., 2019. A socio-ecological approach to improve industrial zones towards eco-industrial parks. *J. Environ. Manage.*, *250*, p.109507.

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## Objective 6: Maximize wider uptake of IS2H4C's results during and after the project's execution

**IS2H4C** will adopt a multi-level outreach strategy (details in Section 1.2-2.2) that will ensure an effective exploitation and protection of research outputs in commercial (e.g., patent licensing, trade secret) and non-commercial activities (e.g., training materials, policy feedback). To boost all Dissemination, Exploitation and Communication (DEC) activities and their impact, an Open Science (OS) approach will crosscut the development, balanced with a FAIR data management strategy based on the principle "as open as possible and as closed as necessary." This will enhance the research's impact while protecting the commercial interests.

| Specific activities, KPIs and validation means |
|---|
| - Elaborate exploitation plans for the project's Key Exploitable Results (KERs), devising the upcoming development activities to continue advancing the technology and business readiness level of the different innovations and identifying the possible intellectual property (IP) protection options and exploitation routes.<br>- Expand the **H4C** to additional parties within the **IS2H4C** hubs and in new hubs across EU.<br>- Actively disseminate the project results for conveying project KERs to targeted audiences and build an active R&D community intricately linked to related initiatives. The ultimate purpose is to create interactive co-learning spaces for the R&D and business community through specific webinars, conferences, etc.<br>- ***EU Community of Practice of H4Cs*** to diffuse best practices and achieve knowledge exchange.<br>- Cooperate through technical and scientific information exchange with other ongoing and future projects working on the issue, so, synergies can be identified and boosted.<br>- Communicate with society at large key messages about the project and its objectives, the EU R&D funds, and how they contribute to a better and more sustainable EU.<br>- Assist learning process: Using online serious games of IS[11] and circular economy transition[12][13] and to-be-developed digitaltwins of four demo **H4C** of the project in ***Sustainable and Resilient Circular Economy Laboratory (SRC-Lab)***[26] to educate students and teachers and to train policymakers, company managers, and local communities to support learning process on **H4C**s in the context of ***EU social innovation.*** |
| - Produce at least **30 scientific papers** to be published in open access high-impact peer-reviewed journals;<br>- Organise at least **10 thematic workshops framed** within dissemination & capacity building activities;<br>- Produce at least **1 guideline** supporting the **H4C** European Community of Practice (ECoP) and engage at least **15 new H4C across EU**, who also should benefit and multiply the local/regional synergies in the co-implementation of the identified innovations and solutions within the next five years.<br>- Produce at least **1 policy paper** highlighting regulatory hurdles and correspondent contingency measures<br>- Develop **4 digital-twins of demo H4Cs** by an online game company and embedded in the SRC-Lab<br>- Using the digital-twins, **8 sessions of game play** for company managers and governmental bodies and **8 sessions of game play** for students in educational settings will take place in the SRC-Lab.<br>- Train at least 350 students, 175 scientific assistants in workshops and knowledge-sharing with 15 new **H4C** across EU<br>- Reach 100+ companies engaged in **IS2H4C** Hubs and 500+ companies through consortium partners, sister projects and advisory board members. |

[11] Fraccascia, L. et al. (2021) An industrial symbiosis simulation game: Evidence from the circular sustainable business development class. J Ind Ecol. 25; 1688– 1706.

[12] University of Twente (2023, April). Minor Circular Economy Transition. Retrieved April 1 2023.

[13] University of Twente. (2023). Sustainable and Resilient Circular Economy Laboratory Homepage. Retrieved April 10, 2023.

**Validation:** Dissemination, communication, and exploitation activities performed in WP8, D8.1, D8.2, D8.3, D8.4 and D8.5. The leadership of WP8 by EEIP will ensure the technical success of the dissemination, communication and exploitation of the results which will be enhance by the network of the project's partners as CHX leading the stakeholder engagement using their network and technical partners across the project that are already coordinating and participating in key projects enhancing the synergies. UT will use its vast expertise and online infrastructure at the development and execution of online games to disseminate learning outcomes.

## 1.2. METHODOLOGY & OVERALL CONCEPT #@CON-MET-CM@# #@COM-PLE-CP@#

### OVERALL CONCEPT

IS is a symbiotic mechanism through which traditionally fragmented industries collaborate based on exchange and sharing of energy, water, by-products, infrastructure, and services. The goal is to minimize environmental impact and ideally add economic and social value. IS contributes to the backbone of circular economy seeking opportunities to reduce virgin material consumption and waste generation. **IS2H4C** will apply and develop cutting-edge sustainable technologies in four demo hubs resulting in **H4C** through the utilisation of IS. This process can be elaborated based on (1) three vertical levels, and (2) two horizontal directions. **Vertical levels:** The three vertical levels are micro-level, meso-level and macro-level. First, **IS2H4C** focuses on the symbiotic interactions at the micro level and aims to understand how IS individual behaviours can be stimulated by **enhancing societal engagement**. Second, interorganisational IS networks (ISNs) emerge as a systematic performance at a meso-level based on complex individual interactions. Integrating the material and value chains among a multiplicity of previously independent actors leads to a **complex adaptive system**. It demands for strong power and careful attention to coordinate such a system from an operational perspective. Here, **IS2H4C** innovates Information Technologies (IT)-based solutions on an **integrated digital platform to manage IS networks holistically**. Third, **IS2H4C** investigates how different ISNs in the demo hubs contribute to the advancement of circularity at a macro level. This relates to the future expansion of hubs and entails knowledge generalisation for European **policymaking**. These three levels interact with and influence each other iteratively as **H4C**s develop and evolve. Strategical coordination power is implemented in a top-down manner while systematic emergence occurs bottom-up (Figure 2). **Horizontal directions:** Two fundamental factors for realizing functional IS exchanges are the ***spatial*** (geographical) proximity of industrial actors and the ***temporal*** availability of exchanged items. For example, the utilisation of industrial waste heat is most efficient in proximity due to increasing heat losses with longer transmission distances (spatial). The availability of renewable energy such as wind and solar is intermittent and needs to be utilized when it is available (temporal). Therefore, symbiotic activities will be most successful if spatial and temporal factors are carefully managed. **IS2H4C** includes all relevant IS pathways and provides a comprehensive manual to establish **H4C** including industrial and urban areas. By explicitly considering social engagement and the **humancentred** nature of **H4C**, **IS2H4C** ensures benefits for the surrounding household areas and does not stop at the fence of industrial zones. For instance, green hydrogen systems can reach far beyond the industrial area to fuel households for multiple purposes. In summary, **IS2H4C** addresses fundamental issues of IS development and engage diverse stakeholders from a broad and **multifaceted landscape** including industrial partners and households.

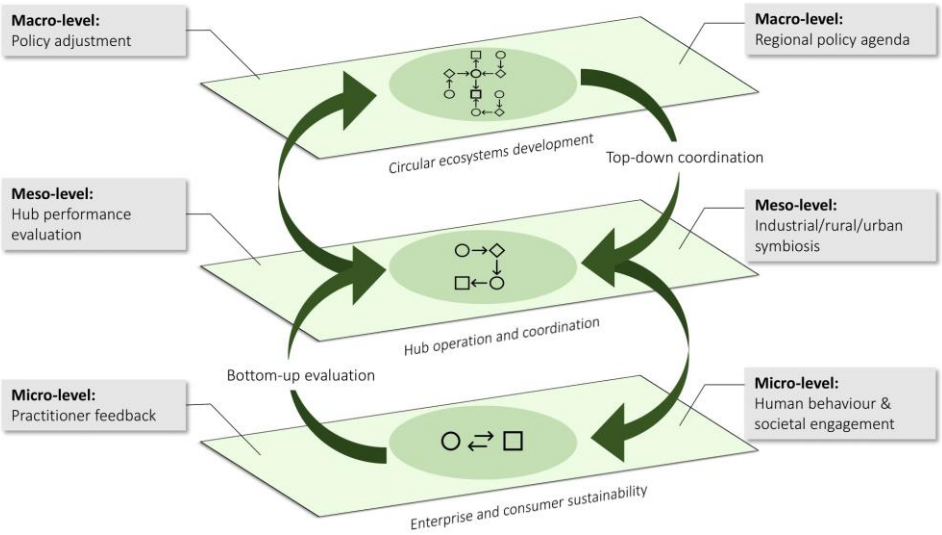


*Figure 2. **IS2H4C**'s layered hub implementation strategy*

### METHODOLOGY

To materialize the vision of a sustainable and resilient Circular Economy based on the concept above, **IS2H4C** has designed a
48-month work plan composed of 8 Work Packages (WPs), which will allow the consortium to achieve the objectives outlined in

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Section 1.1. The plan revolves around 4 highly interconnected working phases: (1) Technology development for H4C; (2) Sustainability of H4C; (3) Deployment and demonstration and (4) Outreach Strategy. These phases are described below followed by a presentation of the four demonstration hubs.

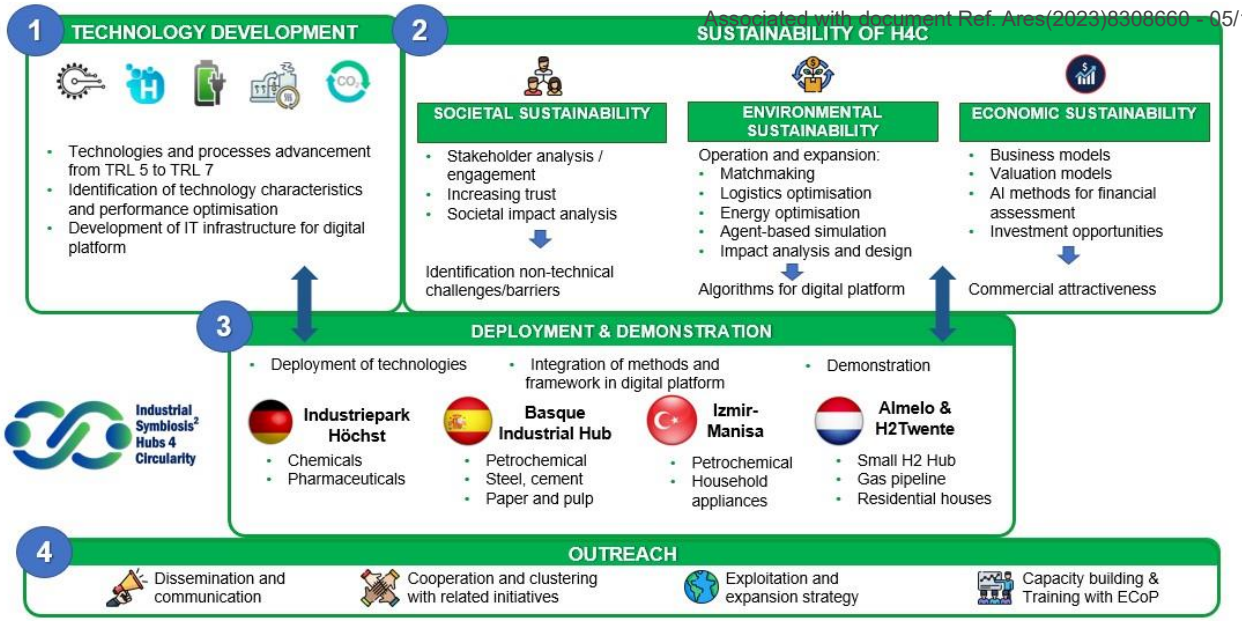


*Figure 3. **IS2H4C**'s Methodology*

## Phase 1: Technology development for H4C (WP2 & WP7)

The first phase is centred on the physical and digital technologies and their technical aspects to prepare for deployment and demonstration in Phase 3. The work is carried out mostly in WP2 and WP7. For technologies below TRL7, technology development will be carried out.

**Identification of technologies characteristics (WP2):** First, the capabilities of the physical technologies available in H4C need to be identified. IS2H4C will catalogue technologies including existing technologies at demonstration hubs and candidate technologies for deployment in Phase 3. The latter entails technologies that still need to be advanced to TRL7 during the project but also technologies that are commercially available at TRL7 and above. After the set of technologies is identified, simulation models of the technologies will be developed for validation and determination of the techno-economical and physical characteristics. These models will further act as basis for preparing surrogate models (simplified models focusing on specific characteristics) that can be used in the decision support tools developed in Phase 2.

**Technology development (WP2):** Some of the technologies needed for the demonstration at H4C are not at TRL 7 yet and as such will be scaled up to this level during the project. The TRL levels of the 11 process technologies are: one at TRL3, one at TRL4, four at TRL5, one at TRL6, two at TRL7, one at TRL8, and one at TRL9. As lower TRLs require more work before demonstration in the field, their further development is intensified in WP2.

**Adaption of processes (WP2):** IS2H4C investigates potentials for adaptation of physical processes in the H4C to increase resource utilisation and/or reduce resource consumption and waste emissions. Adaptation and redesign of processes enable the establishment of synergies at an advanced level. IS2H4C will analyse the processes at the demonstration hubs to identify opportunities enhancing the level of circularity. The partners in WP2 will propose solutions in terms of alternative process (co-) designs. Afterwards, the designs are evaluated regarding sustainability and efficiency gains, and promising solutions will be demonstrated in Phase 3, WP6.

**DigitalH4C – IT infrastructure (WP7):** Apart from the physical technologies, IS2H4C will develop IT infrastructure in the form of a digital collaboration platform to facilitate the operational management of the hub and interaction of hub participants. It will also support the environmental and economic analysis of the H4C in terms of tracking the exchange of resources as well as data analysis and visualisation tools. The goal is to develop a user-friendly, scalable, and adaptable digital platform (usable as a web application or mobile application) that can be used in different hubs with different industries and stakeholders. Input from WP3 regarding the needs of stakeholders will be considered such that citizens are engaged to the hub.

State-of-the-art software development techniques will ensure a scalable and secure platform, which includes using a microservices architecture (for scalability) and implementing security-by-design. Ultimately, the platform will provide a secure basis for hub management while ensuring data protection and compliance. **IS2H4C** will build on previous platforms implemented by the partners (as SHAREBOX, Incubis, the BiowasteHub, among others) taking them to the next level in terms of performance and functionality, not only providing a user interface and collaboration spaces for hub participants, but also including the results from Phase 1 and 2 in the backend. The actual implementation of the outputs from Phases 1 and 2 are carried out in Phase 3, while developing the general IT infrastructure is part of Phase 1.

## Phase 2: Sustainability of H4C (WP3, WP4 & WP5)

To realise the environmental and socio-economic impacts of the **IS2H4C** value chain, WPs 3,4 & 5 will ensure the sustainability of the **H4C** concept in three dimensions: societal, environmental & economic. **IS2H4C** will develop models, methods, and guidelines for development and management of **H4C**. The outcome of PhaseAssociated with document Ref. Ares(2023)8308660 - 05/12/2023 2 will support well-informed decision-making, design,
and operation of **H4C**. Establishing this strong methodological foundation is expected to enhance the probability of success beyond the project's completion, particularly for Digital**H4C** (WP7). **IS2H4C** will provide a methodological basis for extending the hubs to additional parties and guidelines for the design of new **H4C**. For this, **IS2H4C** approaches the dimensions as follows:

**Societal sustainability (WP3)**: WP3 identifies relevant stakeholders that are affected by the **H4C** and their needs, including companies, citizens living in the surrounding area of the hub, policy makers and the public. WP3 also identifies non-technical (regulatory, ethical, societal) challenges and barriers that arise in connection with the establishment of **H4C**. Relevance will be given to the local community actors through social living labs and involving them at local and EU level (desk research, interviews, discussion panels). These activities will also provide the opportunity for the definition of needs, knowledge exchange between different stakeholders as well as definition of best strategies based on the stakeholders' needs. Apart from the stakeholder analysis and engagement, WP3 establishes measures for increasing trust among stakeholders (through standardisation and policymaking) and analyses the regional/societal impact of the **H4C** (Societal Readiness Level, regional input-output models). The results will be collected in a general guideline for integrating non-technological aspects into **H4C**.

**Environmental sustainability (WP4)**: WP4 addresses environmental sustainability by establishing a digital database of resources and developing methodologies for operating and expanding **H4C**s. This includes 2 IS recommenders (matchmaking algorithms) for identifying and facilitating industrial symbiosis, logistics optimisation models, energy management algorithms including energy market interaction, digital product passport for tracing resources, and agent-based simulations for developing hubs bottom-up. The environmental impact of the concept is assessed, and expansion scenarios are analysed to provide a basis for future plans on adopting **H4C** as sustainable regional development models.

**Economic sustainability (WP5)**: WP5 develops business and valuation models to ensure commercial attractiveness of **H4C**. This includes impact valuation using KPMG's True Value methodology, AI models for financial credit risk assessments and valuation of business opportunities, and design of IS impact metrics. It proposes financial and business models focused on sustainability and societal impact, enabling simulation of scenarios with incentives. The results support a comprehensive financial plan for implementing the concept and determining its viability.

*A range of AI methods, including supervised, unsupervised, reinforcement learning, and recommender systems, will serve diverse purposes: (1) Enhancing the recommender system algorithm for IS collaboration (D4.2); (2) Developing a modeling system that characterizes resources, energy flows, and IS (D4.1); (3) Evaluating financial investments, credit risks, and the value of sustainable and circular business opportunities (D5.3). Importantly, these three groups of AI applications can utilize not only individual enterprises but also encompass process/energy technologies and infrastructure within the H4C data. Moreover, AI will be used in the DigitalH4C platform to streamline IS identification, coordination, and*

*evaluation (WP7). Throughout all AI-based methodologies, IS2H4C will align with EU norms concerning AI utilization, data protection, and GDPR regulations.*[14][15][29]

In *WP4 Models for H4C development* the project will implement techniques from Artificial Intelligence (AI), Operations Research, and Agent-based Simulation for the development of an agent-based simulation tool that enables the analysis of future scenarios of hub evolution; the AI will simulate different scenarios and their potential outcomes, based on different inputs and assumptions in order to help inform decision making. In addition, in *WP5 Business and financial models*, AI models will be developed capable of assessing financial investment, credit risk evaluation, and valuation of sustainable and circular business opportunities, whereby fair and accurate AI frameworks will be used that are capable of assessing the risk level of retail and commercial customers in H4C settings and accurately predicting the valuation of H4C's assets. These intelligent components will be deployed in the Digital Collaboration Platform in *WP7 Digital Collaboration Platform for H4C.*

It will ensure that these developments adhere to legislative requirements of the EC's Artificial Intelligence Act (published on 21 April 2021), which governs the development and use of AI within the EU and outlining rules & actions based on the principles of transparency, accountability, and human oversight. We will ensure that our AI developments and associated tools are:

(1) lawful - respecting all applicable laws and regulations

(2) ethical - respecting ethical principles and values; we will follow the Ethics Guidelines for Trustworthy Artificial Intelligence to ensure we deliver trustworthy AI (and covering such issues as data privacy, transparency, accountability, and bias)

(3) robust - both from a technical perspective, while taking into account its social environment and context, as well as reliable and explainable. To ensure the reliability, it is important to ensure its accuracy and reproducibility, as well as its ability to handle and inform about potential failures, inaccuracies, and errors. This can be achieved through testing the AI system with various scenarios to identify and address any weaknesses or vulnerabilities. Additionally,

[14] https://commission.europa.eu/law/law-topic/data-protection_en

[15] https://digital-strategy.ec.europa.eu/en/policies/european-approach-artificial-intelligence

29 https://gdpr-info.eu/

monitoring the AI system for unexpected behaviour or changes in the problem domain can help to ensure its ongoing reliability. In some cases, a hybrid approach that combines the strengths of AI with human oversight may be necessary to address complex or ambiguous situations. In relation to explainable AI, we will explore such strategies as: decision trees or linear regression models; breaking down the model's decision-making process; explaining the models limitations; providing context; providing access to the training data, among others.

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## Phase 3: Deployment and demonstration (WP6 & WP7)

After deployment of the technologies and preparation of the digital collaboration platform, Phase 3 includes the actual demonstration in the four demo **H4C** (WP6) and implementation of the Digital**H4C** (WP7). The data from the demonstration will be used for analysis and dissemination (WP8) as well as feedback into WP2,3,4&5 developed in Phase 1 & 2. The demonstration will support the validation of the **IS2H4C**'s concept including technologies and a suite of tools in the digital collaboration platform. The four **H4C** are in different regions of EU, have different maturity levels of **H4C** (regarding degree of collaboration, industrial sectors, existing infrastructure), and have geographic diversity (in terms of combination of rural, urban, and industrial areas). As such, these hubs present unique challenges that can help to aggregate the findings to extend the reach of the hub and further enhance local/regional synergies. These insights will be used in Phase 4 for expansion planning and exploitation activities.

## Phase 4: Outreach (WP 8)

The aim of this last phase is to maximise the impact of the **IS2H4C** results and ensures the achievement of the expected outcomes. Methodologies developed in Phase 2 and the data obtained from the demonstrations in Phase 3 support exploitation plans in Phase 4. The associated strategy will consist of (further elaborated on Section 2.2):

**Dissemination and communication:** A detailed dissemination and communication strategy will be developed targeting specific audiences, including industry stakeholders (Section 2.1). This fosters a collaborative innovation ecosystem around the project, enhanced by clustering and cooperation with related projects and sister initiatives. The connection with them will be a key pillar considering the participation of key partners in the consortium, such as CIRCE who are coordinators of projects like REDOL.

**Expansion:** of the **H4C** to additional parties within the **IS2H4C**' hubs as well as in new hubs across the EU. The guideline supporting the **H4C** EU Community of Practice (ECoP) will focus on the engagement of at least 15 new **H4C** across the EU, who also should benefit and multiply the local synergies in the co-implementation of the identified innovations and solutions within the next five years. The participation of Crowdhelix will significantly enhance this specific objective supported by their network across the EU focused on Impact in Industrial Circularity.

**Exploitation:** To ensure the post-project final TRL step towards market deployment, commercialisation and industrial uptake of the developed solutions, **IS2H4C** will define attractive business models and feasibility studies for the adoption of **IS2H4C**'s innovations. The necessary IP provisions will also be defined, as well as the post-project exploitation strategy to be followed in terms of technical, financial, and legal preparedness.

**Capacity Building:** Tailored skills development activities will be organised to foster knowledge transfer, including the development of e-learning resources and thematic webinars/workshops. These activities will be targeted both at students and working professionals to contribute to improve the attractiveness and awareness of career opportunities in **H4C**s.

## Description of the IS2H4C demonstration hubs

The four demo **H4C** including their facilities, partners, focus industries and TRLs, and surrounding ecosystems are presented below. A flow diagram for each hub is presented to display what processes already exist (**BLUE COLOR**), which synergies are demonstrated within the project (**GREEN COLOR**), and which processes/synergies will be studied as hub expansion plans (**ORANGE COLOR**). The flow diagrams visualise the demonstrated synergies mentioned in Objective 1 and studied expansion

opportunities. The reader is referred to the workplan, specifically WP2 and WP6, for more details of the demonstrated synergies and hub expansion studies.

**German hub: Industriepark Höchst (Process4Sustainability), Germany** Associated with document Ref. Ares(2023)8308660 - 05/12/2023

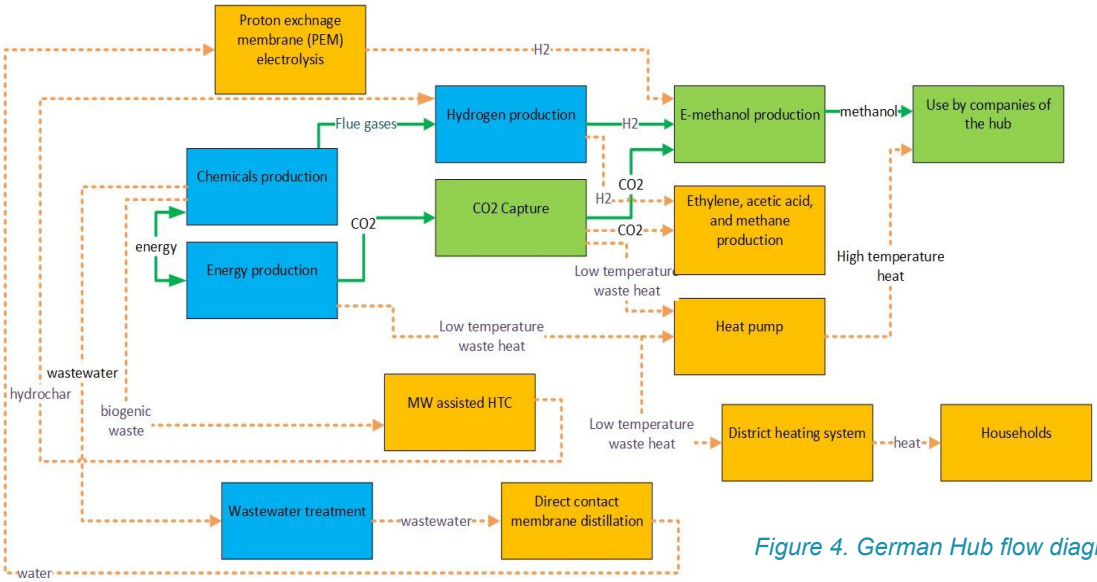


*Figure 4. German Hub flow diagram*

***Description***:Industriepark Höchst has a long tradition as one of the largest, well-established sites for the chemical and pharmaceutical industry in Europe. In total, 90 companies from the pharmaceutical, chemical, biotechnology and services sector are located on its site close to the city of Frankfurt. At Industriepark Höchst, the existing Process4Sustainability cluster aims to reach $CO_2$ neutrality for individual companies located in Industriepark Höchst by further developing and demonstrating CCS, CCU, $H_2$, bio-based resources and heat pump technologies to establish PtL (power to liquid), PtG (power to gas), PtX, electrolysis of water, $H_2$ for FC-trains, and $CO_2$ as raw material for chemicals, amongst other innovation ideas. On its way towards climate neutrality and new business models, the hub nowadays integrates innovative start-up companies, pilot and demonstration plants for PtG, PtL and power to chemicals (PtC), and first-of-its-kind plants for sustainable technologies.

**IS2H4C** will use the existing infrastructure (see Fig. 4) chemical production, energy production, wastewater treatment and hydrogen production. The flow diagram shows the synergies to be demonstrated stated in Objective 1, namely the use of captured $CO_2$ from energy production and $H_2$ from electrolysis for e-methanol production to replace traditional methanol used by the industry partners in the hub. **IS2H4C** will also study the following possibilities for hub expansion: 1) Use of captured $CO_2$ and $H_2$ for ethylene, acetic acid, and methane production. 2) Use of waste heat in district heating systems for urban symbiosis and boosting waste heat by heat pumps for high-temperature use in the industry, both replacing the need of fossil fuels for heat production. 3) Wastewater utilisation for hydrogen production using membrane distillation and electrolysis (11). 4) Microwave-assisted Hydrothermal Carbonisation based on biogenic waste from the chemical industry as input to hydrogen production. Due to the history of the industrial site, many established processes already offer a high TRL, such as waste-water management, energy efficiencies and shared operations from different companies at the site. $H_2$ is an intermediate technology with TRL6 as of today. New approaches, such as carbon capture not being present at this stage. Overall, with this deployment TRL level of the site is expected to go from 5 to 7, by the end of the project.

***IS2H4C partners***: From the Process4Sustainability cluster, the following partners will join the **IS2H4C** directly Infraserv Höchst (Service operator of industrial site), CIRCE, AEE and INEGI. The other partners BAYER, Celanese, Kuraray, Sanofi support the proposal with expansion strategies.

**Basque hub: Basque Industrial Hub for Circularity, Spain** Associated with document Ref. Ares(2023)8308660 - 05/12/2023

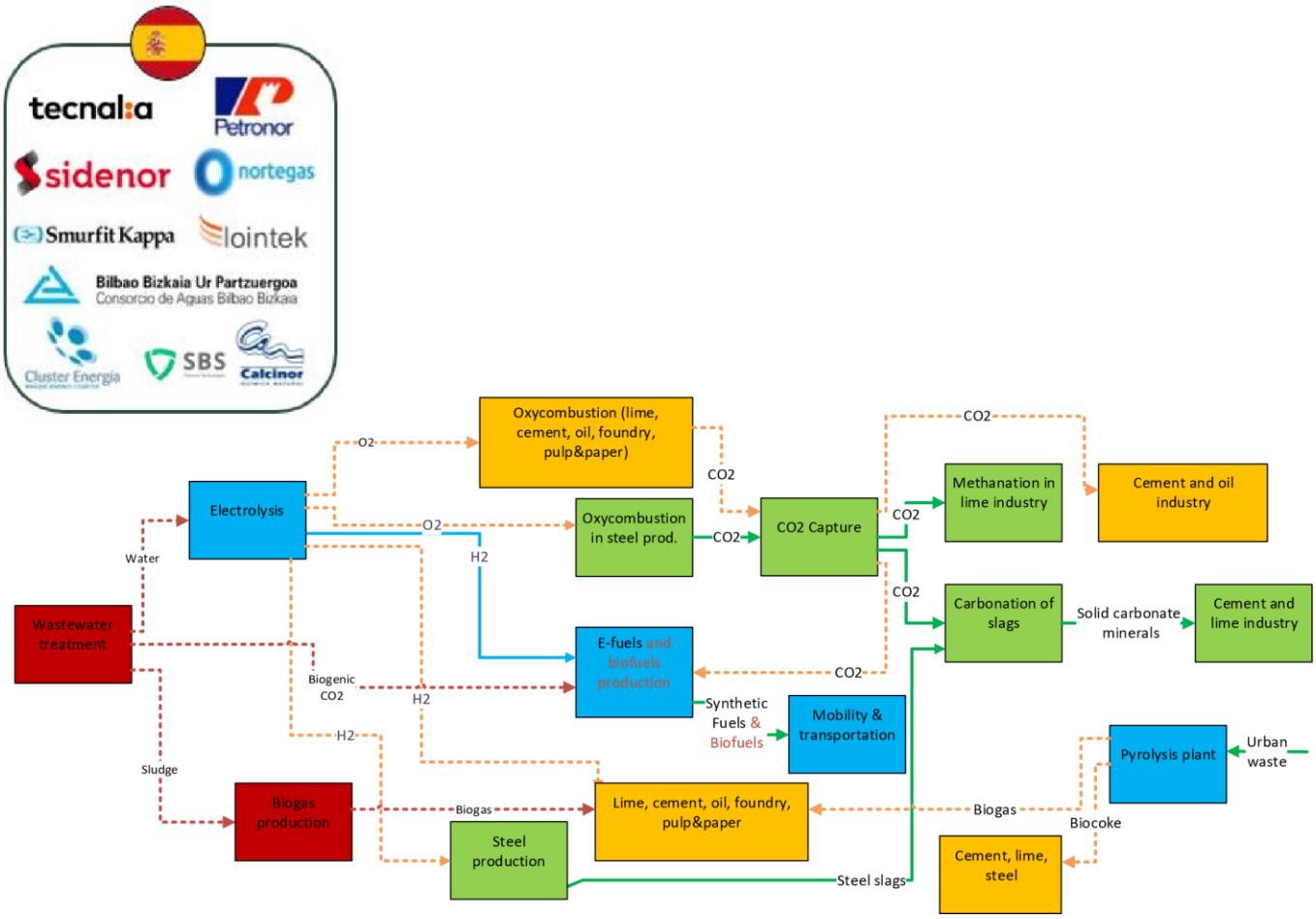


*Figure 5. Basque hub flow diagram*

In the Basque hub, three extra expansion synergies will be studied in order to complete the hub with (waste)water use and bioenergy production scenarios (these synergies are shown in red color). These technologies will not be deployed but only studied as potential hub expansion scenarios.

***Description***: The Basque Industrial Hub for circularity (BI**H4C**) constructs upon the foundations settled to speed up the decarbonisation of the industry by 2050 (Basque Government's NZBIS initiative[16] and Basque $H_2$ corridor[31]). The Basque Country, a strong innovator region with a high industrial concentration, has already identified a series of symbiosis (image below) between different sectors, that will present a relevant consumption, materials and emissions reduction.

The hub includes steel, cement, lime, oil refining and pulp & paper industries (see Fig. 5) as well as already existing technologies for wastewater treatment, waste pyrolysis, electrolysis, e-fuel production in transportation. **IS2H4C** will demonstrate (see Objective 1) using either $O_2$ resulting from electrolysis for oxycombustion or hydrogen in steel production (and evaluate oxycombustion in other industries as expansion plan) to reduce fossil fuel consumption. Captured $CO_2$ will be utilized in lime production (demonstrated) as well as cement and oil industry (expansion study). Furthermore, steel slag and $CO_2$ will be used in carbonation to produce construction products for the cement industry (demonstrated). In an expansion study, **IS2H4C** will investigate how biocoke and biogas, produced from urban waste in the pyrolysis plant, can replace coal-based coke in the cement, lime and stell industry and natural gas in the process industries (lime, cement, oil, pulp and paper, foundry and steel), respectively. Finally, the use of $H_2$ in these industries is also considered additional hub expansion synergy. The BI**H4C** starts with TRL5 (existing $H_2$ pipeline and companies with electrolysers) and will account for a systemic TRL 7 by the end of the project. The technologies (CCU, oxycombustion and $H_2$) and suitability of the use and production of

[16] Grupo SPRI. (2023, February 7). Net-Zero Basque Industrial Super Cluster - SPRI Group. SPRI. https://www.spri.eus/en/support/net-zero-basque-industrial-super-cluster/ [31] Basque Hydrogen Corridor BH2C (n.d.) About us. Retrieved February 22, 2023, from from http://www.bh2c.org/en/about-us

these circular subproducts and gases, will be validated in real industrial environments (TRL7), and accompanied with the potential collaboration and business schemes between these industries, the Hub will be able to achieve a systemic higher TRL (TRL7).

***IS2H4C partners:*** TECNALIA, PETRONOR, NORTEGAS, Smurfit Kappa Nervion, Basque Energy Cluster, SIDENOR, SIDENOR I+D SA, LOINTEK, CALCINOR, Consorcio aguas Bilbao, SBS

## Turkish Hub: Iz-Manisa area, Turkey

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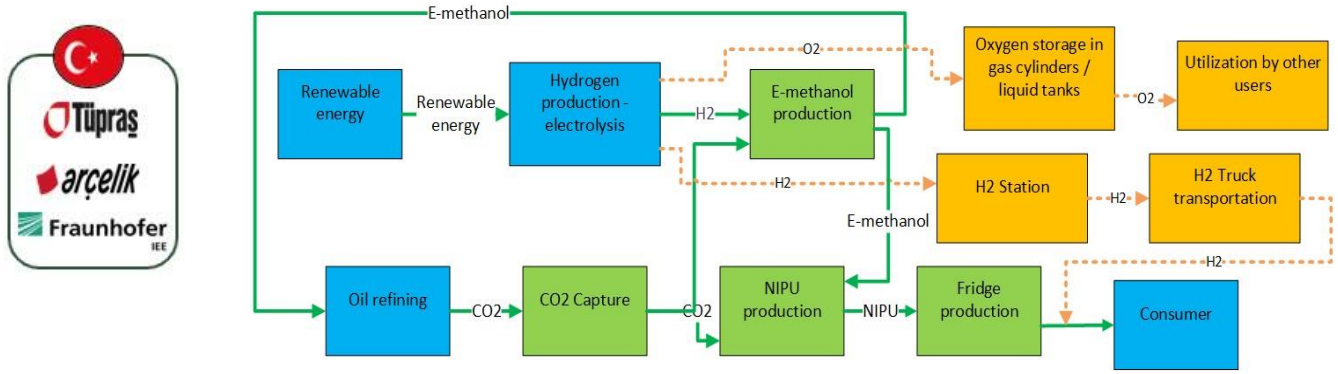


*Figure 6. Turkish hub flow diagram*

***Description:*** The Turkish hub is an industrialized port area in the Izmir-Manisa area at the Aegean coast containing oil and gas (Tüpras) and household appliances companies (Arcelik). **IS2H4C** will consider the existing oil refineries, fridge production facilities and $H_2$ production from renewable energy (see Fig. 6). The demonstration in the project (see Objective 1) includes the utilisation of the produced $H_2$ for e-methanol production replacing traditional methanol as input to the oil refineries. Additionally, captured $CO_2$ will be used as input to non-isocyanate polyurethane production which again is used in fridge production instead of traditional polyurethane. Possibilities for expansion strategies studied in the Turkish hub include storage for $O_2$ from electrolysis offered to other users and provision of $H_2$ via a fuel station and $H_2$ trucks to consumers reducing the need for non-renewable resources. TUPRAS, the biggest industrial enterprise in Türkiye, aims to install and operate 120 MW electrolyser capacity until 2040 and 1.5 giga-watt (GW) electrolyser capacity until 2050 (producing around 135,000 million ton (Mt) green $H_2$ per year). As outlined in the recent "Strategic Transformation Plan", the company is committing to reduce its carbon emissions (6 million ton $CO_2$/ year) gradually, and to becoming carbon neutral by 2050, which requires focusing on energy efficiency and expanding into new areas. Moreover, TUPRAS plans new renewable energy investments (solar power plant) to meet the need for clean energy sources for green hydrogen production up to 2050's targets. TUPRAS has four different refineries which are in İzmir, İzmit, Batman and Kırıkkale. Refineries have, high purity and good quality hydrogen producer units, consumer units and purge systems. Production and consumption units of pure hydrogen through steam methane reforming (99.99 vol.%) and hydrogen rich gas (~91 vol.%) exists in three refineries (Kırıkkale, İzmir, İzmit) of Tüpraş which are all located in Turkey. TUPRAS has a demo scale (TRL6) CC unit in Izmit refinery which has been developed and installed in the EU project MOF4AIR. In **IS2H4C**, the CC unit is moved to Izmir Refinery to capture $CO_2$ on site. It has capture capacity of 10-15 kg $CO_2$/h. Water electrolysis at TRL9 will be implemented to produce green $H_2$, which is further used for e-methanol. E-methanol is also an intermediate product for NIPU production. Fraunhofer has a TRL8 technology for NIPU production, which will be demonstrated in refinery conditions. In total, the NIPU process will be developed from TRL3 to TRL7.

***IS2H4C partners****:* Tüpras, Arçelik, Fraunhofer

## Dutch Hub: Almelo and H2Hub Twente

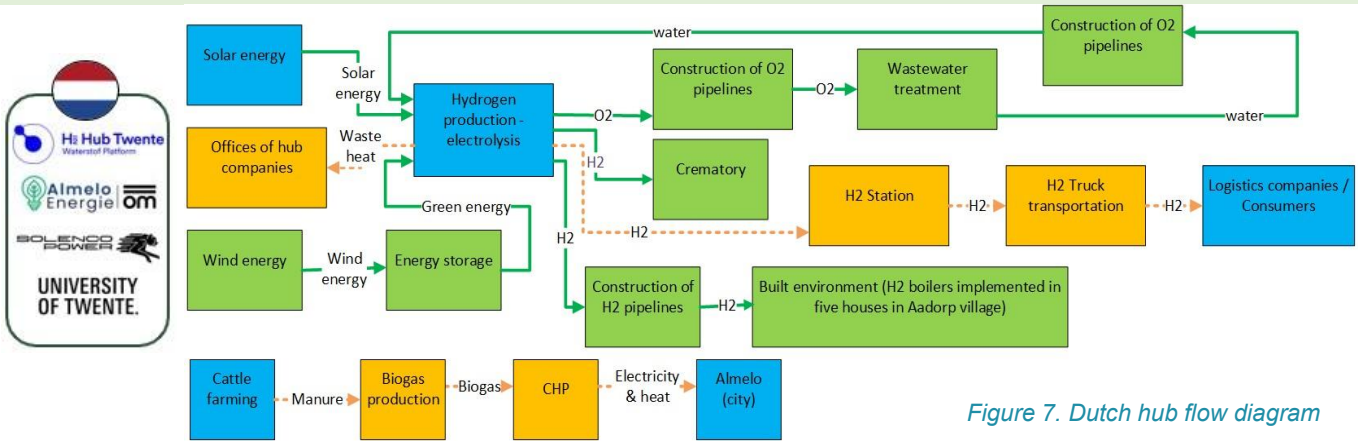


*Figure 7. Dutch hub flow diagram*

***Description:*** H2Hub Twente is a physical place where entrepreneurs, governmental organisations, researchers, and knowledge institutions collaborate to develop applications of hydrogen technology. Knowledge and experience are gained through research and infrastructure test projects. Applications include setting up (pilot) projects in the field of generation and storage, mobility, hightemperature combustion, industrial processes, and applications in the built environment. Almelo Energy is the first of a kind Positive Energy District in Twente, the Netherlands, to lead the energy transitions in the neighbourhoods they serve. Positive Energy Districts include the introduction of natural gas free neighbourhoods, will allow for local balancing reducing energy distribution over longer distances, facilitate peak shaving and load shifting using concepts of demand response and reduced

curtailment of RES, and district-level self-consumption. The Dutch Hub has existing infrastructure for $H_2$ production from solar energy, wastewater treatment, office buildings and a crematory. **IS2H4C** will demonstrate the following synergies (as described in Objective 1): using $O_2$ from $H_2$ production in wastewater treatment and treated water again as input to the $H_2$ production closing the loop and avoiding the use of other water resources (in a region prone to water scarcity), utilizing $H_2$ instead of natural gas in the crematory, establishing energy storage for wind energy to be used in green $H_2$ production, and distributing $H_2$ via pipelines to five residential houses for $H_2$ boilers in the village Aadorp substituting natural gas for heating purposes. **IS2H4C** will also explore opportunities for the Dutch hub: 1) Using manure from existing cattle farming for biogas production which can replace natural gas in a combined heat and power plant (CHP) for electricity and heat distributed in Almelo. 2) Recovery of waste heat from electrolysis for space heating in office buildings removing the need for natural gas boilers. 3) Distribution of $H_2$ via fuel stations and truck transportation to consumers and logistics companies replacing fossil fuels. Waste heat recovery (from TRL5 to TRL7), oxygen use in wastewater treatment (from TRL8 to TRL9), use of water from wastewater treatment in electrolysis (from TRL6 to TRL8).

***IS2H4C partners:*** In the consortium: UT, H2Hub Twente, Solenco Power, Almelo Energie. In the H2Hub: Schroder Vastgoed, Powerspex, Cogas, VDL, Gas Unie, Siemens energy, Gemeente Almelo, Coteq, Alliander, Sensata Technologies, Ondernemers Bedrijvenpark, Dorpscooperatie Aardorp, DEMCON, Remeha, and Province Overijssel.

## 1.3. RELATED RESEARCH AND INNOVATION ACTIVITIES

A summary of the key projects in which the partners have participated, and the relevant outputs that will be used in the **IS2H4C** are listed in Table 1. Additionally, **IS2H4C** will prepare a detailed and well provisioned clustering strategy so the research plan is executed in coordination with other projects (those funded on the same topic, the ongoing ones, or in complementary topics). Moreover, it will exploit all the relevant links of the partners to maximise the outreach to the industry beyond the involved companies, thanks to partners participating in similar projects.

*Table 1. Related R&I projects and initiatives*

| Project (partners) | Link to the project proposal |
|---|---|
| REDOL (2022-2026); (CIRCE, EEIP, ICCS) | REDOL redesigned 5 value chains (packaging, plastics, textiles, electronic waste, construction) resulting in 12 circular products. The processing routes method utilised in REDOL will enhance **IS2H4C**'s logistics. |
| SHAREBOX (20152019); H2020 SPIRE (IRIS, UT) | SHAREBOX developed a secure ICT platform for the flexible management of shared processes resources that provided plant operations and production managers with the robust and reliable information needed in real-time.<br>The experience gathered will enhance the development of the digital collaboration platform in **IS2H4C**. |
| CORALIS (2020-2024); (CIRC, ICCS, EEIP) | Execution of real IS initiatives and the prevention of obstacles faced by these initiatives, addressing technical, managerial and economic factors that set the basis for definition of the IS readiness level. |
| DIGINTRACE (20232026); HE (ICCS) | The traceability methodologies & data gathering towards sustainability and circularity gathered will be provided to **IS2H4C** for the development of the digital collaboration platform & the traceability of value chains. |
| COREALIS (ICCS) | COREALIS introduces circular economy models into the strategy and operations of ports to create efficient connections with the hinterland transport network and enhance decision making. The decision-making knowledge gathered will be provided to **IS2H4C** for the development of the digital collaboration platform. |
| CIRC4LIFE (ICCS) | CIRC4LIFE demonstrates an eco-innovative circular economy approach for sustainable products and services through their lifecycles. The ICT platform (developed by ICCS) facilitates data exchange between stakeholders along the value chain and could be a reference for some of the functionalities of the the **IS2H4C** Digital Platform. |
| **H4C** EUROPE (2022-2026); HE (CIRCE) | Provides a community, knowledge platform and exchange structures that will help existing and future hubs (such as **IS2H4C** Hubs) in creation, management, and growth |

| | |
|---|---|
| FLEXINDUSTRIES (2022-2026); HE (CIRCE) | Builds a holistic multi-disciplinary and multi-scale approach fostering its 7 multi-sector energy intensive industries. The methodology utilised to select the most suitable Energy Efficiency Measures could be useful for **IS2H4C**'s resource and waste management. |
| RE4Industry (20202023); H2020 (CIRCE, EEIP) | Supported energy intensive industries in the identification and integration of renewable energy solutions through a network of stakeholders and market actors. The logistics behind RE4Industry and the methodology to identify solutions could enhance **IS2H4C** results. |
| NEFI (2017-2026) (MUL) | The project's cluster demonstrates that energy technologies developed in Austria can enable 100% renewable energy supply for selected industrial sites and that energy transition towards a low-carbon energy system can be reached. MUL can provide useful insight to optimise green energy technologies and accelerate the transition in the hubs. |
| AI-CUBE (2020-2023); H2020 (IRIS, ZLC) | Defined a roadmap in AI and use of big data for the process industries, which could be useful to optimise data exchange between industries in the **IS2H4C** hubs. |
| FuelSOME (2022-2026) (AEE) | Conceptualisation of new multi-fuel supply chain pathways and development of a software-based modelling and KPI based optimisation of a multi-fuel supply backbone concept to ensure sustainable supply pathways for the FuelSOME concept. |

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The core concept is to design, synthesise and test the MOF-based materials against flue gas compositions of MOF4AIR (2019-2023); refinery and waste-to-energy industries for $CO_2$ capture using adsorption technology on real industrial H2020 (TUP) environment (TRL6). Tüpraş will be hosting one of the demonstration units in its premises after carrying out necessary infrastructural studies and conduct elaborate performance test studies in 2023.

## 1.4. INTER-DISCIPLINARY APPROACH, SOCIAL SCIENCE AND HUMANITIES

Addressing the above challenges requires a holistic and interdisciplinary approach. **IS2H4C** is aware of this complex socioecological scenario where these value chains operate. Therefore, **IS2H4C** engages multidisciplinary academic partners, producer and processor companies, as well as experts in technology and innovation trends and cross-regional exploitation. The wide variety of profiles involved in the consortium and their expertise is framed in three main areas. **(1) Scientific and technical disciplines:** mathematicians, industrial processes experts, industrial parks managers, technical and life sciences experts, environmental scientists, and ICT engineers, brought mainly by the academic and research partners as well as industrial partners in **IS2H4C**. **(2) Social science and humanities disciplines:** sociology, social science and technology studies, economics, and policy studies (TUDO, UT, and KPMG). **(3) Business disciplines:** EEIP, KPMG, CHX and UT address economic and financial aspects of **H4C**s, such as finance and investment, human resources, skills and labour market, end-users' demands and behaviours. The **IS2H4C**'s methodology will foster cross-fertilisation among disciplines, stakeholders, and regions, enabling a close interaction between actors throughout the whole innovation process leading to multi-actor and interdisciplinary results.

## [17] .6. AN INTEGRATED APPROACH TO DATA MANAGEMENT AND OPEN SCIENCE

The **IS2H4C** consortium will systematically follow best practices for Open Science (OS) to improve the quality, efficiency, and responsiveness of research. These practices include the **FAIR Data Principles** (Findable, Accessible, Interoperable, and Reusable data) and the Responsible Research and Innovation (RRI) methodologies. **IS2H4C** will share and transfer results between relevant actors from the point of generation, facilitating the availability of research outputs, research continuation and contributing to the fulfilment of the project's objectives. UT, as Data Manager, will elaborate and distribute a comprehensive Data Management Plan (DMP) by M6, providing guidelines for data management under the premise of "protecting and ensuring that any relevant information that may compromise the added value of exploitable results and outcomes will not be disclosed." This DMP will allow the project partners to manage research outputs following the FAIR principles (Article 17 of the HE Model GA) and OS practices while ensuring the adequate exploitation of the IP generated and other justified legitimate interests. The DMP will cover the

## 1.5. GENDER DIMENSION

Aligned with the principles of the Processes4Planet, **IS2H4C** ensures gender equality and diversity by integrating sex and gender analysis into research and innovation activities. This analysis will identify potential barriers or inequalities towards a successful and inclusive outcome from the following three perspectives. (1) From a **behavioural perspective**, gender norms impact actor's behaviours in the industry, which may lead to gender-based differences in decision-making around sustainability and circularity. (2) From a **representation perspective**, a gender analysis can identify potential barriers or opportunities for promoting circularity. Understanding these issues help us to analysis the underrepresentation of women and gender-specific barriers in accessing funding or training opportunities. (3) From a **decision-making** perspective: Female decision-makers may bring different perspectives on issues related to sustainable development, social responsibility, and community engagement, it could ensure that there are opportunities for female decision-makers to participate in the governance and decision-making processes. **IS2H4C** actively promotes gender equality and diversity in the industry ensuring women and other underrepresented groups have equal opportunities to participate in and benefit from circular initiatives. In support of the European Strategy for Gender Equality 2020– 2025, **IS2H4C** will implement a **Gender Equality Plan** to promote gender equality and adapt spaces for learning to enable improvements at an organisational level. The plan will consist of four tasks: (i) Gathering and utilising data samples that are balanced in terms of gender and other inclusion considerations (age, disabilities, etc.). GDPR and consent will govern data collection and use; (ii) Tracking diversity-related issues by evaluating if the **H4C** establishment benefits all stakeholders, regardless of gender and background and by building human-centric AI frameworks that are transparent, reliable, and impartial (WP1); (iii) Offering inclusive possibilities for commercialising energy based on ethnicity and race, social class, human physical parameters, gender identity, sexual orientation, disability, and age; and (iv) Ensuring diversity and inclusiveness in the project management and within the WP teams. **IS2H4C** contains a diverse group committed to inclusive policies. The current members are individuals from different gender, nationalities, and ethnicity. Moreover, **IS2H4C** is committed to promoting diversity in hiring Ph.D. students and postdoctoral research and supporting the industrial partners of the consortia in adopting similar strategies. Therefore, **IS2H4C** considers the gender dimension in all tasks related to data collection, hubs perspectives evaluation, societal impacts, and market uptake of proposed innovations. Moreover, communication, dissemination, and exploitation activities will also be gender-sensitively addressed and mainstreamed. To increase the visibility of minorities and provide role models for future generations, efforts will be made to encourage balanced participation in every event organized. Likewise, educational materials and capacity building initiatives will contribute to reducing the gender gap in industrial sector by providing cutting-edge skills increasingly rewarded by the labour market.

complete execution and foreseen tasks of the project and provide recommendations for post-project management data strategies according to each partner's role and type of generated data. By embracing OS as a part of its research lifecycle, **IS2H4C** will improve the reproducibility and transparency of the project's outputs.

**Open Access (OA) to scientific publications** ease the access for knowledge-based industry and accelerating the innovation pace and delivery time to market. The **IS2H4C** consortium is committed to ensure full and immediate Open Access to peerreviewed scientific publications related to their research outputs, according to Article 17 of the HE Model Grant Agreement. The knowledge, outputs, tools, and instruments gathered from this deployment will be free and available OA for the public (considering IP restrictions) following the "as open as possible and as closed as necessary" principle at the latest upon and will be deposited in a trusted repository (ZENODO), under the Creative Attribution International Public License (CC BY). Likewise, public deliverables will be made available as soon as approved by the EC. Parallelly, the consortium will be encouraged to publish on the Open Research EU (ORE) platform.

**Type of data and research outputs:** **IS2H4C** will generate data and research outputs from technological, societal, and financial disciplines Technological data will be associated with the types of technologies developed/integrated, main metrics related to their technical development/integration, and their performance/operation capabilities. The societal and financial data will be collected based on the engagement and participation in the digital platform, adoption of the use of new technologies or practice of industrial symbiosis in any form with the hubs or in their surroundings, and business and financial institutions (e.g., collaboration rate to the **H4C**). A complete definition of the data sets (name, purpose,

formats, use of existing data, expected size, utility, etc.) will be provided by M6 and updated during the reporting period and whenever significant changes arise.

**Findability of data and research outputs:** Specific metadata templates will be defined to describe, discover, and trace existing data collected and/or generated by the project. Specific metadata fiches will be made publicly accessible. The metadata will be used to describe datasets, including specific keywords to allow for refined search across datasets. The bibliographic metadata will at least include the following: (i) the terms "European Union" and "Horizon Europe", (ii) the **IS2H4C**, acronym, and grant number, (iii) the publication date and length of embargo period if applicable, and (iv) the persistent identifier. The data generated during the project will be deposited in ZENODO, a cloud-based repository federated in the EOSC and comply with FAIR data principles. It facilitates finding, accessing, re-using, and interoperability of data sets. Furthermore, the repository will integrate with data from similar projects (previous and ongoing) for easier accessibility, linear, tree, and graph data structures defined by the existing ontologies and standards.

**Accessibility of data:** Data and research outputs open to the public will be fully and freely accessible, according to the FAIR principle, utilising tools such as: open access peer-reviewed journals, scientific magazines, on-line platforms, and other digital tools, printed and digital dissemination materials. If parts of datasets cannot be shared, the reasons for this will be clearly mentioned (e.g., ethical, rules of personal data, intellectual property, commercial, privacy-related, compliance with GDPR).

**Reusability of data:** To facilitate the validation of scientific publications, **IS2H4C** enables the disclosure and reusability of the data, acknowledging the convenience of enabling the review, scrutiny, and verification from external scientific actors. Creative Common Licensing will be used to protect the ownership of datasets. Despite the data openness, an embargo period might be applied to the data if used in Green OA journals. The datasets located in ZENODO will have unlimited access, although for some data restrictions to the re-use policy may apply. This can be the case for green OA embargo periods or Creative Commons licensing restrictions (attribution, prohibition of use for commercial purposes, to use the same license as the original).

**Curation and preservation:** The coordination and monitoring of the DMP will be carried out by UT as Data Manager, being the specific implementation responsibility of each WP leader. UT will deliver the DMP first version by M6 and update it during the project life. UT will produce a code of conduct, and explain it in the Kick-off meeting, so all partners are aware of the DMP and the OA practices of the project. An analysis on the security levels of the different repositories used by the project will be performed in the DMP, to ensure they meet a minimum-security level (from the common project consortium workspace, probably Microsoft OneDrive, to the public repositories chosen for making the data available, probably ZENODO).

**Regulatory and Ethical aspects:** **IS2H4C** will adhere to legislative requirements of the General Data Protection Regulation. It will also comply with the EC's Artificial Intelligence Act, which outlines new rules & actions aiming to turn EU into the global hub for trustworthy AI, whereby by setting the standards, EU can pave the way to ethical technology worldwide & remains competitive along the way. Following the Ethics Guidelines for Trustworthy AI trustworthy: i) lawful - respecting all applicable laws and regulations, ii) ethical - respecting ethical principles and values and iii) robust - from a technical perspective while considering its social environment. #@CON-MET-CM@# #@COM-PLE-CP@#

# 2. IMPACT

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## 2.1 PROJECT'S PATHWAYS TOWARDS IMPACT

### 2.1.1 STAKEHOLDER ANALYSIS

**IS2H4C** has identified stakeholder groups that will benefit from the results. Attention will be paid also to sectoral associations existing in the EU, such as covenant of companies for climate and energy and CircLean[18].

- **Industrial clusters as potential new H4C:** The outcomes, knowledge and data generated by **IS2H4C** will become a demonstrator for successful circular hubs in the future. The Digital Collaboration Platform (WP7), the methodologies for deployment of SoA technologies, the models for sustainable financial

[18] CircLean. (2022, January 20). The project - CircLean. CircLean. https://circlean-symbiosis.eu/the-project/

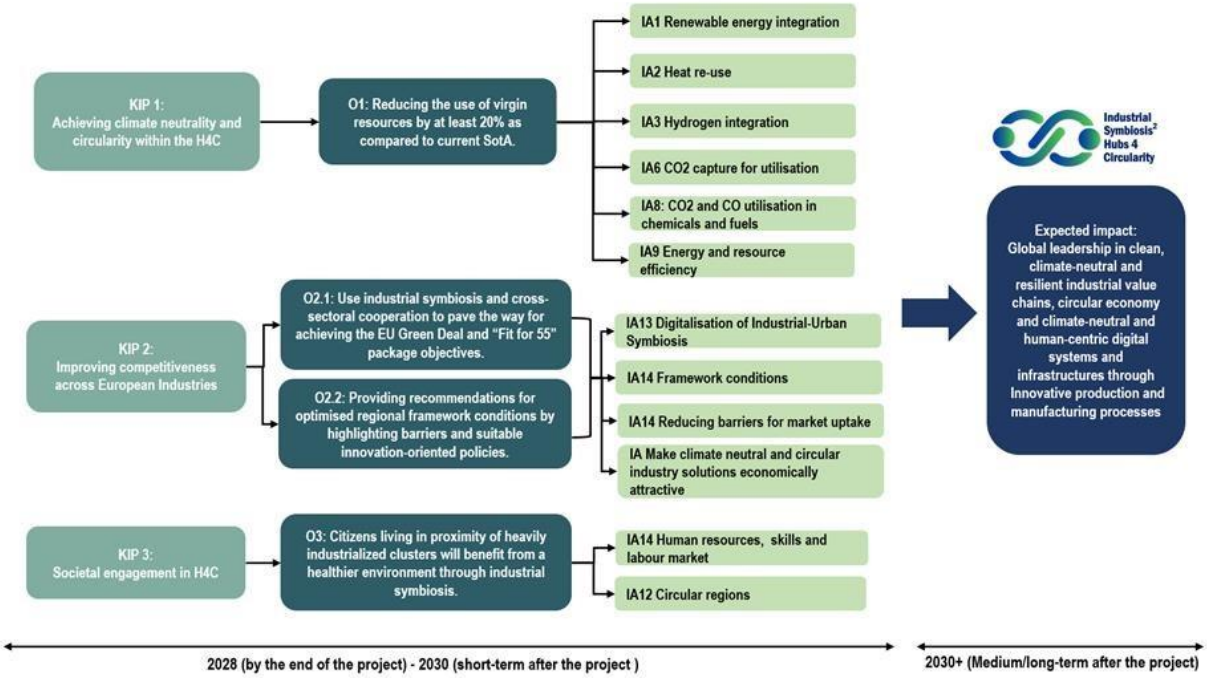
performance, and the low-waste and highly productive processes in **IS2H4C** will impact society positively by increasing competitiveness of other hubs.

- **Green Energy Companies/Green tech manufacturers:** Green tech and energy companies will be one of the main beneficiaries of the transition, since **IS2H4C**'s hubs will broaden and facilitate the use of green fuels/renewable energies in various economic and industrial sectors (chemical, refining, steel, pulp & paper, household appliances, among others). **IS2H4C** also contributes to lower the bar of regulatory acceptance by engaging in policy-making processes and subsequently reducing costs of production and distribution.
- **Financial Institutions/Investors:** **IS2H4C** represents an opportunity for future economic returns due to expected increase in overall demand and availability of sustainable value chains. Also, it will help financial institutions to better understand the current and future sustainability trends staying competitive for future common challenges
- **Research and Academia:** Academic partners outside the project will benefit from the OS approach and the specific cooperation and dissemination activities targeted (i.e., workshops, events, scientific papers). The ground-breaking work of **IS2H4C** will boost the quality of scientific production around the topic, multiplying its effects.
- **Policymakers and Standarisation Organisations:** R&D conclusions drawn from the project results will directly feed policy initiatives at EU and national level, from the perspectives of industry and sustainability policies. The proposed IS methodologies will eventually contribute to a paradigm shift towards **H4C**. In this regard, a partnership with standardisation committees is required to foster the development of **H4C** and their extensions by verifying legal compliance and the accurate/correct operation of all technologies (both individually and collectively related to the **H4C**).
- **General Public:** The project deepens the citizen's understanding in relation to CE, its benefits over current linear business models, and the positive impacts (economic, environmental and societal) of being part of an IS environment including, most importantly, how they can be involved. This is crucial for informed citizens, in view of their potential role in **H4C**, particularly if they live in zones where hubs are and/or will be implemented.
- **Sectoral associations and P4Planet partnership:** Sectoral associations are prestigious organisations that can serve as gateways to disseminate and promote the results of the project. Other type of industry associations and public-private partnerships will benefit from the project. Especially, the project will cooperate with P4P as it will become one of the first funded **H4C** under its portfolio.

### 2.1.2 EXPECTED OUTCOMES AND IMPACTS

**IS2H4C** will make a **tangible and enduring impact** in paving the way towards the development of Hubs for Circularity (**H4C**) based on the circularity and resilience requirements of existing and future industrial zones and their surrounding ecosystems. This impact will be secured through 3 realistic and credible **Key Impact Pathways** (KIPs) to achieve the expected outcomes of the HORIZON-CL4-2023-TWIN-TRANSITION-01-37 call along with the expected long-term impacts of "Climate neutral, circular and digitised production (2023/24)" Destination of the Cluster 4 Work Programme. Project's pathways towards impacts will be directly coupled with appropriate and quality DEC measures to maximise the expected outcomes and impacts. A description of each KIP that will be implemented within **IS2H4C** can be found below, detailing the expected status of the innovative solutions at the end of the project (Project results), the short-term effect of the uptake, dissemination, and deployment of these results by direct target groups (outcomes), and finally the wider effects in the long term (impacts). In addition, the inter-relation between the **IS2H4C** KIPs and the Innovation Areas (IA) established by Process4Planet are illustrated (Figure 8).

*Figure 8 Relation between KIPs, outcomes, IAs and expected impacts.*




## Key Impact Pathway 1. Achieving climate neutrality and circularity within the Hubs

**RESULTS—By the end of the project (2028):** **IS2H4C** expects to comply with outcome 1 "***O1. Reducing the use of virgin resources by at least 20% compared to current SoA"*** and parallel with the Climate Neutrality Pathways of the Strategic Research and Innovation Agenda (SRIA), which are: **(i) to integrate renewable energy/feedstock**; **(ii) to reduce emissions through energy efficiency**; and **(iii) to reduce emissions through CO/$CO_2$ capture and use**. These actions are implemented in the **H4C** that are active in many industrial sectors (steel, pulp & paper, refining, pharmaceuticals, chemicals, lime, bio-based energy, water, & electrodomestics), & each **IS2H4C** hub will tackle the IAs using different strategies to achieve the project's goals.

**IA1:"Renewable energy integration"** - Waste management services from the **German Hub** will help local chemical and pharmaceutical firms achieve carbon neutrality. The Industriepark Höchst biogas plant can produce 30,000 $m^3$/d of biogas from over 100 types of waste (food waste, pharmaceutical industry waste, oil, grease trap waste, biogenic refuse from biochemical processes, solvent mixtures, sewage sludge, alcohols, and others) and generate electrical (5.1 MW), thermal (3.9 MW), and biomethane (10 MW) power. Utilising $H_2$, biogas, green electricity, process electrification, and methane synthesis will reduce energy demand by 139.45 GWh/year at the hub. **Petronor (Basque Hub)** expects to decarbonize their refinery through pyrolyzing urban garbage to produce biogas. The plant will produce 4GWh of biogas from 10 KT of urban garbage, replacing 2.9 million m3 of natural gas and eliminating 5.5 million kg of $CO_2$. **Tüpras (Turkish Hub)** applies Hydrogen manufacturing unit will generate 3,000 tons H2/year with a future capacity of 20MW. Carbon capture will be used to produce e-methanol replacing traditional ones saving 4000 tons carbon emission.

**IA3: "$H_2$ integration"** - By 2027, **Industriepark Höchst** expects to complete the 1 MW Proton Electrolyte Membrane (PEM) electrolysis plant to refill cars, trucks, buses, and rail vehicles and stabilize the local electrical system. The Basque Hub will install 2.5 MW, 10 MW, and 100 MW electrolysers by 2027. The new infrastructure will provide H2 to the lime sector (CALCINOR) and steel ladle preheating units (SIDENOR), reducing emissions of the Basque Hub by 47% (considering that all the NG, the gas from the refinery and the coke used nowadays is replaced by H2). The Basque Hub also plans to finish the 2.8 M L/year transport efuels ($H_2$ and $CO_2$) pilot plant. They plan to create 8.4 million kg of $H_2$/year to save 4.375 million kg of $CO_2$/year. **H2Hub Twente's** green $H_2$ will heat Almelo's residential homes (130MWh per year). This reduces 5 dwellings' heating demand. Replacement reduces heating loss by 17%. 550 Almelo dwellings might reduce their carbon impact by 65% by replacing natural gas with hydrogen. With sustainable energy sources, this can reach 100% without major infrastructure changes. **IA9: "Reduce emissions through energy efficiency"** - **Petronor** aims to explore energy efficiency improvements in their refineries to enhance 25% by 2025 and 10% from 2030 to 2050. The Energy STAR Guide for

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Petroleum Refineries recommends focusing on utilities, heat exchanges, and fired heaters for their low-investment potential.

**IA6 and IA8: "Circularity of carbon" and "Reducing emissions though CO/$CO_2$ capture and use" - German Hub** has identified 4 main sources responsible for 87% of total $CO_2$ emissions (4.6 million tons of $CO_2$ in 2021), which are: natural gas/methane (46.2% $CO_2$ emissions), fossil ethylene (24.8% $CO_2$ emissions), fossil methanol (11.4% $CO_2$ emissions) and fossil acetic acid (4.5% $CO_2$ emissions). By 2027, green electricity, electrification of heat, CCU-produced synthetic methane, $H_2$, methane pyrolysis, and biomass/biogenic methane will replace all natural gas. This will eliminate 787,500 tons of $CO_2$ for Industriepark Höchst. Calcinor also plans to decarbonise its lime producing processes in the **Basque Hub**: (i) Carbon capture: Calcinor needs 500 $Nm^3$ $CO_2$ and 2,000 $Nm^3$ $H_2$ to create 1KT of lime/day (normal operation). (ii) Petronor's pyrolysis process will produce biocoke, a fossil coal-free alternative to coke. Petronor anticipates producing 3 million kg of biocoke from 10 million kg Associated with document Ref. Ares(2023)8308660 - 05/12/2023

of urban garbage. (iii) Oxycombustion: Calcinor and Sidenor will use $O_2$ in the steel ladle preheating units to minimize fuel use by 25%. Smurfit Kappa will use paper waste and lignin to produce 26.7 MJ/L of biofuels (replacing 123 million $m^3$ of natural gas and lowering 61.5 million kg of $CO_2$) and chemicals by enzymatic hydrolysis and depolymerisation. It expects to create 214,000 $m^3$ (214.2 million kg) of synthetic gas to eliminate 10.6 million kg of $CO_2$. Finally, (iv) biogas production from paper sludge anaerobic digestion will be considered. In the **Turkish Hub**, 1,000 tons/year of CO2 capture capacity and 3,500 tons of hydrogen lead to the production of 5.000 tons of NIPU resulting in 500.000 fridges. The total NIPU-related carbon reduction is 32,500 tons/year. This contributes to 100% resource saving in the Manisa factory of Arcelik. Furthermore, 9.000 tons natural gas from methanol and 40.000 tons of natural gas from polyurethane production are saved, with a total of 49.000 tons of natural gas savings

**IMPACTS—Medium/long-term (2030+):** The **IS2H4C** consortium will comply with the **Expected Impact "Global leadership in clean, climate-neutral and resilient industrial value chains"** by accomplishing the following actions in the medium/long-term after the project: **By 2030:** Industriepark Höchst aims to substitute fossil ethylene with synthetic ethylene from CCU and plastics recycling and cracking. This measure will reduce their $CO_2$ emissions by 4.8%. **By 2040:** Industriepark Höchst aims to reach carbon neutrality in their entire processes by replacing fossil methanol with synthetic methanol from CCU and biogenic compounds, and by replacing fossil acetic acid by synthetic acetic acid from CCS and recycling. **By 2050:** Basque Hub: **1) Oil industry** (Petronor): To produce 10 times more biogas compared to 2028 (40GWh of biogas, from 100 KT of waste). **2) Steel industry** (Sidenor): Achieve a production of 23 KT of $H_2$/year, which is the estimated amount of $H_2$ needed to reach $CO_2$ neutrality.
**3) Lime industry** (Calcinor): To generate 667 KT of capture $CO_2$.

**External barriers and requirements:**

1. The main barrier identified in relation to this KIP involves **lack of infrastructure**. Due to the dual approach of the proposal involving not only policymaking/governance but also heavily relying on the technical aspects of the transition, some European regions may lack the necessary infrastructure to apply the guidelines developed in IS2H4C. To address this issue, the consortium will consider logistics aspects within the pathway evaluations carried out in WP2, WP3 and WP4, and provide alternative routes to overcome lacking infrastructure. The project will cluster with active initiatives should regions need coherent bioeconomy strategies and thus increase the intensity of bio-clusters and provide the labour force with new sets of skills needed for the circular bio-based transition
2. **Lack of awareness:** certain European regions may not be aware of the benefits of industrial symbiosis or may not fully understand how to implement it effectively. The capacity building activities designed for this project (e-learning resources, workshops/webinars and IS online game, see Section 2.2), can help address this barrier.
3. **Resource Matching:** Finding suitable partners for resource exchange can be difficult. Industries need to identify compatible partners with complementary resource needs and available surpluses. The Digital Collaboration Platform and the Circular Industry Helix (CHX) will facilitate networking and matchmaking between partners, and also will help to find new suitable collaborators.
4. **Risk Aversion:** Industries may be risk-averse and hesitant to engage in new collaborative efforts, especially if they are concerned about the security of their proprietary information or potential

liabilities. The consortium has a preliminary draft of IP strategy for the exploitable results (See Section 2.2) to be generated within IS2H4C, which guarantee the security of sensible information.

*Table 2. KIP 1 implementation indicators*

| Time | Indicators of scale and relevance |
|---|---|
| Year 1 (Outcomes | **German Hub:** To produce electrical (5.1 MW), thermal (3.9 MW) and biomethane (10 MW) power from biogas; to produce 1 MW of $H_2$ for transportation.<br>**Basque Hub:** To produce 4GWh from urban garbage; to have the capacity to produce 112.5 MW of $H_2$ overall (from the 3 electrolysers); to produce 2.8 M L/year of e-fuels for transportation.<br>**Dutch Hub:** To replace the heating demand of 5 houses in Almelo with $H_2$ and reducing heat loss by 17%.<br>**Turkish Hub:** To produce 1000 tons of captured $CO_2$ annually and 3500 tons of $H_2$ to subsequently produce polymers and build bio-fridges (Arçelik). |
| Year 5+ (Impac | **German Hub:** To reach climate neutrality by replacing fossil methanol and acetic acid.<br>**Basque Hub:** To upgrade their infrastructure capacities to produce 40 GWh of biogas (oil sector), 23 KT $H_2$/year (steel sector), 667 KT of captured $CO_2$ (cement sector).<br>**Dutch Hub:** Replacement of natural gas for heating purposes in Almelo (10,000 houses) will be initiated using biogas and $H_2$<br>**Turkish Hub:** To eliminate the use of gas in the polymerisation process, by producing and using polymers and optimize production processes and transportation logistics to reduce carbon emissions in the production of refrigerators. |

## Key Impact Pathway 2. Incentives for Climate Neutral, Circular Industry Solutions, and IS

**RESULTS—By the end of the project (2028):** Activities carried out in the project duration – and especially focussed on the more technical approach of the project such as those proposed in WP2, WP4, WP5, and WP6 (and described in the previous KIP) - will directly contribute to the creation of a more efficient and sustainable industry across Europe. However, to accomplish the sustainability transition of EU processes industries, unprecedented levels of investment are required. Considering this, and complying with outcomes ***O2.1 "Use Industrial Symbiosis and cross-sectoral cooperation to pave the way for achieving the EU Green Deal objectives"*** and ***O2.2 "Providing recommendation for optimised regional framework conditions by highlighting barriers and suitable innovations-oriented policies"***, **IS2H4C** contributes to inspiring and de-risking investment by designing and implementing an **effective policy framework,** providing **adaptable business models**, and boosting **digitalisation**. In doing so, **IS2H4C** ensures a safe transition to IS for the process industries involved and improves their competitiveness. The **policy framework** provided by **IS2H4C** by the end of the project will have the purpose of addressing prices, energy and waste regulations and standardisation, which subsequently will contribute to ***"make climate neutral and circular industry solutions economically attractive (IA)"***, and also to ***"reduce barriers for market uptake (IA14)"***.

The **policy framework** will be divided into 3 phases: (i) **Raising awareness:** one of the keyways to promote standardisation and regulation is to raise awareness of the benefits of IS. Complemented by the dissemination, communication and exploitation activities (Section 2.2), the policy framework to be developed will include best practices and recommendations derived from the project's results and share them with relevant stakeholders from different industries, government and academia. (ii) **Collaboration: IS2H4C** provides spaces for collaboration between industry, government and standardisation organisations (e.g., CEN, EUR-ISA), to develop guidelines that are applicable across different industries and sectors. And, (iii) **Training and capacity building:** Companies and organisations may need support in order to adopt standardised Industrial Symbiosis practices. To this end, **IS2H4C** organises workshops and conferences (see Table 5) to provide this support and ensure that companies have the knowledge and skills needed to implement sustainability practices effectively. In order to contribute to **"Reaping the full benefits of digital technologies (IA13)"**, the **IS2H4C** Digital Collaboration Platform (WP7) will integrate intelligent matchmaking decision support and tracking tools, a marketplace and a collaboration module, which: (i) facilitates resource matching between companies and industries, (ii) improves information

sharing (e.g., energy consumption, emissions, waste generation), (iii) enables collaborative planning and decision making to optimise the use of resources and minimise waste, and finally, and, (iv) enhances monitoring and tracking of resource flows, which can help identify areas for improvement. Moreover, to assess financial investment, credit risk evaluation and valuation of sustainable and circular business opportunities, **IS2H4C** will utilise AI frameworks to accurately predict the valuation of **H4C** assets (WP5).

**IMPACTS—Medium/long-term (2030+):** In medium/long-term, **IS2H4C** will make significant contributions to overall European industrial sustainability, competitiveness and resource independence through the implementation of its proposed methodology for policymaking and digitalisation. This wider impact will revolve around lowering the use of non-renewable materials, reduction of GHG, enabling climate neutrality, increasing resource efficiency, enabling the development of innovative value chains across Europe, and overall substitution of fossil feedstocks. Moreover, as carbon emissions will be minimized using renewable energy additional savings are achieved, since according to the World Bank, carbon pricing schemes that incentivize carbon reduction could generate $23 trillion in new revenue by 2030[19]. Also, sustainability boosts productivity and competitiveness, as studies have shown that sustainable enterprises outperform their peers by 15%[20]. Finally, throughout IS, and waste management, the waste disposal costs can be reduced, economically impacting the **H4C** and their surroundings. The EU Environment Agency predicts that reducing food waste alone may save 131 kg per EU inhabitant per year [35,21].

1. **External barriers and requirements:** One of the main hurdles faced by this KIP involves **regional differences in the progress of policymaking** regarding IS transitions. These differences may even extend through micro-meso-macro levels within one country. Some regions may already have strong policymaking in place requiring only slight improvements or dissemination, while other regions may have only planned policies, with none actually implemented. As such, important mitigation actions such as regional peer-learning across the EU are essential to balance out differences and coordinate actions beyond a single border.
2. **Economic Incentives:** The economic viability of industrial symbiosis projects can be a challenge. Industries may be hesitant to invest in the necessary infrastructure without clear financial incentives or a guaranteed return on investment. The successful deployment of the 4 H4C presented in this project, will prove the efficiency of IS and its solutions. By presenting the results obtained to policymakers and standardization bodies, it is expected to convince them of the utility of making this transition, and further support the development of economic incentives.

Table 3. KIP 2 implementation indicators

| Time | Indicators of scale and relevance |
|---|---|
| **Year 1-2 (Outcomes)** | • Development and implementation of the policy framework and adaptable business models<br>• Introducing the **IS2H4C** Digital Collaboration Platform |
| **Year 3-5 (Impacts)** | • 100% of hubs use the digital platform, & benefit from matchmaking recommenders and decision support tools offered.<br>• 100% of consortium enterprises (including those directly related to hub's activities) adopt business and financial models based on technological integration and process optimisation in an IS environment. |

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## Key Impact Pathway 3. Non-Technological Elements Essential for Successful Innovation Deployment

**RESULTS—By the end of the project (2028):** All innovations outlined will only be successful if they are supported by nontechnological elements such as (i) societal acceptance, (ii) skills enhancement and (iii) market conditions, according to ***"Human resources, skills and labour market (IA14)"***. Although the concept of IS has its roots in the late 1980s, a significant portion of society may not be aware of what it is

[19] The World Bank (2022, May 24) Global Carbon Pricing Generates Record $84 Billion in Revenue. Retrieved March 25, 2023.

[20] Clark, G.L. et al. (2015) From the stockholder to the stakeholder, How sustainability can drive financial outperformance. University of Oxford. Retrieved March 25, 2023 [35] European Commission (n.d.) Food waste. Retrieved April 1, 2023

[21] European Commission (2023, February 22) Circular economy: definition, importance and benefits: Retrieved April 1, 2023

and the benefits it brings. Hence, there needs to be a concerted effort to increase awareness among the general public. Through DEC activities, **IS2H4C** engages with the surrounding communities of each hub to (i) understand and identify their specific needs, drivers and demands, (ii) promote environmentally responsible behaviour, and (iii) build trust and confidence in the concept of IS among different stakeholders. Besides DEC activities, **IS2H4C** will develop a full stakeholder analysis (see Section 2.1.1 for preliminary analysis) to identify relevant groups that are affected (directly or indirectly) by **H4C** innovations and involve them in decision-making processes. Process4Planet has identified that skills and education are a bottleneck to develop, implement and unfold the full potential of the new technological solutions. For that reason, **IS2H4C** will implement training and capacity building activities which will provide useful knowledge that helps the community to better adapt to the sustainability transition. Also, **IS2H4C** believes that making process industries more attractive for the younger generation is crucial to enhance deployment of digital technologies, which is why interactive and innovative activities such as IS online games and hackathons (further explained in WP8), will be implemented.

**IMPACTS—Medium/long-term (2030+):** The results are expected to have a positive impact on society at different levels. For instance, by collaborating with villages and surrounding communities, **IS2H4C** enhances social cohesion, stimulates innovation exchange of ideas, knowledge and technology among the different industries and sectors, which subsequently will lead to development of new products and processes that are more sustainable, efficient and tailormade for each **H4C**. Additional societal benefits of the **IS2H4C**, contributing to ***"Circular regions (IA12)"***, include (i) improving public health due to air and water quality (see KIP 1), (ii) generation of at least **40,000** new jobs related to **H4C** in general and increase productivity, and (iii) creating a balanced economic growth, social well-being and environmental protection.

**External barriers and requirements:**

1. The main external hurdle faced by this KIP is the potential **lack of interest** from the public and other stakeholders. **IS2H4C** considers co-definition of solutions as a critical aspect for its success, the consortium has elaborated a deep and thorough dissemination and communication plan that seeks involvement of all relevant stakeholders. Through this action, risk posed by this barrier is mitigated.
2. **Resistance to Change:** People and organizations often resist change, especially when it disrupts existing routines and practices. Adopting industrial symbiosis may require significant changes in operations and relationships, which can face resistance. The activities of the DEC strategy are designed to inform and engage the local community and potential stakeholders, in order for them to observe the benefits of this transition.
3. **Competitive Concerns:** Some industries may view resource sharing as a competitive threat rather than an opportunity. Concerns about losing a competitive edge or revealing proprietary information can deter collaboration. The solutions to be developed within each H4C are aimed for the industrial community welfare, where all gain valuable insight and tools to reduce their emissions and waste and subsequently reducing costs for everyone.

*Table 4. KIP 3 implementation indicators*

| Time | Indicators of scale and relevance |
|---|---|
| **Year 1-2 (Outcomes)** | • Full stakeholder analysis<br>• Implementing training and capacity building activities<br>• Implementing IS online games & development of 1 Hackathon (topic to be discussed). |
| **Year 3-5 (Impacts)** | • Generation of at least 40.000 new jobs among **H4C**[22].<br>• By instituting energy-efficient practices, a typical household can save up to €500/year; for businesses, these savings can be substantial[23]; therefore, hubs and their environments can reap these benefits.<br>• By implementing IS, the costs associated with energy, $CO_2$ capture, water consumption, and waste disposal will be shared and, as a result, lowered to the economic and societal benefit of all involved. |

[22] Hub, I. S. K. (n.d.). European Commission Adopts Circular Economy Action Plan | News | SDG Knowledge Hub | IISD.

[23] Energy saving actions by EU citizens could save enough oil to fill 120 super tankers and enough natural gas to heat 20 million homes - News - IEA. (n.d.). IEA.

## 2.2 MEASURES TO MAXIMISE IMPACT – DISSEMINATION, EXPLOITATION AND COMMUNICATION #@COM-DIS-VIS-CDV@#

**IS2H4C** will establish a dedicated plan for the dissemination, use and valorisation of research and innovation results, critically contributing to the delivery of expected outcomes and impacts described in Section 2.1. It will also include the communication activities to make the society aware of the use of EU funds for R&I tackling societal challenges, the impacts of IS in the EU and enhancing the knowledge exchange between science, industry and society. WP8 will provide a complete dissemination and exploitation as well as a communication plan by M6 (with updates during the project) to lead these activities and coordinate partners to deliver highest impact. Here, a brief introduction to the main elements of this plan is offered.

### DISSEMINATION AND EXPLOITATION PLAN

**IS2H4C**'s dissemination and exploitation plan will ensure that the project results and outputs are effectively and efficiently disseminated and protected to guarantee their replicability and exploitation. This involves the implementation of dissemination and exploitation activities, as well as knowledge and IP management.

**DISSEMINATION PLAN.** **IS2H4C**'s dissemination strategy will boost knowledge transfer and visibility of results obtained during and after the project. Stakeholders identified in Section 2.1.1. will be actively engaged to enable the uptake of the information and data generated by **IS2H4C**, thus maximising outcome and impact. The dissemination plan will be adaptable in its objectives and activities to better adapt to the project's evolution. Moreover, to facilitate adequate open cooperative work and knowledge transfer, this plan will reflect the project's "Open Science" approach as described in Section 1.6. It will consist of three main phases:

i. **Community building (M1-M12).** The channels for dissemination will be set up for the correct diffusion of messages to the targeted stakeholders. It will be focused on generating a community of interested stakeholders around the project, informing and generating awareness on the basics: **IS2H4C**'s objectives, expected results and impacts, etc.

ii. **R&D diffusion and cooperation (M12-M36).** As the results of the project emerge, they will be spread through several tools, including newsletters, social media posts, scientific articles, and conferences. In addition, cooperation with related projects and initiatives will be sought during this phase.

iii. **Exploitation enablement (M24-M48)**. It will support the actual exploitation of the project results via workshops, capacity building activities and marketing towards final end-users (commercial results).

As shown in Table 5, various online and offline channels will be established to disseminate the knowledge resulting from the project's activities, including **IS2H4C**'s vision. Channels and tools such as the website, newsletters, press releases and social media fall under the communication strategy but will support the dissemination strategy.

*Table 5. Dissemination activities*

| Activity | Target group(s) | KPI |
|---|---|---|
| **Scientific publications** | Academia | At least 30 peer-reviewed papers |
| Partners are expected to submit at least 30 peer-reviewed scientific papers by the end of the project. OA to these articles will bring additional benefits like greater transparency, opportunities for new scientific collaborations, and increased efficiency as research does not need to be repeated but can be shared, reproduced and replicated. Relevant High-Impact Journals may include the following (listed below). After the project, effects will be measured according to citation-based metrics such as citation counts (aimed at Field-Weighted Citation Impact >1).<br>- Applied Energy - European Journal of Operational Research - Renewable Energy<br>- International Journal of Hydrogen Energy - Production and Operations Management - Energy Policy<br>Sustainable Production and Consumptio - Renewable and Sustainable Energy Reviews - Energy Economics | | |

| | | |
|---|---|---|
| **Capacity building activities** | Industry, Academia | At least 5 e-learning technical guidelines. At least 3 workshops<br>1,200 students engaged |
| As part of T8.2, EEIP, in cooperation with the partners will coordinate the development of different training activities oriented to both students and workforce, including the preparation of e-learning resources and the arrangement of specific workshops/webinars. All contents will be available on **IS2H4C**'s website up to two years after the project end. Academic and industrial partners will involve and train BSc/MSc/PhD students among their activities in the project. It is expected to engage at least 1,200 students through the IS Online games and the on-site visits, mainly supported by the coordinator UT, all the academic partners of the project and **IS2H4C**'s hubs. | | |
| **Conferences and events** | All stakeholders | Attend at least 8 and present in at least 4 conferences/year |
| EEIP, as D&C Manager, will orchestrate the organisation of yearly dissemination events (4) to showcase project results. **IS2H4C**'s Advisory Board, together with other renowned R&D/policy experts, will be invited to participate in roundtable discussions. Moreover, the consortium will attempt to establish links with EU funded projects under the same HORIZON-CL4-2023-TWIN-TRANSITION-01-37 call and any other relevant calls. Preliminary, the project will seek to collaborate with the projects HEAVENN, REDOL, CORALIS, INITIATE, Hubs4Circularity Community Practice on (i) common dissemination activities, (ii) joint technical workshops, (iii) unified policy, standardisation and R&I recommendations. To improve the outreach of the project. Co-organisation of joint events/workshops will be sought to obtain more resources and expand their scope and impact. In addition, partners will attend other events/conferences/trade fairs to disseminate project's results, including: (i) Conference on Sustainable Development of Energy, Water & Environment Systems (SDEWES) (ii) Smart Energy Systems International Conference (iii) Offshore Energy Exhibition and Conference (OEEC) (iv) Envirotech for Shipping Forum (v) World Hydrogen Congress (vi) World Hydrogen Energy Conference (WHEC) (vii) International Conference of Industrial Ecology (viii) Mobile World Congress (ix) Cloud Expo EU. CHX will organise 1 helix event (M48) and 1 online clustering event (M12). | | |

**"Circular Industry" helix on the Open Innovation platform Crowdhelix (CHX is a IS2H4C partner):**

IS2H4C will use and enrich the already existing international virtual community/ecosystem[24]. The Circular Industry Helix will be an engine for impact, exploitation, clustering and dissemination activities. The Circular Industry Helix is an international Open Innovation community of specialists and stakeholders from domains such as materials science, process engineering, mineral waste recycling and industrial innovation, as well as relevant policy makers, standardisation agents and citizen interest groups. The helix is currently composed of 79 organisations from 32 countries. IS2H4C benefits from the CHX network that gathers **600+ member organisations** from **57 countries** (all EU member states). It has the potential to reach **11,000+ research and innovation collaborators** from its existing membership base. This network, supported by a customised technology platform, language processing and machine learning tools, has **40+** virtual thematic areas/clusters (called "helixes") that cover all research and innovation fields. Through the helix, the consortium will build a network with at least **150** new organisations (M36), including IS2H4C **partners**, members from existing **relevant helixes** (i.e., *Energy, Solar Energy, Circular Plastics, Water, Hydrogen, Raw Minerals, Digital, Manufacturing, Just Transition helixes,* etc.), **researchers**, **industry actors**, **ethic and policy makers, civil society**, participants of **other relevant projects** and **relevant actors/associations** across **Europe/International** (e.g. EU Factories of the Future Research Association (EFFRA), EU Partnership Made in Europe, Clean Steel Partnership (CSP), A.SPIRE-Processes4Planet, EIT Digital, IT InnoEnergy, EIT Climate-KIC, S3P-Industry, Process4Sustainability, Alliance for a Competitive European Industry (ACEI), EU Energy Research Alliance EERA, Cecimo, Orgalim, Clean Hydrogen partnership, EU Energy Network EnR, Energy Efficiency Financial Institutions Group (EEFIG)). To ensure sustainability, CHX will commit to maintaining the helix for at least **2 years after the project** has concluded. When the project ends, the helix will continue to be hosted on CHX's platform, and everyone signed during the project up will be able to retain their **accounts for free** (as external organisations), as well as have the option for their organisations to join as full members on the network. The CHX Helix Manager will identify a member from the helix, that would be interested in strategically leading this helix after the end of the project. The new helix will be supported by CHX, for details see Figure 9.

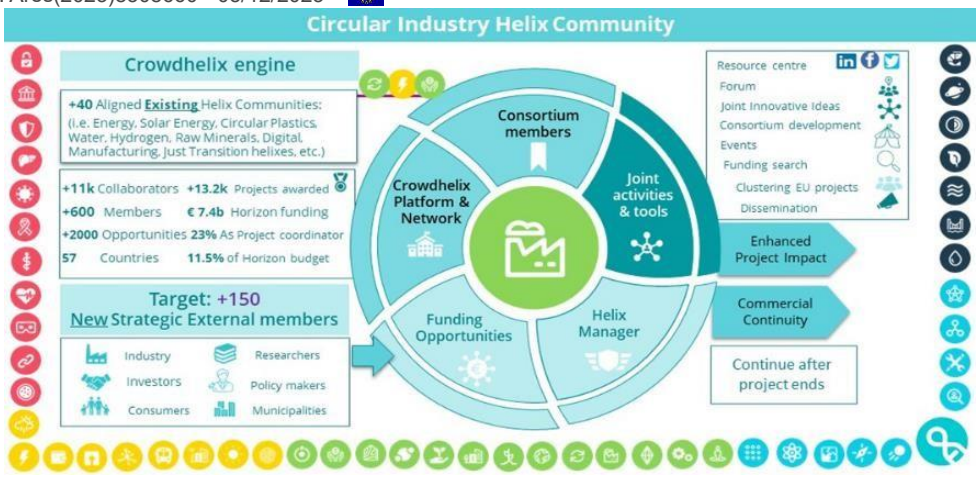


*Figure 9. **IS2H4C** Circular Industry Helix Ecosystem: Virtual Community*

**IS2H4C Clustering plan:** IS2H4C cross-projects collaboration plan is divided in 3 main sections: collaboration board, joint activities, and future collaboration plan.

[24] Circular Industry Helix - https://crowdhelix.com/helixes/circular-industry/info

1) Projects with similar aims and synergies will be identified for a collaboration board (**1.1**), including:
a) the projects funded under the same call (sister projects), b) other European projects and c) national and regional projects. **IS2H4C** will collaborate with **relevant European projects** [REDOL (CIRCE, EEIP), CORALIS (CIRCE, EEIP, SIDENOR), ReSoURCE (CHX), HEAVENN (Hydrogen Valley), INITIATE (engaging with the steel industry sector), Hubs4Circularity Community Practice (engaging with other **H4C** in EU), Covenant of companies for climate and energy, CircLean, projects funded under the Digital EU Programme, cluster 5-destination 'Efficient, sustainable and inclusive energy use", Industries in energy transition', etc.], **programmes** (e.g. EU Innovation Fund, Life Plus Programme, activities of the EIB, EEFIG, eeef) and **relevant actors/associations** across **EU/International** (mentioned above). All the projects and key stakeholders will be contacted, and the Collaboration Board (CB) will be consolidated (**1.2**). The CB will define the Key Performance Indicators **(1.3)** and the main ways of a fluent and frequent communication **(1.4)**. A common identity (logo & templates for communication and reporting) will be designed **(1.5)**. Joint working groups - project members are organised **(1.6)**.

*Figure 10. **IS2H4C**'s Clustering Plan*

2) **IS2H4C** CB members will promote and encourage collaborative activities with the different projects identified above (mainly with the sister projects). These activities include: regular meetings (sister projects will be invited to **IS2H4C** Kick Off meeting) (collaborative ideas and work are expected to emerge thanks to these meetings/workshops) **(2.1)**, joint participation and attendance to webinars and local info days (organised by projects), CHX will organise 1 online clustering event (M12) **(2.2)**, active participation in projects social media sites (Facebook, Twitter, LinkedIn, and similar tools) **(2.3)**, development of joint scientific publications **(2.4)**, supporting future legislation (European Industrial Strategy, Circular Economy Action Plan, Zero Pollution Action Plan, etc.) **(2.5)** and joint participation in scientific conferences and business tradeshows **(2.6)**.

3) Future collaboration plan, a helix event will be organised (M48) to promote the development of interdisciplinary research and innovation ideas, to identify target calls aligned with the consortium strengths.

**Measuring IS2H4C dissemination and project outputs:** *Measuring IS2H4C dissemination using Public Data Sources:* CHX, in collaboration with all partners, will measure the dissemination of **IS2H4C** outputs (Table 6). Using repositories of academic research papers, it will study classic measures of reach, such as numbers of article citations, in addition to using topic analysis to understand degrees of interdisciplinary dissemination. In addition, other forms of dissemination such as reporting in news articles, blog posts and social media will be evaluated using techniques from data science.

*Measuring IS2H4C outputs:* CHX, in collaboration with all partners, will measure **IS2H4C** results (Table 6). This may include, but is not limited to, aggregating additional and external data sources to inform on broader project outputs, modelling to determine causality of outputs, and statistical analysis to conclude on the robustness of outputs measurements.

## EXPLOITATION PLAN

**a)Business case.** The business case for **IS2H4C** is based on the development and deployment of H4C in four different locations across Europe, which is expected to create new market opportunities for the

participants and contribute to the wider deployment of sustainable and circular solutions in the EU. Consequently, the **targeted market** for **IS2H4C** is primarily the process industries, including oil refining, steel, pulp and paper, lime, petrochemical, household appliances, and chemicals and pharmaceuticals industries. The estimated market size for these industries in the EU and globally is significant, with a growing demand for more sustainable and circular solutions to address environmental challenges and reduce waste. The solutions developed by IS2H4C are designed to match the **market and user needs in a cost-effective manner**, with a focus on resource efficiency, use of renewable energy, and waste prevention. Moreover, **IS2H4C** 's expected **market position and competitive advantage** will be based on the project's unique approach to developing H4C, which is expected to become a reference for promoting sustainable regional development in Europe.

**German hub.** E-methanol, ethylene, acetic acid, and methane production will trigger investments on CC and electrolysis. Potential capacity expansion of these technologies will trigger infrastructure investments and call for participation of other hub companies in these investments for sharing costs. As above-mentioned materials are widely used by diversified industries and have increasing prices, hub companies using these materials will benefit from traditional material purchase cost reductions. Considering a scenario of higher carbon taxes, hub companies will economically benefit from replacement of traditional production techniques with the green ones. As the processes of hubs require high temperature of heat, R&D investments for developing more efficient heat pumps (such as to reach higher temperatures of 250-300 degrees) will also be triggered, leading to significant energy cost savings in future. Sustainable action must take ecological, economic and social aspects into account. The measures have to support companies goals to achieve a climate neutrality(in Germany Net Zero CO2-Emission in 2045!). The measures must also make economic sense (global competitiveness). Therefore investment and production costs must be carried out in accordance with the specifications of the respective companies, and acceptance, quantities and prices on the market for sustainable generated chemicals / heat must be determined.

**Basque Hub.** The different gases and subproducts of the industries will be offered based on the demand and production profiles. PETRO will potentially provide $O_2$ and $H_2$ to the identified sectors (steel, foundry, cement and lime) and the process $CO_2$ from cement and lime will be used for the production of e-fuels by PETRO. This symbiosis will provide a cost reduction and efficient business scheme. The market segments to target will be the industries defined (cement, lime, steel). The production and distribution costs with the CAPEX and OPEX of the technologies anticipate an investment needed of 150M€ of these industries in $H_2$. It is estimated that the $H_2$ could be available at 3,5€/kg by 2028 (after project) and $O_2$ 8€/l which would be more profitable than having a $O_2$ on-site production system (approx. cost in steel industry of 1M€).

**Turkish hub.** Capacity of electrolyser will increase from 5MW to 20MW triggering $H_2$ investments in the hub and subsequently will trigger investments in wind and solar energy to feed the electricity need of the electrolyser. Next, oxygen supply will increase which might attract investments of companies widely using oxygen in their processes (e.g., chemicals, pharmaceuticals, pulp&paper production). Further, oxygen stored in gas cylinders can be also used in hospitals for medical/therapeutic usage. As the CC technology will be deployed, extension of CC capacity will lead to production of more e-methanol and polymers, which consequently will attract a large number of companies from different sectors. From CC, production of other materials can also be considered like in the case of German hub (e.g., ethylene, acetic acid). In particular, higher amounts of NIPU production will allow ARCELIK to spread the use of NIPU in its other premises and higher amounts of e-methanol will allow TUPRAS to have significant cost savings from traditional methanol purchase. The use of rigid polyurethane foam, which will be obtained with the NIPU output, in Arçelik refrigerator products is completely in line with Arçelik's sustainability strategy. At the end of this project, the NIPU technology can be realized and commercialized by serial production of maximum 1000 units of refrigerator with 10% ramp-up by year for the Northern and Western Europe market where low carbon emission and sustainability targets considered

**Dutch Hub.** Waste heat Recovery and Oxygen use in water treatment in combination with effluent usage in H2 production The hub offers a range of services for stakeholders interested in hydrogen technology, including infrastructure testing, consulting, and training programs. To ensure energy supply continuity for electrolysis, the hub can trigger investments in solar and wind energy. Increased oxygen supply may attract companies that use oxygen as their main resource. Successful testing of $H_2$ in the built environment could lead to an investment in infrastructure change. The $H_2$ station's location between the North Sea Baltic and Rhine Alpine corridors may attract more logistics companies operating at international level. A biogas digestor using cattle manure can help reduce nitrogen emissions for cattle farmers while producing renewable energy. This would reduce the Dutch government’s pressure on cattle farming sector (about

nitrogen emissions) which is one of the largest industry in the region. The heat from biogas production can be used for district heating in Almelo city.

**Strategic Context for all hubs.** Symbiosis and circularity are increasingly critical to creating a sustainable world shortly. In recent years, we have seen increased water scarcity in the Netherlands. With the ambition to increase green H2 production by multiple gigawatts in Europe, we will witness a continental increase in water demand. On top of that, we must make the most of the green energy we harvest to avoid wastage. By using effluent from the water treatment facility next door, we can limit the use of tap water or purified water acquired elsewhere in electrolysis. At the same time, the water treatment facility should be able to run more efficiently using the nearly pure oxygen that we produce as a "by-product" in electrolysis, increasing our overall efficiency by wasting less energy. Another "by-product" of electrolysis is heat produced by the alkaline stack. Under normal operating circumstances, this heat would go to waste. We aim to store this heat in geothermal storage on site, where we can recover it by water-water heat pumps to heat the buildings' radiators, tap water, and other heating systems and keep the inflow to the alkaline stack at the optimal operating temperature.

**Brief economic analysis.** By demonstrating and implementing the abovementioned techniques, we aim to optimize our energy consumption by effectively improving the COP of the entire facility. In turn, this will lead to lower, or preferably no, external consumption of gas and electricity, reducing the operational costs of the building significantly. Furthermore, by using the proposed symbiosis with the wastewater treatment facility, we can get cheaper feedstock for our electrolysis process and reduce the cost of wastewater treatment while optimizing the potential losses of electrolysis.
***Commercial approach***. The knowledge of the subjects will be valuable for businesses in the region of Twente and on a national and European level. Having this knowledge, we can support companies in Twente to move forward efficiently in the energy transition. Furthermore, corporate members stand to benefit from the knowledge they receive by taking part in projects, ensuring their future in the energy transition, and educating their workers in these new fields.
***Management approach.*** The proposed projects will be managed through the H2Hub Twente by dedicated project leaders overseen by the Hub Manager, as we have been doing successfully in recent years. Each project leader will get support from corporate members and knowledge institutions who are affiliated with the project's content or stand to benefit from it mainly in the future. This way, we can bundle strength and knowledge to ensure the proposed outcome outlined in the project proposal.

**b)Exploitation plan for industrialisation.** The main goal of the exploitation plan is to effectively use the project's results through scientific, economic, political or societal exploitation routes, aiming to turn R&I actions into concrete value and impact for society and industry. **IS2H4C** will produce different types of Key Exploitable Results (KERs), individual innovations key for the overarching H4C's exploitation plans. See WP8 for further details on how UT as exploitation manager will handle exploitation activities.

*Table 6. Preliminary identified KERs*

| KER (ownership) | Added value | End users | IP Strategy | Exploitation route |
|---|---|---|---|---|
| Model for self-sustaining **H4C** | Boosting the systemic TRL of Hubs | R&D, **H4C**, Green Energy Companies, Green tech manufacturers, Investors, Public Bodies & Policymakers | Open access | Further R&D, joint exploitation agreement, engineering services, licensing. |
| Digital Platform for information sharing | Secure information sharing between the Hubs | | Utility mode l, patenting | |
| Novel green technologies | Reduction in GHG emissions | | Open access | |
| Serious online games | Knowledge dissemination | General Public & all stakeholders | Patenting, trade secret | Further R&D, developing /selling own services |
| Virtual Hubs | Simulating circular economy | **H4C** managers & all stakeholders | Open access | Further R&D |

| | | | | |
|---|---|---|---|---|
| Financial canvas | Identifying financial risks and investment opportunities | Investors & all stakeholders | Open access | Further R&D, developing/selling own services |

| | | | | |
|---|---|---|---|---|
| Smart decision support tools | Support informed decisionmaking processes | **H4C** managers & all stakeholders | Patenting, trade secret | B2B, B2C, engineering services, licenses, database for **H4C** managers |
| Handbook Integrating social innovation into Hubs4Circularity | Guideline for successful stakeholder engagement | **H4C** managers & all stakeholders | Open access | Further R&D, European Communities of Practice |
| Sustainable Technologies | Integrate technologies to provide Hubs and its surroundings green energy and IS environments | Investors, general public & all stakeholders | Patenting, trade secret | B2B, B2C, engineering services, licenses, and R&D |

document Ref. Ares(2023)830866
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By M6, the proposal's preidentified KERs will be updated and reassessed. KERs will be classified among commercial and noncommercial KERs, and prioritised according to three main parameters: degree of innovation, potential profitability, and impact on the industry. The Exploitation Manager, together with the project partners (owners of KER), will define the focus of each KER exploitation strategy. **IS2H4C**, an IA action, is expected to achieve final TRLs between 6&7, meaning that commercial KERs will still be a few years before their exploitation. The exploitation strategies will be tailored considering the overarching EU masterplan regarding the development and deployment of IS. The exploitation strategy will be crosscut by the IP strategy (further explained in the next section), the DMP & OS practices, and in close cooperation with the dissemination & communication strategies.

**Follow up plan**. The follow up plan is focused on achieving a higher degree of technical, social and financial maturity of the four H4Cs in the project, as well as to continue contributing to the expansion of the concept across other industry clusters in Europe. Since the beginning, during the project, the consortium is already working on the greatest potential barriers to be addressed in the adoption of the IS concept. The execution of WP3 will help to detect additional barriers and adapt the exploitation strategy to better overcome them. One of the key obstacles to reaching higher TRLs is **matching value chains,** as this requires integrating various sectors and industries into a cohesive system that maximizes the circularity and sustainability of resources. To overcome this obstacle, the IS2H4C project will focus on developing customized business cases and financial models that distribute investment costs and risks among the involved actors and ensure the replication of these models in diverse geographical contexts. This will involve engaging with industrial integrators, policymakers, and other stakeholders to build consensus and commitment towards the project's goals. Additionally, the digital tools DigitalH4C and the Circular Industry Helix will facilitate this engagement. To address **Securing industrial integrators and user acceptance**, the IS2H4C project will engage with industrial partners and end-users throughout the project, gathering feedback and insights to refine and optimize the developed solutions. This will include developing training and capacity building programs for stakeholders, as well as establishing a community of practice (ECoP) to promote knowledge sharing and collaboration. Another critical requirement is that the technologies developed and deployed during the project enhance their **robustness** to reach higher TRLs. This will involve a particular pathway for each one of them, involving the further testing and validation of the systems, ensuring their technical and sustainability performance, and their compliance with regulations and standards.

To ensure the commitments of the industrial partners after the end of the project, the IS2H4C project will develop a **post-project exploitation strategy** that includes technical, financial, and legal preparedness. This will involve defining attractive business models and feasibility studies for the adoption of IS2H4C's innovations, as well as establishing IP provisions and other measures to protect the project's intellectual property. The project will also work to identify and engage with potential investors, customers, and other stakeholders to ensure the long-term viability and sustainability of the developed solutions.

## COMMUNICATION ACTIVITIES

A strategic communication plan will start at the outset of the project and continue throughout its entire duration. The aim is to provide universally comprehensible information to the public at large about the project goals and results. An integrated communication campaign will be designed and launched, led by EEIP, using a variety of tools and tactics to communicate the project's success stories, significance, and

the overall operational framework that is understandable to wider public audiences. Through the exploitation of mainstream communication channels, the consortium will increase awareness and enhance societal perception on how R&I through IS can positively impact the society, while increasing the visibility and information flow on the vital role EU funded research in realizing and achieving ambitious EU goals. The communication plan will be continuously updated, implementing communication measures and tools to maximise social acceptance considering cognitive and awareness raising issues as well as risk perceptions. **IS2H4C**'s strategic communication plan will include (i) the development of a Visual Identity (project logo and brand for all communication materials), (ii) a dedicated website operational from M3, (iii) social media channels such as LinkedIn running from M3, (iv) biannual newsletters, (v) press releases, (vi) participation and presentation of the project and its results in innovation and networking events, technological fairs, (vii) participation in other networks and groups, not directly linked to the project, where consortium partners have strong links and involvement, (viii) in-house presentations to existing clients and collaborators and brainstorming for further extending **IS2H4C**'s results to other applications and markets. The following demonstrates the relationship between tools, channels, target groups and monitoring indicators / KPIs / Targets:

*Table 7. Communication activities*

Associated with document Ref. Ares(2023)8308660 - 05/12/2023

| Channels | Tools | Target groups | KPI/target |
|---|---|---|---|
| **Digital means** | Website | All | 5,000 visits/year |
| | LinkedIn/Twitter | Industry, Academia | 500 followers |
| | Newsletters | All | 2 newsletters per year, 10% opening rate |
| | Press-releases | General public | 300 media outlets/articles covered |
| **Digital materials** | Project video and infography | All | 1,250 views |
| | Highlight videos | All | 1,250 views |
| **Events** | Conferences, fairs and exhibitions | All | Presentation on 8 conferences/fairs/exhibitions |

## INTELLECTUAL PROPERTY MANAGEMENT

The overall IPR strategy of the project is to ensure that partners can fully exploit their market position, involving IP tools. The partners will sign a CA prior to the signature of the GA with the EC in which all the rules for ensuring confidentiality (procedure for granting publication of the results), ownership regime and IPR issues (background included and excluded, ownership, licensing terms, etc.). An initial analysis of the background IP has been performed at this stage, so it is ensured that the knowledge already held by the partners (data, know-how and any other type of tangible or intangible information) and needed for a proper project implementation and exploitation is available. Within the KER's exploitation plans, a more in depth Freedom-to-operate (FTO) analysis will be performed (including patent analysis if needed). With regards to the foreground IP and the ownership of results, the preliminary KER definition (Table 6) provides an overview on the ownership of such results, and the joint exploitation strategy which the owning partners are envisaging after the end of the project. This will be finalised towards the end of the project, when the Results Ownership List will be released as final summary. Not just the identified KERs will be subject to IP management; additionally, the IP strategy of the project will allow to set specific restriction levels for the research outputs, including data, managed under the DMP (open by default). In alignment with the MGA obligations, partners will ensure IP protection while also looking for a balance between that and the open access to data and results. The discussion on the best suited protection measures (patents, copyright, trade secret, etc.) will be conducted during the exploitation workshops led by UT during General Assembly

meetings and in the questionnaires distributed as part of the exploitation strategy. Possible use of Horizon IP Scan[25] services will be evaluated.

[25] Horizon IP Scan (europa.eu)

## 2.3 SUMMARY

To develop sustainable technologies and demonstrate synergies in H4C: **German Hub:** CC, green $H_2$ generation, heat recovery **Basque Hub:** pyrolysis of urban waste, green $H_2$, oxycombustion, CC. **Turkish Hub:** green $H_2$, wind energy, CC, polymer production. **Dutch Hub:** green $H_2$,solar energy.

To ensure the resilience and sustainability of H4Cs: To achieve **30%** carbon, **10%** energy and **20%** overall resource reduction

To facilitate market penetration of H4C through: **Social innovation:** stakeholder interviews, discussion panels, training and education, tailor-made business cases **Financial innovation:** innovative circular models, cooperation and competition, de-risking strategies.

To create **regional development models**

To develop a **Digital Collaboration Platform**

To maximise **wider uptake** of IS2H4C's results

**KIP1:** Hubs results will comply with **"O1. Reducing the use of virgin resources by at least 20% compared to SotA"**
**German Hub results:** (i) To replace all natural gas with green power, electrification of heat, production of synthetic methane (CC and water electrolysis) and biogenic methane (reducing 46.2% $CO_2$ emissions; 1.98 million tons of $CO_2$) and utilise it in chemical and pharmaceutical industries. (ii) To produce electrical (5.1MW), thermal (3.9MW) and biomethane power (10MW) through biogas (from 100 types of waste). (iii) To reduce fossil fuel consumption by recovering heat waste through heat pumps.
**Basque Hub results:** (i) To analyse the obtention of biocoke (3 million kg) and biogas (4 GWh) from carbonatation of steel slags to replace pet coke and natural gas in the lime, steel, cement, pulp & paper, oil refining and foundry industries (reduction of 5.5 million kg of CO2). (ii) To validate stablished H2 production aims, based on the production of green H2 from electrolysis (capacities 2.5, 10 and 100 MW) from renewable sources (reducing 47% of emissions). (iii) To utilise O2 (from the electrolysis) for oxycombustion (reducing 25% the use of fossil fuels for combustion). (iv) To capture CO2 emitted from the Lime process industry and utilise H2 (from electrolysis) to potentially produce e-fuels (2.8 M L/year). (v) To use the rejected steel slag from the steel industry (and reducing industrial waste), to produce added value products (through carbonatation) for the construction sector
**Turkish Hub results:** (i) To produce e-methanol from H2 (electrolysis), wind energy and CO2 (captured from refining activities). (ii) To produce non-isocyanate polyurethane (NIPU) (16 metric tons/year) to produce bio-fridges (53% reduction of CO2)
**Dutch Hub results:** (i) To replace natural gas with H2 (electrolysis, 130 MWh/year) in five houses (17% loss of heat is reduced) and three crematoriums in Aardorp. (ii) To recover waste heat (electrolysis) and utilise it in H2Hub Twente offices. (iii) To utilise O2 (electrolysis) for wastewater treatment.

**KIP 2:** Complying with **O2.1 "Use IS and cross-sectoral cooperation to pave the way for achieving EU Green Deal objectives"** & **O2.2 "Providing recommendation for optimised regional framework conditions by highlighting barriers and suitable innovation policies"**
**Results:** (i) to implement an effective policy framework, (ii) adaptable business models, (iii) boost digitalisation (Digital Collaboration Platform).

**KIP3:** Complying with **O3. "Citizens living in proximity of heavily industrialised environment through IS"**
**Results:** (i) training and capacity building, educational games, lower energy bills (more than 500€ in savings), job creation (40.000+)

- Other circular hubs - Green energy companies/green-tech manufacturers - Financial investors/institutions
- Research & academia - Policymakers & standardization organizations - General public

**KERs:**
-Model for self-sustaining H4C
-Digital Platform for information sharing
-Novel green technologies
-Serious online game
-Virtual Hubs
-Financial canvas
- Smart decision support tools - Handbook integrating social innovation into H4C -Sustainable technologies

**Dissemination: (i) Scientific publications:** at least 15 papers**; (ii) Capacity building activities:** at least 5 e-learning technical guidelines and at least 3 workshops with 1.200 students engaged. **(iii) Conferences and events:** Attend at least 8 and present in at least 4 conferences/year (to seek cooperation with REDOL, CORALIS, INITIATE, H4C Community Practice.; (iv) To build **the IS2H4C virtual community** and connect with at least 150 new organisations
**Communication:**
Website (1000 visits/year), LinkedIn (500 followers), Newsletters (2/year), press releases (300 articles), project video (1250 views), Highlight videos (1250 views).

Hub will substitute fossil ethylene with synthetic ethylene (CCU) (reducing CO2 emissions by 24.8%) By 2040: The German Hub will reach carbon neutrality in their entire processes. By 2050: The Basque Hub will (i) increase biogas production in the oil industry by 10 (compared to 2028), (ii) achieve a production of 23 KT of H2/year in the steel industry (enough to reach carbon neutrality) and (iii) generate 667 KT of CC CO2 and 4 KT of H2/year in the lime industry. **KIP2:** Contribute to European industrial sustainability, competitiveness and resource independence by progressing in policymaking and digitalisation, lowering non-renewable materials, reduction of GHG, and enabling the development of innovative value chains.
**KIP3:** Achieving a positive impact on society at different levels and specifically to the areas surrounding the H4Cs by (i) enhancing social cohesion and stimulating innovation and subsequently leading to new product/processes development or optimisation, (ii) Improving public health due to improved environmental conditions, increases productivity and job generation (40.000 at least), (iii) creating a balanced economic growth and social well-being.

**2028** **2031** **2033+**

**Target groups:**

**Expected impacts**

Complying with **Expected Impact "Global leadership in clean, climate-neutral and resilient industrial value chains" KIP1:** By 2030: The German

# 3 QUALITY AND EFFICIENCY OF THE IMPLEMENTATION

Associated with document Ref. Ares(2023)8308660 - 05/12/2023

## 3.1.1 INTRODUCTION TO THE WORK PLAN

**IS2H4C will be implemented over 48 months** as an appropriate timeframe based on a consortium that is experienced with IS implementations. The work plan is **structured into 8 Work Packages (WPs).** The project management and governance activities will drive the entire work plan (WP1), which will be cross-cut with impact-oriented dissemination and exploitation activities (WP8). The scientific and technical activities will be undertaken in an interrelated workflow spanning WP2 to 7. As described in section 1.2, the different WPs (2, 3, 4, 5) represent interconnected compartments of work which are eventually integrated into the H4C demonstration (WP6) and the digital collaboration platform built in WP7. **The technical management of the project (WP1), undertaken by UT, will ensure the full coherence and alignment of the work** comprised by each WP towards fulfilling the project objectives. **WP2,** led by **AEE**, will identify and evaluate the circularity pathways within each H4C (German, Dutch, Basque and Turkish) to further develop and integrate advanced technologies. To include the participation of the surrounding ecosystems into the hubs, **WP3**, led by **TUDO**, will analyse the local setting, identify relevant stakeholders to engage with and collect useful insight for the improvement of the hub. **WP4**, led by **UT**, will utilise AI, operation research and agent-based simulation to ensure efficient operations and expansion of each H4C and overcome obstacles along the value chain. **WP5**, led by **KPMG**, will conduct a comprehensive analysis of existing business models in CE and IS, identify their strengths and weaknesses, and develop a new business model that capitalizes on emerging industry trends and technologies. Integrating the research and knowledge gathered from previous WPs, **WP6**, led by **PROV**, will test and demonstrate the technologies developed in **WP2** in each H4C. The technologies' performance will be evaluated to identify possible optimisation areas. Finally, **WP7**, led by **IRIS**, will build a userfriendly, scalable and adaptable **Digital Collaboration Platform for Circularity Hubs (DigitalH4C)** which will facilitate the sharing or resources and data among participating entities, fostering collaboration and transparency.

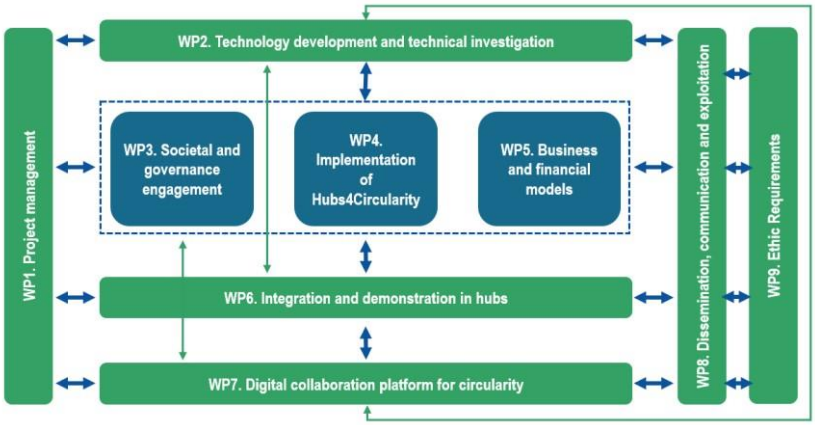


*Figure 11. **IS2H4C**'s PERT*

*Table 8. List of work packages*

| WP | Title | Leader | PMs | Start | End |
|---|---|---|---|---|---|
| 1 | **Project Management** | UT | | 1 | 49 |
| 2 | **Technology development and technical aspects** | AEE | | 1 | 49 |
| 3 | **Societal engagement and governance** | TUDO | | 4 | 49 |
| 4 | **Modelling and optimisation for H4C development** | UT | | 4 | 49 |
| 5 | **Business and financial models** | KPMG | | 1 | 49 |
| 6 | **Integration and demonstration in hubs** | PROV | | 1 | 49 |
| 7 | **Digital Collaboration Platform for H4C** | IRIS | | 1 | 49 |
| 8 | **Dissemination, communication, and exploitation** | EEIP | | 1 | 49 |
| 9 | **Ethics Requirements** | UT | | 1 | 49 |

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# GANTT CHART

| | Year 1 | | | | | | | | | | | | Year 2 | | | | | | | | | | | | Year 3 | | | | | | | | | | | | Year 4 | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 |
| **WP1. Project management** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T1.1 Decision-making management | | | D1.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T1.2 Administrative and financial management | | | D1.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T1.3 Contractual management | | | D1.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T1.4 Quality management | | | D1.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T1.5 Data management | | | | | | D1.2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D1.2 |
| **WP2. Technology development and technical investigation** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T2.1 Identification, classification, and analysis of key technologies | | | | | | D2.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T2.2 Development of energy technologies | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D2.2 | | | | | | | | | | | | | | | | | | | |
| T2.3 Development of waste recovery and utilization technologies | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D2.3 | | | | | | | | | | | | | | | | | | | |
| T2.4 Development of carbon capture & methanation technologies | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D2.4 | | | | | | | | | | | | | | | | | | | |
| T2.5 Producing e-methanol from captured carbon | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D2.5 | | | | | | | |
| T2.6 Synthesis of dicarbamates from CO2, methanol and diamines | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D2.6 | | | | | | | |
| T2.7 Designing and developing NIPU production from captured carbon | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D2.7 | | | | | | | |
| T2.8 Performance analysis & optimisation via simulations | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D2.8 |
| **WP3. Societal engagement and governance** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T3.1 Addressing stakeholders´ needs and interests | | | | | | | | | | | | | D3.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T3.2 Establishing the working structure for integrating topics | | | | | | | | | | | | | | | D3.2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T3.3 Development of socio-economic indicators to measure regional impact of hubs | | | | | | | | | | | | | | | | | | D3.3 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T3.4 Developing a management guideline for integrating non-technological aspects into the hubs | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D3.4 |
| **WP4. Models for H4C development** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T4.1 Modelling and geolocation of resources, energy, and Industrial Symbiosis | | | | | | | | | | | | | | | | | | | | | | | | | D4.1 | | | | | | | | | | | | | | | | | | | | | | | | |
| T4.2 Matchmaking for industrial symbiosis | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D4.2 | | | | | | | | | | | | | | | | | | | |
| T4.3 Hub logistics and supply chain management | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D4.3 | | | | | | | | | | | | |
| T4.4 Energy system operation strategies and optimisation | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D4.4 | | | | | | | |
| T4.5 Agent-based analysis of H4C | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D4.5 |
| T4.6. Impact assessment and decarbonization pathways for H4C | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D4.6 | | | | | | | | | | |
| T4.7 Top-down design of virtual hubs and their integration on expansion pathways for long-term impact | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D4.7 |
| **WP5. Business and financial models** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T5.1 Objective setting, scoping and baselining | | | | | | D5.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T5.2 Measuring and valuation | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D5.2 | | | | | | | D5.3 |
| T5.3 Designing IS impact measurement approach | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D5.2 | | | | | | | D5.3 |
| T5.4 Develop a comprehensive financial plan for the H4C business models | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D5.3 |
| T5.5 Business & financial models for society | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D.5.2 | | | | | | | D5.3 |
| **WP6. Integration and demonstration in hubs** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T6.1 Definition of the current state-of-the-art and validation of success | | | | | | | | | | | | | | | | | | D6.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T6.2 Commissioning, set up and initial testing of technologies | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D6.2 | | | | | | | | | | | | |
| T6.3 Monitoring of operational performance and evaluation | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D6.3, D6.4 |
| **WP7. Digital collaboration platform for circularity** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T7.1. DigitalH4C Platform architecture design | | | | | | | | | | | | | D7.1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T7.2. Online marketplace | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D7.2 | | | | | | | | | | | | | | | | | | | |
| T7.3. Virtual collaboration tools | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D7.3 | | | | | | | |
| T7.4. Integration of intelligent matchmaking and decision support components | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D7.3 | | | | | | | |
| T7.5. Integration of monitoring and evaluation components | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D7.3 | | | | | | | |
| T7.6. Cyber security, data protection and compliance | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D7.4 |
| T7.7. Evaluation and version release deployment of Digital H4C | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D7.5 |
| **WP8. Dissemination, communication and IP** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T8.1 Dissemination and communication plan & visual identity | | | | | | D8.1 | | | | | | | | | | | | | | | | | | | D8.2 | | | | | | | | | | | | | | | | | | | | | | | | D8.3 |
| T8.2 Collaborative ecosystem: Building the Circular Industry Helix | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D8.8 |
| T8.3 Clustering with other EU projects and initiatives | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D8.4 |
| T8.4 Exploitation plan and IPR management | | | | | | D8.5 | | | | | | | | | | | | | | | | | | | 8.6 | | | | | | | | | | | | | | | | | | | | | | | | D8.7 |
| T8.5 Impact analysis | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D8.4 |
| **WP9. Ethics requirements** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| T9.1 Ethics requirements | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D9.1 |

### 3.1.2 SUBCONTRACTING, PURCHASE AND OTHER COSTS CATEGORIES ITEMS

The beneficiaries must base their contracts/subcontracts according to the principles for best value for money and absence of any conflict of interest (according to Articles 6.2 and 9.3 of GA). Beneficiaries that are 'contracting authorities' or 'contracting entities' (within the meaning of the EU public procurement Directives 2004/18/EC and 2004/17/EC or any EU legislation that replaces these Directives) must moreover comply with the applicable national law on public procurement.

*Table 3.1.g 'Subcontracting costs' items*

| **1. UT** | **Cost (€)** | **Justification** |
|---|---|---|
| Subcontracting 1 | 80,000 | WP8 – Online games: UT develops 4 Excel-based Circular Hub online games which need to be taken into the digital-twin version by a company which can develop interfaces where the games can be run. An average of 20.000 € are assumed per digital twin game. Online digital-twin hub games provide an experiential e-learning environment to boost the learning process of students, practitioners, society, & policymakers. |
| **Total** | 80,000 | |
| **3. TUDO** | **Cost (€)** | **Justification** |
| Subcontracting 1 | 80,000 | WP3 - The subcontractor will analyse potential for job creation and inclusiveness in the four demo hubs. It will develop recommendations for regional stakeholders on how to unfold potential for creating jobs and monitor job creation and inclusive growth progress |
| **Total** | 80,000 | |
| **6. FHG** | **Cost (€)** | **Justification** |
| Subcontracting 1 | 44,000 | WP2 - The budget is used to commission a plant manufacturer to work out a concept for scaling up monomer synthesis so that the industrial implementation can be estimated. |
| **Total** | 44,000 | |
| **18. TUPRAS** | **Cost (€)** | **Justification** |
| Subcontracting 1 | 400,000 | WP6 – Turkish Hub Infrastracture (Civil, electrical, mechanic support, piping, utility supply etc.)<br>Installation of three main demo units into refinery site will require site modifications. Concrete ground for units, electricity grid connections to power the units, piping for flue gas and H2, piping connections between the different modules, storage tanks for intermediate and final products, cooling water, dry air connections are some of the main works needs to be done for site preparation. |
| **Total** | 400,000 | |
| **22. SIDENOR** | **Cost (€)** | **Justification** |
| Subcontracting 1 | 35,0000 | WP6 - H2 trials monitoring and tests |
| **Total** | 35,0000 | |
| **13. H2HT** | **Cost (€)** | **Justification** |
| Subcontracting 1 | 80,000 | WP6 Dutch Hub - A certified party is required for connection/commissioning. Rate of at least €100 per hour. Company subcontracting support |
| **Total** | 80,000 | |
| **16. ALMELO** | **Cost (€)** | **Justification** |

| | | |
|---|---|---|
| Subcontracting 1 | 80,000 | Cost for the grid operator will include: feasibility study and project management; modification of existing infrastructure to make it suitable for hydrogen á 500m in Aadorp €80k; |
| Subcontracting 2 | 25,000 | WP6 – Dutch Hub - Adjustment of house connections and replacement of meters in 5 homes |
| Subcontracting 3 | 75,000 | WP6 – Dutch Hub - Construction of a replacement gas network for natural gas distribution of 500m |
| Subcontracting 4 | 300,000 | WP6 – Dutch Hub – Work needed for the construction of connection H2-Hub – Aadorp at 3000m |
| **Total** | 480,000€ | |

All depreciation costs for equipment, infrastructure or other assets in the project are in compliance with Article 6 and will be recorded in the appropriate beneficiary's accounts, purchased in accordance with Article 6.2.C of the grant agreement and written off in accordance with international accounting standards and the beneficiary's usual accounting practices.

*Table 3.1.h. 'Purchase costs' items (**T&S**: Travel and Subsistence, **OGS:** Other goods, works and services, **Remaining costs**: The record must list cost items in order of costs and starting with the largest cost item, up to the level that the remaining costs are below 15% of personnel costs).*

*Table 15. 'Purchase costs' items*

| **5. INEGI** | **Cost (€)** | **Justification** |
|---|---|---|
| OGWS | 31,250 | ., WP2+WP4: SW license; computing consumables.<br>WP8: Conference fees, gold open access publications. |
| Remaining | 20,500 | |
| **Total** | 51,750 | |

| **8. ICCS** | **Cost (€)** | **Justification** |
|---|---|---|
| Equipment | 55,000 | Monitoring and sensing technologies (40k) network and tracking devices (15k) for the development of monitoring and traceability test modules in the hubs. |
| Remaining | 12,000 | |
| **Total** | 67,000 | |
| **9. AEE** | **Cost (€)** | **Justification** |
| Equipment | 55,000 | Membrane Destillation (MD) Membranes and Module supplies (2 Modules Lab, 2 Modules pilot plant ≥10m²),<br>Reagents and Chemicals for MD operation and Laboratory Supplies |
| Remaining | 37,000 | |
| **Total** | 92,000 | |
| **13. H2HT** | **Cost (€)** | **Justification** |
| Equipment | 814,000 | Additions for the call in 2025-2026: 1) wind turbines of in total 70kW to add to the already available sunroof generation necessary to stabilize green hydrogen generation, to be depreciated in 10 years with no residual value, total costs estimated at €360,000 = 4x€36,000=€144,000; 2) energy storage of 1,4 MWh necessary to stabilize the feed of the green energy generation, to be depreciated in 10 years with no residual value, total costs estimated at €1,200,000 = 4x€120,000=€480,000;and 3) a hydrogen filling station for trucks (separately funded), and 4) double pipeline to the nearby wastewater treatment plant, €190,000. |
| Remaining | 10,000 | |
| **Total** | 824,000 | |
| **18. TUPRAS** | **Cost (€)** | **Justification** |
| Equipment | 1,300,000 | e-Meth production unit 2.5 M € (depreciation cost : 500K €) + Carbamate Production unit 300K € + NIPU Production Unit 300K €+ Electrolyser (depreciation cost 200K €) |

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| | | |
|---|---|---|
| OGWS | 86,000 | Consumables (chemical feedstock (diamines),absorbent- for capture (zeolites, activated carbon, silica), catalyst for chemical conversions, spare parts) 50K €+ C-capture unit transport form Izmit Refinery to Izmir Refinery 5K €+ Technology transport cost from DE to TR 20K € (estimated, including customs /shipping) + Auditing 5K €+ Conference fee 6K € |
| Remaining | 25,000 | |
| **Total** | 1,411,000 | |
| **22. SIDENOR** | **Cost (€)** | **Justification** |
| OGWS | 257,000 | Equipment & installation of Heater NG/$O_2$/$H_2$ – 160.000€ and Cover modification - 40.000€. Fuels (14kNm3 $O_2$ y 27*2 KNm3 H2): $O_2$ 0.22 €/Nm3 * 14.000 = 3.080 € & H2 1 €/Nm3 * 27.000 *2 = 54.000 € |
| Remaining | 12,000 | |
| **Total** | 269,000 | |
| **24. CALERA** | **Cost (€)** | **Justification** |
| OGWS | 76,000 | WP6 - To set-up the prototyping in CALERA's plant; shipping (10,000€) engineering works (5,000), foundation reinforcement (30,000), Cost of new pecking for a possible furnace with gas without sugar to capture $CO_2$ (6,000), heating system for the methanation reactor -Larger diameter & heat-insulated pipes (12,000) & Fuels, gases (15,000) |
| Remaining | 12,000 | |
| **Total** | 88,000 | |
| **25. LTK** | **Cost (€)** | **Justification** |
| OGWS | 180,000 | WP6 - The material cost covers the modifications to be implemented in the prototype to reach correct process pressure and temperature improving heating and pressure regulations and control. Heating and cooling equipment as well as insulation concepts have to be also reconsidered and implemented. |
| Equipment | 60,000 | WP6 - Lointek together with Tecnalia have developed a $CO_2$ Capture prototype based on membrane and PSA as well as a methane reactor to convert capture $CO_2$ with green hydrogen, proven with real industrial gases of Calcinor. |
| Remaining | 12,000 | |
| **Total** | 252,000 | |
| **26. SBS** | **Cost (€)** | **Justification** |
| OGWS | 70,000 | Materials for the compression of CO2 to be transported. Transportation buffers of CO2 at high pressure. Materials for the carbonation pilot tests. . |
| Remaining | 12,000 | |
| **Total** | 82,000 | |
| **27. IH** | **Cost (€)** | **Justification** |
| Equipment | 1,410,000 | Equipment needed for the deployment of the German Hub WP6 – in Industriepark Höchst for any pilot plant .; (1) 650.000 for preparation of infrastructure for hosting CC and E-methanol container, (2) 160,000€ for Carbon Capture demo (depreciated from 400000k sales price) and , (3) 600,000€ e-methanol demo (depreciated from 1.500.000 e-methanol plant sales price) |
| Remaining | 66,000 | |
| **Total** | 1,476,000 | |
| **29. IRIS** | **Cost (€)** | **Justification** |
| OGWS | | |

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| | | |
|---|---|---|
| Remaining | | |
| **Total** | | |
| **30. WDK** | **Cost (€)** | **Justification** |
| OGWS | | |
| Remaining | | |
| **Total** | | |
| **31. SOLENCO** | **Cost(€)** | **Justification** |
| OGWS | 49,000 | 5xDemonstrators: i) Hydrogen pressure handling and monitoring manifold for the pneumatic system (7,5k€ incl. pressure reducer, electromagnetic valves, pressure transmitters, fittings), ii) Power electronics for communication with Digital**H4C** (2,3k€ incl.microcontroller, I/Os for digital & analog signal monitoring & control, cabling, resistors). |
| Remaining | 10,000 | |
| **Total** | 59,000 | |

## *3.2* CAPACITY OF PARTICIPANTS AND CONSORTIUM AS A WHOLE

**IS2H4C** consortium brings together 32 partners from 10 member states (Netherlands, Spain, Portugal, Belgium, Turkey, Austria, Germany, Greece, Ireland and Belgium) and 10 members from 8 countries for the International Advisory Board (France, Austria, Sweden, Italy, Netherlands, Denmark, Canada, USA) which have been selected for their outstanding R&D, innovation performance and track record. Industry, financial, logistics, technological and academic partners lead the way in their respective areas of action across the Industrial Symbiosis network. Each member will have a distinctive role and will ensure that the results will be exploited in an efficient way. **IS2H4C** involves a multidisciplinary group comprising four types of entities: (i) Research centres/universities (33% of the consortium, accounting for 44% of the budget), (ii) large industries (24% of the consortium, 20% of the budget), (iii) SMEs (22% of the consortium, 19% of the budget), (iv) NGOs (21% of the consortium, 17% of the budget) which operate at different stages of the project. **UT**, **IS2H4C**'s project coordinator, has participated in several circular economy projects funded by different entities such as: (i) Designing sustainable second generation bioenergy supply chains: an agentbased modelling approach (Tech4People call); (ii) Secure Management Platform for Shared Process Resources (SHAREBOXH2020); (iii) Engineering an Information Platform for Sustainable, Smart, and Resilient Circular Construction Ecosystems (BMS Signature proposal), and more recently (iv) Serious gaming for Circular Economy Transition (SG4CET, WSV innovative teaching proposals). **UT** is #1 at Times Higher Education ranking for SDG 9 "Industry, Innovation and Infrastructure". UT is #6 in the world and #1 in Europe by a total of **31 Scopus-indexed articles** in the field of **"Industrial Symbiosis"**, being **Devrim Yazan**[26] (project coordinator) ranked #2 author in the world by number of articles in this field (author #1, Luca Fraccascia, is member of the advisory board). Devrim is an associate professor at the Industrial Engineering and Business Information Systems (IEBIS) Department.

[26] https://scholar.google.com/citations?user=ihB58yAAAAAJ&hl=en&oi=ao

Devrim has a PhD in Innovation Management and Product Development and a vast experience in industrial symbiosis. He is a member of A.SPIRE's Industrial and Urban Symbiosis working group contributing to SPIRE's 2050 roadmap *"Processes4Planet".*

*Figure 12. Geographical location of **IS2H4C** consortium*

## GERMAN HUB PARTICIPANTS (Germany)

**PROV** (German Hub representative) has bundled its interdisciplinary research and education projects in the field of industry and sustainability at the **Center for Industry and Sustainability (ZIN)** where upscaling and testing of new technologies is being held, specifically in the areas of green energy supply, green raw materials, circular economy and new business models, CCU & CCS, sustainable industrial area management and industrial-urban symbiosis. **Industriepark Höchst** and **ZIN** participate in the "*Process4Sustainability"* Cluster supported by EIT Climate-KIC, a EU knowledge and innovation community, working to transition to a zero-carbon economy. Process4Sustainability is linked and will enhance the replication considering that the Process4Sustainability cluster is based on more than 90 companies (e.g. BAYER, ARCUS, Kuraray, Celanese, among others) and the provider of more than 22,000 jobs. **IS2H4C** will have on board 2 of these companies to deploy and test **IS2H4C**'s innovative technologies. **IH** supplies companies in the process industry with products and services in the fields of energy, facility management, environmental protection, safety and waste disposal. IH is a leading service provider and operator of integrated facilities for the chemical, pharmaceutical and related process industries. IH supplies companies in the process industry with products and services in the fields of energy, facility management, environmental protection, safety and waste disposal. IH is investing considerable resources in making Industriepark Höchst climate-neutral, and is planning to expant the power supply to obtain the necessary green electricity for decarbonisation or to build more electrolysers to supply $H_2$ for PtX. **CIRCE** is a technology centre seeking to provide innovative solutions for a sustainable development. They work towards improving the competitiveness of enterprises through generation of technology transfer by means of R&D activities and market-oriented training within the field of resource sustainability and effectiveness, energy grids and renewable energies. CIRCE has been involved in IS and CE projects specifically in the areas of KPIs measuring performance and evaluation of symbiotic solutions based on environmental, economic ans technical feasibility criteria. In **IS2H4C**, CIRCE will contribute to the technological development of solutions in the hubs. **INEGI** is contributing to definitions and simulation of energy technologies, especially based on hydrogen, within different hubs and their simulation. One area of particular focus will be the potential for recovering heat from the hydrogen production plant, including the electrolyzers and compressors. Further, INEGI will also work on Waste heat recovery, heat pumps and heat-to-cold solutions. **AEE** is one of the leading EU institutes of applied research in renewable energy and resource efficiency. They specialise in decarbonisation, resource efficiency and digitalisation of Industrial systems.

## BASQUE HUB PARTICIPANTS (Spain)

**TEC** (Basque Hub representative) is the largest applied research and technological development centre in Spain, and a member of the Basque Research and Technology Alliance. **IS2H4C** has the participation of **Nora Fernandez**, projects' coordinator in the energy and environmenttransition unit division at Tecnalia. She is a specialist in energy transition technologies and roadmaps, and in their engineering optimisation through simulation tools. She, and has also experience in designing, manufacturing and testing of renewable energy technologies. **ACE** is a non-profit organisation within the framework of the Basque Government's policy to promote competitiveness of the industrial sector. **SKN** is one of the leading suppliers of corrugated cardboard packaging in the world, with operations in 23 EU countries and 13 countries in America. Their role involves (1) research in the areas of paper waste for Biofuels production (Ethanol), sludge to produce biomethane, and lignin for biofuels production; (2) providing consumption/generation data and validating industrial feedstock, subproducts, energy and potential synergies quantification; (3) investigating governance and collaborative business models, potential co-investment schemes with other industries. **NORTEGAS** is the second biggest natural gas company in Spain, with more than 8,100 km of distribution infrastructure of natural gas and LPG within the Basque country, Asturias and Cantabria. During the project, they will be in charge of (1) validating transport and distribution modes of $H_2$, $CO_2$ and $O_2$, (2) investigating the potential use of waste for biomethane production and (3) governance and ownership of biomethane plants. **CALCINOR** is a holding company made up of nearly 50 companies and around 1,500 workers, with a production presence on three continents and serving markets on five continents. CALCINOR has a diversified business structure, which ranging from the mining activity of various deposits to industrial processes based on latest technologies. In addition, **LTK** is a private company solidly established in the international market for engineering, supply, manufacturing and service in the capital goods sector for the oil refining, chemical, petrochemical and energy industry. CALCINOR and LTK will be in charge of the validation of the carbon capture and methanation technology. With a production capacity of 12 million tons/year, **PETRONOR** is the biggest refinery in Spain. Among their objectives as a company, the following stand out: continuous quality improvement and commitment to minimise environmental impacts. Within this deployment, PETRONOR provides e-fuels and electrolysis infrastructure, a key asset for the decarbonisation of the Basque country. **SIDENOR** is a leader in the steel sector and supplier of calibration products. They will contribute to the demonstration of oxycombustion and $H_2$ in the steelmaking process, support the investigation of the potential carbonation of steel waste for other sectors, and finally will provide consumption/generation data and validate industrial feedstock. **CABB** is the public utility in charge of supply and sanitation, for approximately one million inhabitants. CABB owns and operates Galindo WWTP (wastewater treatment plant) which produces around 93 $Hm^3$ per year of treated water. The sludge produced in this plant is burned in the incineration industrial furnaces located at the WWTP. Due to this incineration $CO_2$, among other gases, are expelled to atmosphere and could be recuperated. Finally, **SBS**, develops technologies that provide solutions to waste management problems. SBS is the High Efficiency Contact technology, a patented technology that allows the thermal treatment of all types of waste, improving the process conditions to maximize the desired products (greater efficiency) with a lower consumption of energy resources (greater efficiency) developing a more sustainable heat treatment processes, such as pyrolysis, gasification, drying or roasting.

## TURKISH HUB PARTICIPANTS (Turkey)

Operating four oil refineries with a total of 30 million tons annual crude oil processing capacity, **TUPRAS** is the first producer in Turkey's refining sector and largest industrial enterprise of the country with its subsidiary in sea transportation, Ditaş, its subsidiary in railway transportation, KUAŞ (Körfez Ulaştırma A.Ş.) and affiliate in fuel oil distribution sector, Opet. The company, prioritising long-term sustainable investments with low carbon emissions in its plans, takes steps to fulfil its responsibilities, especially in achieving the goals set within the scope of environmental policies such as the Paris Agreement. TUPRAS approaches the investments as the key to a sustainable high performance. With this approach, thanks to the investments to save energy and reduce emissions, annual energy savings and reductions in $CO_2$ emissions are achieved at TUPRAS. In **IS2H4C**, they will provide $H_2$ and CC processes to develop polymers. In the hub, TUPRAS will be site demonstrator of the innovative technologies including the valorisation of captured $CO_2$ and green hydrogen in the production of fossil-free polymers. **ARCELIK** is the third largest company in home appliances sector in terms of total sales in EU. ARCELIK innovates

products and services to reach a closed circular economy by increasing products' recyclability rates, the usage of recycled content in products and their packaging, reusing products and parts, eliminating potentially harmful substances, and by effectively managing the end-of-life processes including take-back and recycling. BioFridge is made using both biobased polyurethane insulation material (Bio-Cool) and biocomposite raw materials that contain organic material such as soy and castor oil. In this project, ARCELIK will use a polyurethane insulation material made from captured carbon dioxide in refrigerators instead of a petroleum-based polyurethane, in addition to the use of recycling and biomaterials to reduce the carbon footprint. **FHG** is the world's leading applied research organisation on fields such as bioeconomy, AI, next generation computing, resource efficiency and climate technologies, and $H_2$ technologies. In **IS2H4C**, FHG will produce e-methanol from CC to support the polymer production process of TUPRAS.

## DUTCH HUB PARTICIPANTS (The Netherlands)

**H2HT** is a physical place where entrepreneurs, governmental organisations, research & knowledge institutions collaborate to develop applications of $H_2$ technology. Knowledge & experience is gained through research and infrastructure test projects. Applications include: Setting up (pilot) projects in the field of generation & storage, mobility, high-temperature combustion and industrial processes. **ALMELO** as the first-of-a-kind Positive Energy District in Twente will allow for local balancing reducing energy distribution over longer distances, facilitate peak shaving and load shifting using concepts of demand response and reduced curtailment of renwable energy resources, and district-level self-consumption. **SOLENCO** is the manufacturer of scalable, modular, green $H_2$ conversion systems to store renewable energy for days, weeks and even months, to produce power and heat. SOLENCO enjoys worldwide rights over its patented technology for a reversible fuel cell, a bipolar electrode and a unique application. Solenco will contribute to the Dutch Hub by providing knowledge regarding $H_2$ processes.

## OTHER PARTICIPANTS

**IS2H4C** is supported by a strong pillar of research centers and universities, leaders in circularity in EU as: **MUL** addresses social challenges in the areas of resources, climate, energy and environment is a key priority, therefore they work together to develop technical and scientific to support the adaptation of social behaviour. Within **IS2H4C**, **AEE** work alongside MUL to provide key knowledge for the identification and analysis of technologies. **ZLC** is a research and educational institute established in Logistics and Supply Chain Management, which is also the role they play in **IS2H4C**, by the Government of Aragon in Spain. TUDO (German Hub participant) is particularly distinguished in the research fields of (i) society and transformational processes and (ii) innovations for sustainability. **IS2H4C** has the participation of **Dr. Michael Kohlgrüber** associate researcher at the new scientific institution of TUDO *"Social Investigation Centre Dortmund (SFS german acronym)"*. **IS2H4C** is supported by a deep experience group of entities. **EEIP** is specialised in smart B2B content marketing and communication, dissemination and exploitation activities. EEIP is an active partner of HE projects such as CORALIS, RETROFEED, RE4iNDUSTRY and DEESME. Together, EEIP and **CHX** collaborate in the implementation of activities for WP8 (DEC) to enhance **IS2H4C**'s expansion across EU. CHX is a pan-EU Open Innovation Network with more than 600+ member organisations, selected research organisations, SMEs, and industries, from 57 countries, and is present in all EU member states. The network membership delivers 13,000+ collaborative international projects (23% as project coordinator), highlighting the quality of its membership base. **IMMIB** is a professional establishment in affiliation with Ministry of Trade, Republic of Türkiye. They have a vast experience in fairs and exhibitions, B2B meetings, market research activities, seminars, trainings, workshops, educational activities. Other key entities of the consortium are: **IRIS,** a leading EU company in manufacture of photonic and artificial intelligence solutions for the control of industrial and production processes in real time. IRIS is a current member of SPIRE and has participated in diverse projects such as SHAREBOX and SCALIBUR, where they integrated AI solutions such as advanced machine learning, agent-based modelling and big production data, among others. In this deployment, they are in charge of developing the digital collaboration platform, to optimise the data transfer between the companies involved. **WDK** will support the platform development with their experience in management software in WP7. **ICCS** is a research institute linked to the National Technical University of Athens and is the research host of the School of Electrical and Computer Engineering. ICCS aims to promote research and development in the areas of hardware and software technology, digital communication technologies, information systems and

management decision technologies, among others. **KPMG** operates as a global network of independent member firms offering audit, tax and advisory services; working closely with clients. KPMG Tax Incentives and Assurance teams, in Portugal, have an international network, which can help organisations better understand the complexities of EU law,how this can impact their business, and how to leverage opportunities. KPMG has a multidisciplinary team with technical knowledge acquired across several different focus areas, specifically business development, investment analysis, tax incentives advice, R&D, together with strong knowledge on how to commercially exploit innovative solutions and introduce them into market, in diverse sectors such as environment, climate, biotechnology, and health.

*Figure 13. Role of partners in **IS2H4C***

| PARTNER | UT | PROV | MUL | TUDO | TEC | INEGI | FHG | CIRCE | ICCS | AEE | ZLC | EEIP | IMMIB | H2HT | ACE | CABB | ALMELO | KPMG | TUPRAS | ARCELIK | SKN | NORTE | SIDE | PETRO | CALCI | LTK | SBS | IH | CHX | IRIS | WDK | SOL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ROLE | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Project management | ✓ | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | ✓ | |
| Identification and analysis of technologies | ✓ | ✓ | ✓ | | | ✓ | | | | ✓ | | | | | | | | | | | | | | | | | | | | | | |
| Technological development | | | | | ✓ | ✓ | ✓ | ✓ | | ✓ | | | | | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | | | ✓ |
| Techno-economic analysis | ✓ | | | | | | | | | | ✓ | | | | | | | | | | | | | | | | | | | | | |
| Societal engagement and governance | | | | ✓ | | | | | | | | | | | | | | ✓ | | | | | | | | | | | | | | |
| Development of socio-economic indicators | ✓ | | | ✓ | | | | | ✓ | | ✓ | | | | | | | | | | | | | | | | | | | | | |
| Resource characterisation | | | ✓ | | ✓ | | | | | ✓ | | | | | | | | | | | | | | | | | | | | | | |
| Matchmaking for industrial symbiosis | ✓ | | | | | ✓ | | ✓ | ✓ | | | | ✓ | | | | | ✓ | | | | | | | | | | | | | | |
| Logistics and supply chain management | ✓ | | | | | | | | | | ✓ | | | | | | | | | | | | | | | | | | | | | |
| Decarbonization pathways | | | ✓ | | ✓ | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Development of business model and financial plan | | ✓ | | ✓ | | | | | | | | | | | | | | ✓ | | | | | | | | | | | | | | |
| Demonstration of H4C | ✓ | ✓ | | | ✓ | | ✓ | ✓ | | | | | | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ |
| Development of the Digital Collaboration Platform | ✓ | | | | | | | | ✓ | | | | | | | | | ✓ | | | | | | | | | | | | ✓ | ✓ | |
| Dissemination, communication and exploitation | ✓ | | ✓ | | | | | | | | ✓ | ✓ | ✓ | | ✓ | ✓ | | | | | | | | | | | | | ✓ | | | |
| Expansion strategy for Hubs | ✓ | ✓ | | | ✓ | | ✓ | ✓ | | | | ✓ | ✓ | ✓ | ✓ | ✓ | | | | ✓ | | | | | | | | ✓ | ✓ | | | |

## International Advisory Board

**IS2H4C** is being supported by an international Advisory Board which will ensure the expansion of the project covering different advisory areas as it is possible to be analysed in the table 16. :

*Table 16. **IS2H4C**'s International Advisory Board*

| Advisory Board Entities | Key sector to support - IS2H4C |
|---|---|
| Veolia Environnement S.A | Solid waste and wastewater treatment, pyrolysis, waste energy management |
| EU Sustainable Energy Innovation Alliance | Dissemination of findings, reach-out to entire Europe, event organisations |
| CEN CENELEC | Standardisation activities |
| Boden Industrie Park | $H_2$ for steel, electrolysis, wind energy |
| Sapienza University of Rome | IS information-sharing platforms, circular economy models, consumer acceptance |
| Technical University of Denmark | Energy systems, forecasting, decision support tools, energy markets, district heating |
| Politecnico di Bari | Industrial symbiosis, operations of IS networks |
| University of Windsor | Integrated energy systems, wind energy, waste-to-energy |
| University of Texas, Dallas, USA | Information systems and finance |

Associated with document Ref. Ares(2023)8308660 - 05/12/20

## General

During the execution of the **IS2H4C** project appropriate occupational health and safety will be followed by all researchers. The beneficiaries will ensure that all ethics issues related to activities in the grant are addressed in compliance with ethical principles, the applicable international and national law, and the provisions set out in the Grant Agreement. This includes the ethics issues identified in this report and any additional ethics issues that may emerge in the course of the grant. In case any substantial new ethics issues arise, beneficiaries should inform the granting authority. In the following sections the **IS2H4C** consortium's ethical approach towards, personal data, non-EU countries, Environmental health and safety, Artificial Intelligence as well as Recommendations and adoption of the Ethics Summary Report outcomes will be outlined.

## 4.1 Personal data

**IS2H4C** will adhere to legislative requirements of the General Data Protection Regulation. In case any activity of project requires the collection and processing of personal data during the project lifetime, the beneficiaries collecting, processing and hosting the data will fulfil the specific laws (i.e. General Data Protection Regulation, GDPR, EU2016/679) This will also be safeguarded in close cooperation with the Data Protection Officer (DPO). The beneficiaries confirm that they are aware of the following rules for collection and processing of personal data:

- The host institution must confirm that it has appointed a DPO and the contact details of the DPO are made available to all data subjects involved in the research. For host institutions not required to appoint a DPO under the GDPR a detailed data protection policy for the project must be kept on file.
- A description of the technical and organisational measures that will be implemented to safeguard the rights and freedoms of the data subjects/research participants must be submitted as a deliverable.

Detailed information on the informed consent procedures in regard to data processing must be kept on file.

## 4.2 Non-EU countries

In case any activity of project requires to be performed in a non-EU country, also these activities will be performed according to highest ethical standards, and applicable international, EU and national law, also taking all relevant ethical aspects into account.

## 4.3 Environmental health and safety

**IS2H4C** does not raise considerable risks to health and safety of researchers and participant and does not involve the use of potentially harmful substances, processes, technologies, organisms or materials in a manner that raises significant concerns about individual security or safety or the protection of public health. UT (Coordinator) complies with National and European legislation on Protection of Environment and Occupational Safety. All **IS2H4C**' partners have their own Occupational Health and safety procedures that comply with their national legislation.

## 4.4 Artificial intelligence

The EC's Artificial Intelligence Act will be applied. This act outlines new rules & actions aiming to turn EU into the global hub for trustworthy AI, whereby by setting the standards, EU can pave the way to ethical technology worldwide & remains competitive along the way. Following the Ethics Guidelines for Trustworthy AI trustworthy:

i) lawful - respecting all applicable laws and regulations, ii)

ethical - respecting ethical principles and values and iii) robust -

from a technical perspective while considering its social environment.

## 4.5 Recommendations and adoption of the Ethics Summary Report outcomes

The work package WP9, dedicated to Ethics has been added to the work plan of the project. The **IS2H4C** consortium will appoint an independent external Ethics Advisor (EA) who has proven competences and expertise, providing advice on the ethical aspects appropriate to the complexity of the ethical aspects, in the project context at project start. The mandate of the EA would e.g. include an advisory role and assessments concerning the ethical merits of the work performed and providing independent recommendations. The exact details of the mandate will be outlined in a Memorandum of Understanding. The recommendations of the Ethics Advisor will be followed up closely and all activities performed in relation to ethics will be summarised in a report.

**ANNEX 2**